Expository notes on Brylinski–Getzler

A companion to J.-L. Brylinski and E. Getzler, *The homology of algebras of pseudo-differential symbols and the non-commutative residue*, K-Theory **1** (1987), 385–403

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Abstract

These notes are a reader's companion to the paper [7] of J.-L. Brylinski and E. Getzler. The paper computes the Hochschild and cyclic homology of the algebra $\mathcal{P} = \Psi^{\mathbb{Z}}(M)/\Psi^{-\infty}(M)$ of pseudodifferential symbols on a manifold M of dimension n:

$$\mathrm{HH}_k(\mathcal{P}) \cong H^{2n-k}(S^*M \times S^1; \mathbb{C}),$$

and deduces the existence and uniqueness of the non-commutative residue of Wodzicki and Guillemin. The intended reader knows pseudodifferential operators and the symbol calculus from Paul Loya's notes [23, 24]. We do not assume any knowledge of Hochschild homology, spectral sequences, Poisson geometry, deformation quantization or sheaf theory, and we develop what is needed of each from the beginning, always starting from an example. Then we follow the five sections of [7] in order and write out the arguments the paper compresses. These include the identification of the first differential with the Koszul–Brylinski operator δ, with its constant; Brylinski's symplectic star operator; the commutator identity for conformal Hamiltonian fields; and the degeneration of the spectral sequence. We also prove its convergence. This point is treated only briefly in [7], and it depends on the choice of completed tensor products. We give a complete elementary proof that the residue is, up to a factor, the only trace on $\mathcal{P}$ when M is connected and $n \geq 2$. We work the model cases $\mathbb{R}^n$, the circle, the torus, spheres, the Weyl algebra and differential operators on $\mathbb{R}^n$. The sheaf-theoretic method of [7, §5] is explained through Weil's proof of de Rham's theorem on the circle, and its microlocal inputs are stated precisely and quoted. We record a number of corrections to [7]. The two that change statements are these: the "in particular" description of $\mathrm{HH}_*(\mathcal{P})$ by the cohomology of M in the main theorem is false in general, for instance for $M = S^2$; and the constant in the zeta-function formula for the residue depends on the order of the operator A, not only on $\dim M$. The others concern signs, normalizations and misprints. Section 2.3 lists what is quoted and Section 2.4 lists the corrections.

1 Introduction and roadmap

What this section does. We state the main results of [7], describe the two methods of proof, and explain how these notes are organized and which parts can be skipped on a first reading.

1.1 What Brylinski and Getzler compute

Let M be a closed manifold of dimension n. In Loya's notes [23] one meets the classical pseudodifferential operators $\Psi^m(M)$ of integral order m, with full symbols $p \sim p_m + p_{m-1} + \cdots$ whose terms p_{m-j} are homogeneous of degree $m - j$ in ξ. The operators of all orders form an algebra $\Psi^{\mathbb{Z}}(M) = \bigcup_m \Psi^m(M)$. The smoothing operators $\Psi^{-\infty}(M) = \bigcap_m \Psi^m(M)$ form an ideal. The quotient

$$\mathcal{P} = \mathcal{P}(M) = \Psi^{\mathbb{Z}}(M)/\Psi^{-\infty}(M)$$

is the *algebra of (complete) symbols*. An element of $\mathcal{P}$ is a formal sum $a = \sum_{j \leq m} a_j$ of smooth functions on $\dot{T}^*M = T^*M \setminus 0$ with a_j homogeneous of degree j. The product is given by the composition formula

$$a \# b \sim \sum_\alpha \frac{1}{\alpha!} \partial_\xi^\alpha a \, D_x^\alpha b, \qquad D_x = \tfrac{1}{i}\partial_x, \tag{1}$$

in local coordinates (Section 7). BG write Ψ^∞ for our $\Psi^{\mathbb{Z}}$.

Hochschild homology $\mathrm{HH}_*(A)$ is a sequence of vector spaces attached to an algebra A (Section 3). The first one is $\mathrm{HH}_0(A) = A/[A, A]$, and its dual is the space of *traces* on A. For a

commutative algebra of functions on a manifold, HH_k is the space of differential k-forms (Connes' theorem). Cyclic homology $\mathrm{HC}_*(A)$ is a refinement which, for functions on a manifold, sees de Rham cohomology (Section 4).

The main theorem of [7] reads as follows. Here S^*M is the cosphere bundle, which has dimension $2n-1$, and S^1 is a circle. We explain later where the circle comes from.

Theorem 1.1 (Brylinski–Getzler [7, §4]; Wodzicki [36])**.** *Let M be a closed manifold of dimension $n \geq 1$. With the chain complexes of Section 7.6,*

$$\mathrm{HH}_k(\mathcal{P}) \cong H^{2n-k}(S^*M \times S^1; \mathbb{C}), \qquad \mathrm{HC}_k(\mathcal{P}) \cong \bigoplus_{j \geq 0} H^{2n-k+2j}(S^*M \times S^1; \mathbb{C}).$$

In particular, if M is connected and $n \geq 2$, then $\mathrm{HH}_0(\mathcal{P}) \cong \mathbb{C}$. The residue

$$\mathrm{res}(A) = (2\pi)^{-n} \int_{S^*M} a_{-n}\, \iota_R\Big(\frac{\omega^n}{n!}\Big)$$

is a trace on $\mathcal{P}$. Here a_{-n} is the term of homogeneity $-n$ in the full symbol of A, ω is the symplectic form and R is the radial vector field. Every trace on $\mathcal{P}$ is a multiple of res*.*

The theorem is proved in Theorems 11.3, 11.7, 12.8 and 12.9. The last sentence, uniqueness among *all* traces with no continuity assumption, is stronger than the statement in [7], which concerns continuous traces. Its elementary proof (Section 12) follows the argument used for the algebra of operators in [26]. The case $n = 1$ is different: there are two residues, one for each direction of ξ.

1.2 Two methods

BG give two proofs.

The semiclassical method ([7, §§1–4]). The order of a symbol behaves like a power of Planck's constant: the leading term of $a\#b$ is the commutative product ab, and the next term of the commutator is the Poisson bracket, $\sigma([A,B]) = -i\{a,b\}$. Filtering Hochschild chains by order gives a *spectral sequence* (Section 5), a bookkeeping device that computes homology one order at a time. Its first page E^1 is the space of differential forms on $\dot{T}^*M$ (Connes' theorem). The first differential is the Koszul–Brylinski operator δ, which is built from the Poisson bracket in the way d is built from the product (Section 6). A symplectic analogue of the Hodge star turns δ into d, so the second page E^2 is de Rham cohomology of *homogeneous* forms. A homogeneity argument with the radial vector field puts E^2 on a single line, and this forces the spectral sequence to stop there. The circle in $S^*M \times S^1$ appears because a homogeneous form of degree 0 on $\dot{T}^*M \cong S^*M \times (0,\infty)$ can contain dr/r, and dr/r is not exact among homogeneous forms.

The sheaf-theoretic method ([7, §5]). This is modelled on Weil's proof of de Rham's theorem: prove a "Poincaré lemma" on small open sets and glue (Section 14). For differential operators the Poincaré lemma says that $\mathrm{HH}_*(\mathcal{D}(\mathbb{R}^n))$ is $\mathbb{C}$ in degree $2n$ and zero otherwise. For symbols, BG use the microdifferential operators of Sato, Kashiwara and Kawai on a complexification, and the circle becomes $\mathbb{C}^\times$.

1.3 Reading guide

The notes are long because they supply background. A reader who wants the main line can proceed as follows.

- *Essential:* Section 1, Sections 3.1–3.5 of Section 3, Section 5, Sections 6.4–6.8 of Section 6, Section 7 up to Section 7.5, Section 8, Sections 9.1–9.3 of Section 9, Section 11 and Section 12 up to Section 12.6.
- *Skip on first reading:* Section 3.7 on topological tensor products (accept that the formulas hold); Section 9.5 on smooth quantum algebras; Appendices A and B, which prove Connes' theorem and compare two chain models; and the correction remarks, which can be read at the end.
- *Independent of the main line:* Section 4 and Section 10 (cyclic homology); Section 13 (differential operators); and Section 14 (the sheaf-theoretic method). Section 15 collects worked examples and can be consulted at any time.

Figure 1 shows the logical dependencies.

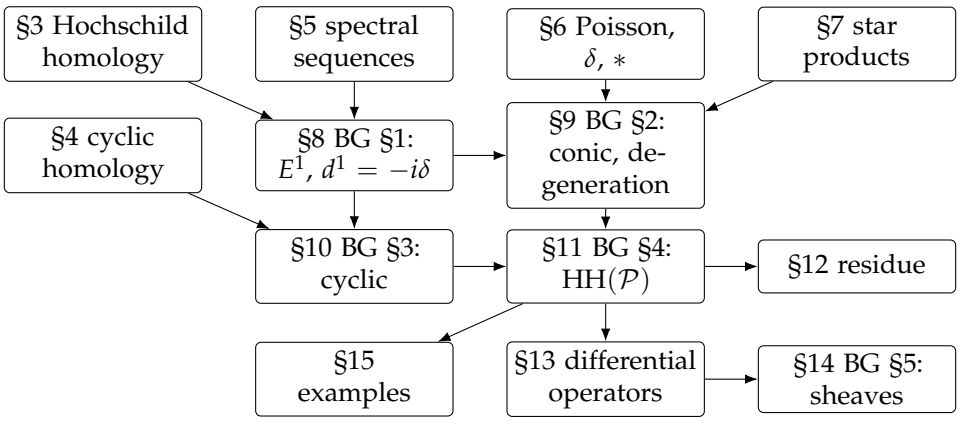


Figure 1: Logical structure of the notes. An arrow means "is used by". Appendix A (Connes' theorem) supports Section 8; Appendix B (the uncompleted chain model) supports Section 11.

1.4 How the notes follow BG

Sections 3–7 are background. Then

[7, §1], homology of quantum algebras	Section 8
[7, §2], δ on conic symplectic manifolds	Section 9
[7, §3], cyclic homology of quantum algebras	Section 10
[7, §4], symbols, residue, higher residues	Sections 11–12
[7, §§4–5], differential operators	Section 13
[7, §5], Poincaré lemmas and sheaves	Section 14

Every statement of [7] that we use is restated, with its hypotheses, before it is proved or quoted.

2 Conventions, dictionary, quoted background and corrections

What this section does. We fix sign conventions, compatible with Loya's notes; give a dictionary between the notation of [7], of these notes and of Loya's notes; list everything we quote without proof; and list our corrections to [7].

2.1 Conventions

We follow [23, §"Pseudodifferential parametrices"]: $D_x = \frac{1}{i}\partial_x$, the left quantization

$$\mathrm{Op}(a)u(x) = (2\pi)^{-n} \iint e^{i(x-y)\cdot\xi} a(x,\xi) u(y)\, dy\, d\xi,$$

and the composition formula (1). The symbol of D_{x_j} is ξ_j. On T^*M, in canonical coordinates (x,ξ), we use

$$\omega = \sum_{j=1}^{n} d\xi_j \wedge dx_j, \qquad \alpha = \sum_j \xi_j\, dx_j, \qquad d\alpha = \omega, \qquad \mathrm{vol} = \frac{\omega^n}{n!},$$

$$\{a,b\} = \sum_j \Big(\frac{\partial a}{\partial \xi_j}\frac{\partial b}{\partial x_j} - \frac{\partial a}{\partial x_j}\frac{\partial b}{\partial \xi_j} \Big), \qquad H_a = \sum_j \Big(\frac{\partial a}{\partial \xi_j}\partial_{x_j} - \frac{\partial a}{\partial x_j}\partial_{\xi_j} \Big), \qquad \{a,b\} = H_a b.$$

Then $\iota_{H_a}\omega = -da$. The radial vector field is $R = \sum_j \xi_j \partial_{\xi_j}$, and $\iota_R\omega = \alpha$. The dilations are $\rho_t(x,\xi) = (x, t\xi)$, $t > 0$. A function or form θ on $\dot{T}^*M = T^*M \setminus 0$ is *homogeneous of degree* p if $\rho_t^*\theta = t^p\theta$; equivalently $\mathcal{L}_R\theta = p\theta$. We write $C^\infty_{(p)}$ and $\Omega^k_{(p)}$ for these spaces. Note that dx_j has degree 0, $d\xi_j$ has degree 1 and ω has degree 1. With these conventions (Lemma 6.1)

$$\sigma_{m+m'-1}([A,B]) = -i\{\sigma_m(A), \sigma_{m'}(B)\}.$$

The cosphere bundle $S^*M = \dot{T}^*M/\mathbb{R}_+$ is oriented by $\iota_R\mathrm{vol}$ (Lemma 6.10). The circle is $S^1 = (0,\infty)/e^{\mathbb{Z}}$, the radial half-line modulo integral dilations. All cohomology has complex coefficients. Hochschild chains are $C_k(A) = A^{\otimes(k+1)}$ (completed when A is topological), and HH_k is homology, never cohomology, unless we say otherwise.

2.2 Dictionary

Table 1 compares notation. The most important difference concerns Hamiltonian vector fields. BG write $H_g f = \{f,g\}$, so their H_g is our $-H_g$. This changes the sign of some of their formulas (Remark 9.5).

2.3 Quoted background

The following facts are used without proof. Every place where one of them is used names it. Everything else is proved in these notes.

(G1) *Classical pseudodifferential calculus* [23, 15, 25]. On a manifold M, compact or not: (a) in a chart, the left quantization identifies operators of order m with compactly supported kernels, modulo $\Psi^{-\infty}$, with classical symbols of order m modulo $S^{-\infty}$, and every formal sum of homogeneous terms is the expansion of a symbol (Borel's lemma); (b) the full symbol of a product is given by (1) modulo $S^{-\infty}$; (c) under a change of coordinates the full symbol transforms by a formal series of differential operators, each lowering the degree of homogeneity, whose top term is the pull-back of the principal symbol; (d) Schwartz kernels are smooth off the diagonal, so an operator $\phi A \psi$ with $\mathrm{supp}\,\phi \cap \mathrm{supp}\,\psi = \varnothing$ is smoothing. We prove the consequence (e), the global star product, in Proposition 7.8.

(G2) *Nuclear spaces* [32, Chs. 43, 50, 51]. For manifolds X, Y, $C^\infty(X) \mathbin{\widehat{\otimes}} C^\infty(Y) = C^\infty(X \times Y)$, where $\widehat{\otimes}$ is the completed projective tensor product. The functor $E \mathbin{\widehat{\otimes}} (\cdot)$, for E nuclear Fréchet, is exact on Fréchet spaces and commutes with countable products.

[7]	these notes	Loya's notes / standard
$\Psi^{\infty}(M)$, $\Psi^{-\infty}(M)$	$\Psi^{\mathbb{Z}}(M)$, $\Psi^{-\infty}(M)$	classical $\Psi^{m}(M)$, smoothing operators
$\mathcal{P} = \Psi^{\infty}/\Psi^{-\infty}$	$\mathcal{P}$, formal sums $\sum_{j\le m} a_j$	full symbols $p \sim \sum p_{m-j}$
$\mathcal{P}_c$	$\mathcal{P}_c$ (M not compact)	compactly supported symbols
$\dot{T}^*M$, S^*M	$\dot{T}^*M = T^*M \setminus 0$, S^*M	cotangent and cosphere bundles
$a \star b$, φ_i	$a \star b = \sum_l P_l(a,b)$	$p\#q$, composition formula
$\hbar$, $A\{\hbar\}$, $\mathbb{C}\{\hbar\}$	$\hbar$, $A((\hbar))$, $K = \mathbb{C}((\hbar))$	semiclassical parameter
$\{f,g\}$, $H_g f = \{f,g\}$	$\{a,b\} = H_a b$, so $H_g^{\mathrm{BG}} = -H_g$	$\sigma([P,Q]) = -i\{p,q\}$
ω, ω^{-1}	$\omega = \sum d\xi_j \wedge dx_j$, bivector π	symplectic form
$\mathcal{R}$, T_t	$R = \sum \xi_j \partial_{\xi_j}$, ρ_t	homogeneity in ξ
$\varepsilon(v)$, $\varepsilon^*(v)$, $\iota^*(Z)$	$\varepsilon(\theta)$, $\iota_{\pi^\sharp\theta}$, $\varepsilon(\iota_Z\omega)$	exterior and interior products
$*$ (an $\mathfrak{sl}(2)$ exponential)	$*$, $\alpha \wedge *\beta = \Lambda(\alpha,\beta)\mathrm{vol}$	analogue of the Hodge star
δ (Koszul)	$\delta = [\iota_\pi, d] = \iota_\pi d - d\iota_\pi$	Poisson "codifferential"
b, B, u	b, B, u	Hochschild and Connes operators
E^{i}_{kl} (k form degree, l homogeneity)	$E^{r}_{p,k}$ (p order, k degree)	spectral sequence pages
$R(P)$	res	Wodzicki–Guillemin residue
Res_m	higher residue functionals	Wodzicki's higher residues
$\mathcal{D}(M)$, $\mathcal{D}_M$	$\mathcal{D}(M)$, sheaf $\mathcal{D}_M$	differential operators
$\mathcal{E}$, $\mathcal{E}^{\infty}$, $\mathcal{E}_{\mathbb{R}}$	same (quoted, Section 14.5)	microdifferential operators
$H^*(X)\{\hbar\}$	$H^*(X)((\hbar))$	Laurent series in $\hbar$

Table 1: Dictionary of notation.

(G3) *Eilenberg–Zilber theorem* [22, §4.2], [33, §8.5]: the shuffle map from the tensor product of two Hochschild complexes of unital algebras to the Hochschild complex of the tensor product is a chain homotopy equivalence, given by universal finite formulas.

(G4) *Basic homological algebra* [33, Chs. 2, 3, 5]: the comparison theorem for projective resolutions; Künneth over a field; the general construction of the pages E^r of a filtered complex.

(G5) *Algebraic topology and Hodge theory* [3, 23]: de Rham's theorem, Poincaré duality, the Künneth formula, the Gysin sequence, Hodge theory on the round sphere (a continuous Green operator for the Laplacian), and Weil's theorem in the general form of Background 14.3.

(G6) *Darboux theorems* [8], [15, §21.1]: local Darboux coordinates on a symplectic manifold, and the homogeneous version on a conic symplectic manifold.

(G7) *De Wilde–Lecomte* [10]: existence and uniqueness of local conic star products, in the form of Background 7.7.

(G8) *Trace class and complex powers* [29, 30]: operators of order $< -n$ are of trace class with trace given by the integral of the kernel on the diagonal, and Seeley's complex powers form a holomorphic family (Background 12.14).

(G9) *H-unitality, excision and Morita invariance* [37, 22]: $C_c^{\infty}(M)$ and $\Psi^{-\infty}(M)$ are H-unital; an H-unital ideal gives a long exact sequence in Hochschild homology; and $\mathrm{HH}_*(\Psi^{-\infty}(M)) \cong \mathrm{HH}_*(\mathbb{C})$. These are used only in remarks.

(G10) *The Milnor sequence* [33, §3.5], used only for periodic cyclic homology.

(G11) *HKR for fibrewise polynomial functions on T^*M.* It is used only for differential operators on a general manifold (Theorem 13.2); we prove it when M is an open subset of $\mathbb{R}^n$.

(G12) *Sheaf theory* [3, 4]: sheafification, hypercohomology, and the localization of Hochschild complexes of sheaves of algebras [31, 6]. It is used in Section 14 only.

(G13) *Microlocal analysis* [28, 16, 17, 6]: the sheaves $\mathcal{E}$, $\mathcal{E}^\infty$, $\mathcal{E}_\mathbb{R}$ and the four results listed in Background 14.6. They are used in Section 14.5 only.

(G14) *Algebraic de Rham theory*: Grothendieck's comparison theorem [12], the Lefschetz principle, and the proposition of Kassel and Mitschi [18] (Background 13.3), quoted by BG from a private communication.

(G15) *Results quoted for comparison only*: the computation of Hochschild homology of deformation quantizations of general symplectic manifolds by Nest and Tsygan [27], and Wodzicki's computations [35, 36].

Connes' theorem $\mathrm{HH}_k(C^\infty(M)) \cong \Omega^k(M)$ is *proved*, in Appendix A, for manifolds without boundary, compact or not.

2.4 Corrections and remarks on BG

The points where we disagree with the text of [7], or where hypotheses are left implicit, are the following. Each is discussed and verified at the place indicated.

1. *Sign in the formula for δ.* The second sum in [7, §1, Thm. (2), and §2] should carry $(-1)^{i+j}$, not $(-1)^{i+j+1}$. As printed the operator is not well defined on forms (Correction 6.4).

2. *Normalization of d^1 and of B.* The constant $\hbar/i$ in $d^1 = (\hbar/i)\delta$ holds for the HKR map normalized by $1/k!$. With the identification drawn in BG's diagram it becomes $\hbar/(ik)$ on k-forms, and B becomes $(k+1)d$ (Correction 8.2).

3. *The $\mathfrak{sl}(2)$ star operator.* The matrix identity used to prove $*^2 = 1$ in [7, §2] is false: the exponential equals -1, so BG's $*$ squares to ± 1. The trigonometric expansion in the proof of their Proposition (1) needs a factor of i removed. The conclusions hold up to non-zero constants (Correction 6.9).

4. *The "in particular" of the main theorem.* The claim $\mathrm{HH}_*(\mathcal{P}) \cong H^{n-*-1}(M) \oplus H^{n-*}(M)$ for $n > 1$ in [7, §4, Thm. (1)] is false in general; it fails for $M = S^2$. The correct statement uses the Gysin sequence and holds when the Euler class vanishes (Correction 11.9).

5. *The filtration for cyclic homology.* The filtration $F_m(\mathcal{P}[u]) = \sum_j u^j F_{m+2j}(\mathcal{P})$ of [7, §4] is not preserved by $u^{-1}B$. It should be $\sum_j u^j F_{m-2j}(\mathcal{P})$ (Correction 11.6).

6. *The zeta-function constant.* In [7, §4, Prop. (2)] the constant in $\mathrm{Res}_{s=0}\,\mathrm{Tr}(PA^{-s}) = c\,R(P)$ depends on the order m of A: it is proportional to $1/m$. The value $c = 2\pi$ asserted in the proof of Prop. (1) depends on normalizations that are not fixed there (Correction 12.12).

7. *Quantum algebras.* The parity condition $\varphi_i(a,b) = (-1)^i \varphi_i(b,a)$ is not satisfied by the left-quantized symbol product and is never needed; and $a \star b - b \star a = (\hbar/i)\{a,b\}$ holds only to first order (Remark 7.5).

8. *Darboux and the conic structure.* The conic Darboux map goes onto an open *conic subset* of $\dot{T}^*\mathbb{R}^n$ and commutes with the dilations. The flow is $T_t = \exp((\log t)\mathcal{R})$, not $\exp(t\mathcal{R})$ (Remark 9.2).

9. *Convergence.* The convergence argument in [7, §§2,4] appeals to Mittag-Leffler without the required estimates, and it depends on the completion used for the chains. We prove convergence for two chain models and show that they give the same homology (Remark 11.5).

10. *Higher residues.* The cochain $T\alpha$ of [7, §4] is defined by a leading-order formula and is not a Hochschild cocycle as written. We give the precise statement, as a functional on $\mathrm{HH}_m(\mathcal{P})$ (Remark 12.17).

11. *Hypotheses.* Uniqueness of the residue needs M connected and $n \geq 2$ (BG say so in the text, not in the proposition). Statements for non-compact M concern $\mathcal{P}_c$ and compactly supported cohomology (Remark 11.10).

12. *Sheaf-theoretic gaps.* In [7, §5] the presheaf $U \mapsto C_k(\mathcal{D}(U))$ is not a presheaf of the kind that Weil's argument needs without a localization step, and the splitting argument for cyclic homology is stated with the arrows reversed. Both points are fixed in Remark 14.5.

13. *Misprints.* The upper limit n for m in the formula for δ in §2; $\alpha = \iota(\omega)$ for $\iota(\mathcal{R})\omega$ in §4; the index k for q in the Poincaré lemma of §5; the words "convergence" and "degeneration" exchanged in the proof of the theorem of §3; "Hochschild cohomology" for homology at the start of §5; and "m sufficiently large" for "$-m$ sufficiently large" in §4. We note these where they occur.

3 Hochschild homology for beginners

What this section does. We define Hochschild homology, starting from its degree-zero part, which is the space dual to traces. We compute it for polynomial rings and for the Weyl algebra by hand, prove the Hochschild–Kostant–Rosenberg theorem for polynomial algebras, and state Connes' theorem for smooth functions. Everything except Section 3.7 is pure algebra over $\mathbb{C}$.

Throughout, A is an associative algebra over $\mathbb{C}$ with unit 1, unless we say otherwise.

3.1 Degree zero: traces

Let $[A, A]$ be the linear span of all commutators $[a, b] = ab - ba$.

Definition 3.1. $\mathrm{HH}_0(A) = A/[A, A]$. A *trace* on A is a linear map $\tau : A \to \mathbb{C}$ with $\tau(ab) = \tau(ba)$ for all a, b.

A linear functional is a trace exactly when it vanishes on $[A, A]$, so traces are the same thing as linear functionals on $\mathrm{HH}_0(A)$. Finding all traces on an algebra is therefore the problem of computing HH_0.

Example 3.2 (Matrices)**.** For $A = M_N(\mathbb{C})$ with matrix units E_{ij}, we have $E_{ij} = [E_{ik}, E_{kj}]$ for $i \neq j$ (any k), and $E_{ii} - E_{jj} = [E_{ij}, E_{ji}]$. So $[A, A]$ contains the matrices of trace zero, and $\mathrm{HH}_0(M_N(\mathbb{C})) \cong \mathbb{C}$, via the usual trace.

Example 3.3 (Commutative algebras)**.** If A is commutative then $[A, A] = 0$ and $\mathrm{HH}_0(A) = A$.

Example 3.4 (The Weyl algebra has no traces)**.** Let A_1 be the algebra of differential operators on $\mathbb{R}$ with polynomial coefficients. It is spanned by $x^a\partial^b$, $a, b \geq 0$, and $[\partial, x^{a+1}\partial^b] = (a+1)x^a\partial^b$. So every element is a commutator and $\mathrm{HH}_0(A_1) = 0$. There is no non-zero trace on A_1.

Example 3.5 (Symbols: a preview)**.** In $\mathcal{P}(\mathbb{R}^n)$ restricted to symbols with compact x-support the same idea works partially. In the left quantization (Lemma 12.2)

$$[x_k, \mathrm{Op}(a)] = i\,\mathrm{Op}(\partial_{\xi_k} a), \qquad [D_{x_k}, \mathrm{Op}(a)] = \mathrm{Op}(D_{x_k} a).$$

So every ξ-derivative and every x-derivative of a symbol is a commutator. By Euler's identity, a function a_j homogeneous of degree j satisfies $\sum_k \partial_{\xi_k}(\xi_k a_j) = (j+n)a_j$. Hence every term of degree $j \neq -n$ is a commutator, and only the term a_{-n} can survive in HH_0. This is the origin of the residue. We return to it in Section 12.

3.2 Degree one: differential forms

To go further we need the chains of degree one and two:

$$C_1(A) = A \otimes A \xrightarrow{b} C_0(A) = A, \qquad b(a_0 \otimes a_1) = a_0 a_1 - a_1 a_0,$$

$$C_2(A) = A^{\otimes 3} \xrightarrow{b} C_1(A), \qquad b(a_0 \otimes a_1 \otimes a_2) = a_0 a_1 \otimes a_2 - a_0 \otimes a_1 a_2 + a_2 a_0 \otimes a_1.$$

One checks $b \circ b = 0$ on C_2 directly; the general case is Lemma 3.8. Then $\mathrm{HH}_0 = A/b(C_1)$, and $\mathrm{HH}_1(A) = \ker(b : C_1 \to C_0)/b(C_2)$.

For a commutative algebra A, the module of *Kähler differentials* Ω^1_A is the A-module generated by symbols db, $b \in A$, subject to linearity in b and the Leibniz rule $d(bc) = b\,dc + c\,db$. For $A = \mathbb{C}[x_1, \ldots, x_n]$ it is the free module on $dx_1, \ldots, dx_n$, and for smooth functions on a manifold it maps onto the ordinary 1-forms.

Proposition 3.6. *If A is commutative, $a_0 \otimes a_1 \mapsto a_0\,da_1$ induces an isomorphism $\mathrm{HH}_1(A) \cong \Omega^1_A$.*

Proof. Since A is commutative, $b = 0$ on C_1, so every 1-chain is a cycle. The map kills boundaries:

$$b(a_0 \otimes a_1 \otimes a_2) \mapsto a_0 a_1\,da_2 - a_0\,d(a_1 a_2) + a_2 a_0\,da_1 = 0$$

by the Leibniz rule. Conversely, define $\Omega^1_A \to \mathrm{HH}_1(A)$ by $a\,db \mapsto [a \otimes b]$. It respects the relations, because

$$a \otimes bc - ab \otimes c - ac \otimes b = -b(a \otimes b \otimes c),$$

and it is A-linear for the module structure $c \cdot [a \otimes b] = [ca \otimes b]$ by construction. The two maps are inverse to each other. □

So $\mathrm{HH}_1(\mathbb{C}[x]) \cong \mathbb{C}[x]\,dx$. In Section 3.4 we show that $\mathrm{HH}_k(\mathbb{C}[x]) = 0$ for $k \geq 2$, and in Section 3.5 that HH_k of a polynomial ring is the space of k-forms. The slogan is: *for commutative algebras, Hochschild homology is differential forms.*

3.3 The Hochschild complex

Definition 3.7. The Hochschild chains are $C_k(A) = A^{\otimes(k+1)}$, $k \geq 0$. The *face maps* $d_i : C_k \to C_{k-1}$, $0 \leq i \leq k$, are

$$d_i(a_0 \otimes \cdots \otimes a_k) = a_0 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_k \quad (i < k),$$

$$d_k(a_0 \otimes \cdots \otimes a_k) = a_k a_0 \otimes a_1 \otimes \cdots \otimes a_{k-1}.$$

The Hochschild boundary is $b = \sum_{i=0}^{k} (-1)^i d_i$, and the bar boundary is $b' = \sum_{i=0}^{k-1} (-1)^i d_i$. Hochschild homology is $\mathrm{HH}_k(A) = H_k(C_\bullet(A), b)$.

Lemma 3.8. *$d_i d_j = d_{j-1} d_i$ for $i < j$. Consequently $b^2 = 0$ and $b'^2 = 0$.*

Proof. The identity is a statement about which adjacent entries are multiplied first; for $i < j-1$ the two products are in different places, and for $i = j-1$ it is associativity. When $j = k$ and $i = 0$ both sides give $a_k a_0 a_1 \otimes \cdots$, again by associativity. Then $b^2 = \sum_{i,j} (-1)^{i+j} d_i d_j$, and the terms with $i < j$ cancel against the terms $d_{j-1} d_i$ with indices $(j-1, i)$, $j - 1 \geq i$, which have the opposite sign. The same argument applies to b'. □

Picture an elementary chain $a_0 \otimes \cdots \otimes a_k$ as $k+1$ beads on a circle. The face d_i multiplies neighbouring beads i and $i+1$, and d_k closes the necklace by multiplying the last bead with the first. The cyclic nature of d_k is why traces appear in degree zero.

3.4 The bar resolution and small resolutions

Let $A^e = A \otimes A^{\mathrm{op}}$. An A-bimodule is the same thing as a left A^e-module, with $(a \otimes b) \cdot m = amb$. Put $B_k(A) = A \otimes A^{\otimes k} \otimes A$ (free A^e-modules), with the bar boundary b'.

Lemma 3.9. *$\cdots \to B_1 \xrightarrow{b'} B_0 = A \otimes A \xrightarrow{\mu} A \to 0$, where μ is multiplication, is exact; it is a free resolution of A as an A^e-module. Moreover $A \otimes_{A^e} B_k \cong C_k(A)$, and the induced differential is b.*

Proof. The map $s(a_0 \otimes \cdots \otimes a_{k+1}) = 1 \otimes a_0 \otimes \cdots \otimes a_{k+1}$ satisfies $b's + sb' = 1$, because $d_0 s = 1$ and $d_{i+1} s = s d_i$. A complex with such a contraction is exact. The isomorphism is $a \otimes_{A^e} (a_0 \otimes w \otimes a_{k+1}) \mapsto a_{k+1} a a_0 \otimes w$, and under it the last face of b' becomes the cyclic face d_k. □

So $\mathrm{HH}_k(A) = \mathrm{Tor}_k^{A^e}(A, A)$. By the comparison theorem of homological algebra (G4), *any* projective resolution $P_\bullet \to A$ by A^e-modules computes the same homology after applying $A \otimes_{A^e} (\cdot)$. Small resolutions give quick computations.

Example 3.10 (Polynomials and Laurent polynomials in one variable)**.** Let $A = \mathbb{C}[x]$, so $A^e = \mathbb{C}[x, y]$. The sequence

$$0 \to A^e \xrightarrow{(x-y)} A^e \xrightarrow{f(x,y) \mapsto f(x,x)} A \to 0$$

is exact: $x - y$ is not a zero divisor, and $f(x, y) - f(x, x)$ is divisible by $x - y$. Applying $A \otimes_{A^e} \cdot$ sets $y = x$, and the map $x - y$ becomes 0. Hence $\mathrm{HH}_0(\mathbb{C}[x]) = \mathbb{C}[x]$, $\mathrm{HH}_1(\mathbb{C}[x]) = \mathbb{C}[x]$, which is $\mathbb{C}[x]\,dx$ by Proposition 3.6, and $\mathrm{HH}_k(\mathbb{C}[x]) = 0$ for $k \geq 2$. The same holds for $A = \mathbb{C}[r, r^{-1}]$: after clearing a denominator $(rs)^N$, $f(r, s) - f(r, r)$ is divisible by $r - s$. Thus $\mathrm{HH}_1(\mathbb{C}[r^{\pm 1}]) = \mathbb{C}[r^{\pm 1}]\,dr$, and $\mathrm{HH}_k = 0$ for $k \geq 2$.

3.5 The Hochschild–Kostant–Rosenberg theorem

For a commutative algebra define the *HKR map* and the *antisymmetrization*

$$\chi(a_0 \otimes \cdots \otimes a_k) = \frac{1}{k!}\, a_0\, da_1 \wedge \cdots \wedge da_k, \qquad \varphi(a_0; a_1, \ldots, a_k) = \sum_{\sigma \in \mathfrak{S}_k} \operatorname{sgn}(\sigma)\, a_0 \otimes a_{\sigma(1)} \otimes \cdots \otimes a_{\sigma(k)}. \tag{2}$$

The factor $1/k!$ is a normalization; it matters for the constants in Section 8 (Correction 8.2).

Lemma 3.11. *For a commutative algebra, $\chi \circ b = 0$, $b \circ \varphi = 0$, and $\chi\varphi(a_0; a_1, \ldots, a_k) = a_0\, da_1 \wedge \cdots \wedge da_k$.*

Proof. Write $T_i = a_0 a_i\, da_1 \wedge \ldots \widehat{da_i} \cdots \wedge da_k$. We compute $(k-1)!\,\chi(b(a_0 \otimes \cdots \otimes a_k))$ face by face. The face d_0 gives $a_0 a_1\, da_2 \wedge \cdots \wedge da_k = T_1$. For $1 \le i \le k-1$ the face d_i gives $a_0\, da_1 \wedge \cdots \wedge d(a_i a_{i+1}) \wedge \cdots \wedge da_k$, and the Leibniz rule $d(a_i a_{i+1}) = a_i\, da_{i+1} + a_{i+1}\, da_i$ turns this into $T_i + T_{i+1}$. The face d_k gives $a_k a_0\, da_1 \wedge \cdots \wedge da_{k-1} = T_k$. Hence

$$(k-1)!\,\chi\big(b(a_0 \otimes \cdots \otimes a_k)\big) = T_1 + \sum_{i=1}^{k-1} (-1)^i (T_i + T_{i+1}) + (-1)^k T_k.$$

The coefficient of T_1 is $1 - 1 = 0$, that of T_j for $2 \le j \le k-1$ is $(-1)^{j-1} + (-1)^j = 0$, and that of T_k is $(-1)^{k-1} + (-1)^k = 0$. For $b\varphi$: the inner faces cancel in pairs, σ against $\sigma \circ (i\ i{+}1)$, because the algebra is commutative; and the face d_0 for σ cancels $(-1)^k d_k$ for the cyclically shifted permutation, whose sign is $(-1)^{k-1} \operatorname{sgn}\sigma$. The third identity holds because each of the $k!$ terms of φ contributes $\frac{1}{k!}\, a_0\, da_1 \wedge \cdots \wedge da_k$ after reordering the wedge product. □

Theorem 3.12 (Hochschild–Kostant–Rosenberg for polynomial algebras [14])**.** *Let $A = \mathbb{C}[x_1, \ldots, x_n]$. Then χ induces isomorphisms $\mathrm{HH}_k(A) \cong \Omega^k_A = A \otimes \Lambda^k \mathbb{C}^n$, with inverse $a_0\, dx_{i_1} \wedge \cdots \wedge dx_{i_k} \mapsto [\varphi(a_0; x_{i_1}, \ldots, x_{i_k})]$.*

Proof. The Koszul resolution. Identify $A^e = \mathbb{C}[x, y]$, with x acting on the left and y on the right. Let $K_k = A^e \otimes \Lambda^k \mathbb{C}^n$, with basis $e_{i_1} \wedge \cdots \wedge e_{i_k}$, and

$$\partial(f\, e_{i_1} \wedge \cdots \wedge e_{i_k}) = \sum_{j=1}^{k} (-1)^{j-1} (x_{i_j} - y_{i_j}) f\, e_{i_1} \wedge \ldots \widehat{e_{i_j}} \cdots \wedge e_{i_k},$$

augmented by $K_0 = A^e \to A$, $f \mapsto f(x, x)$. In the variables $u = x - y$, $v = y$, ∂ is contraction with $\sum_i u_i\, \partial/\partial e_i$. Let $h(f\, e_I) = \sum_i \frac{\partial f}{\partial u_i} e_i \wedge e_I$. Then $\partial h + h\partial$ multiplies $f\, e_I$ by (degree of f in u) $+|I|$, the usual Euler identity. So K is exact except in degree 0, where the homology is $\mathbb{C}[v] \cong A$ through $f \mapsto f(x, x)$. Thus K is a free resolution of A.

Comparison. The map $\Phi(e_{i_1} \wedge \cdots \wedge e_{i_k}) = \sum_\sigma \operatorname{sgn}\sigma\, 1 \otimes x_{i_{\sigma(1)}} \otimes \cdots \otimes x_{i_{\sigma(k)}} \otimes 1$, extended A^e-linearly, is a chain map $K \to B(A)$ over the identity of A: in $b'\Phi$ the inner faces cancel in pairs, and the two outer faces give $\sum_l (-1)^{l-1} (x_{i_l} \otimes 1 - 1 \otimes x_{i_l})$ times the shorter antisymmetrization. By the comparison theorem it induces an isomorphism on homology after $A \otimes_{A^e}$. On the Koszul side $A \otimes_{A^e} K = A \otimes \Lambda \mathbb{C}^n$ with zero differential, since $x_i - y_i$ acts as 0. On the Hochschild side Φ becomes $a_0\, e_I \mapsto \varphi(a_0; x_{i_1}, \ldots, x_{i_k})$. So the classes $\varphi(a_0; x_{i_1}, \ldots, x_{i_k})$ form a basis of $\mathrm{HH}_k(A)$ over the basis of $A \otimes \Lambda^k$, and $\chi\varphi = \mathrm{id}$ by Lemma 3.11. □

3.6 The Weyl algebra: a first Poincaré lemma

The *Weyl algebra* A_n is the algebra of differential operators on $\mathbb{R}^n$ with polynomial coefficients. It is generated by x_i and $\partial_i = \partial/\partial x_i$, with $[\partial_i, x_j] = \delta_{ij}$ and all other generators commuting. Normal ordering $x^\alpha \partial^\beta \mapsto x^\alpha \xi^\beta$ is a linear isomorphism $A_n \cong \mathbb{C}[x, \xi]$.

Theorem 3.13 (Feigin–Tsygan [11]; Wodzicki)**.** $\mathrm{HH}_k(A_n) \cong \mathbb{C}$ *for* $k = 2n$*, and* $\mathrm{HH}_k(A_n) = 0$ *otherwise.*

This is the "Poincaré lemma" of [7, §5] for polynomial coefficients: $\mathrm{HH}_*(A_n) \cong H^{2n-*}(\mathbb{R}^n)$. The proof is elementary.

Proof. Commuting elements of A^e. In $A^e = A_n \otimes A_n^{\mathrm{op}}$ put $U_i = x_i \otimes 1 - 1 \otimes x_i$ and $V_i = \partial_i \otimes 1 - 1 \otimes \partial_i$. In A^{op} the commutator is reversed, so $[1 \otimes x_i, 1 \otimes \partial_j] = 1 \otimes (\partial_j x_i - x_i \partial_j) = \delta_{ij}$, while $[x_i \otimes 1, \partial_j \otimes 1] = -\delta_{ij}$. Hence $[U_i, V_j] = -\delta_{ij} + \delta_{ij} = 0$, and the $2n$ elements $W = (U_1, \dots, U_n, V_1, \dots, V_n)$ commute.

The Koszul resolution. Let $K_k = A^e \otimes \Lambda^k \mathbb{C}^{2n}$ with $\partial(c \otimes e_I) = \sum_j (-1)^{j-1} c W_{i_j} \otimes e_{I \setminus i_j}$ (right multiplication, so ∂ is A^e-linear for the left action), augmented by $c \mapsto c \cdot 1$, that is $a \otimes b \mapsto ab$. Since the W_i commute, $\partial^2 = 0$. Filter A^e by total degree in all generators (the Bernstein filtration). The associated graded algebra is the polynomial ring $\mathbb{C}[x, \xi, y, \eta]$ in $4n$ variables, and the W_i have leading terms $x_i - y_i$, $\xi_i - \eta_i$. The associated graded complex of K is the Koszul complex of these linear forms, which is exact except in degree 0 by the proof of Theorem 3.12 (change variables to $u = (x - y, \xi - \eta)$). Its H_0 is $\mathbb{C}[x, \xi] = \operatorname{gr} A_n$. A complex with an exhaustive filtration, bounded below, whose associated graded complex is exact, is itself exact: a cycle of filtration degree N is a boundary modulo filtration degree $N - 1$, and we induct on N. Hence K is a free resolution of A_n.

The Koszul complex of A_n. Applying $A_n \otimes_{A^e}$ gives $A_n \otimes \Lambda \mathbb{C}^{2n}$, with differential $\partial(a\, e_I) = \sum_j (-1)^{j-1} (a \cdot W_{i_j})\, e_{I \setminus i_j}$, where A^e acts on the right of A_n by $a \cdot (p \otimes q) = qap$. Thus $a \cdot U_i = -[x_i, a]$ and $a \cdot V_i = -[\partial_i, a]$. In normal ordering, $[\partial_i, x^\alpha \partial^\beta] = \alpha_i x^{\alpha - e_i} \partial^\beta$ and $[x_i, x^\alpha \partial^\beta] = -\beta_i x^\alpha \partial^{\beta - e_i}$. So, with $y = (\xi, -x) \in \mathbb{C}^{2n}$ as coordinates on $\mathbb{C}[x, \xi] \cong A_n$, the complex is $\mathbb{C}[y_1, \dots, y_{2n}] \otimes \Lambda \mathbb{C}^{2n}$ with $\partial = \sum_i \frac{\partial}{\partial y_i} \iota_i$, where ι_i is contraction with the i-th dual basis vector, $\iota_i(e_{i_1} \wedge \cdots \wedge e_{i_k}) = \sum_j (-1)^{j-1} \delta_{i, i_j} e_{I \setminus i_j}$.

The polynomial Poincaré lemma. Let ε_i be exterior multiplication by e_i and $h = \sum_i y_i \varepsilon_i$. Since $\frac{\partial}{\partial y_i} y_j = y_j \frac{\partial}{\partial y_i} + \delta_{ij}$ and $\iota_i \varepsilon_j + \varepsilon_j \iota_i = \delta_{ij}$,

$$\partial h + h \partial = \sum_{i,j} y_j \frac{\partial}{\partial y_i} (\iota_i \varepsilon_j + \varepsilon_j \iota_i) + \sum_i \iota_i \varepsilon_i = \sum_i y_i \frac{\partial}{\partial y_i} + \sum_i \iota_i \varepsilon_i.$$

On $f\, e_I$ with f homogeneous this is multiplication by $\deg f + (2n - |I|)$, which is invertible unless f is constant and $|I| = 2n$. Hence the homology is $\mathbb{C}\, e_1 \wedge \cdots \wedge e_{2n}$, in degree $2n$. □

Example 3.14 (An explicit generator of $\mathrm{HH}_2(A_1)$)**.** In A_1, let $c = 1 \otimes x \otimes \partial - 1 \otimes \partial \otimes x - 1 \otimes 1 \otimes 1$. Then

$$b(1 \otimes x \otimes \partial - 1 \otimes \partial \otimes x) = 1 \otimes (\partial x - x \partial) = 1 \otimes 1, \qquad b(1 \otimes 1 \otimes 1) = 1 \otimes 1,$$

so $bc = 0$. The first two terms are the antisymmetrized chain of the HKR theorem; the correction $-1 \otimes 1 \otimes 1$ is needed because x and ∂ do not commute. This is the simplest instance of the mechanism of Section 8: the commutative answer is corrected by the first-order term $[\partial, x] = 1$.

Discussion. Why should the Weyl algebra have homology in degree $2n$ only? Its associated graded algebra is $\mathbb{C}[x, \xi]$, the polynomial functions on $T^*\mathbb{R}^n = \mathbb{R}^{2n}$, whose Hochschild homology is all polynomial forms. Section 8 shows that the non-commutativity acts on these forms by an operator δ, which a "symplectic Hodge star" converts into the de Rham differential. The polynomial de

Rham cohomology of $\mathbb{R}^{2n}$ is $\mathbb{C}$ in degree 0, and the star moves degree 0 to degree $2n$. The Koszul proof above performs exactly this conversion: the complement $e_I \mapsto e_{I^c}$ turns contraction with $\partial/\partial y_i$ into d.

3.7 Topological algebras and Connes' theorem

For algebras of smooth functions the algebraic tensor product is too small: $C^\infty(M) \otimes C^\infty(M)$ consists of finite sums of products, not of all smooth functions on $M \times M$. One uses the completed projective tensor product $\widehat{\otimes}$. For manifolds,

$$C^\infty(M)^{\widehat{\otimes}(k+1)} = C^\infty(M^{k+1}) \qquad \text{(G2)},$$

and the face maps become restrictions to diagonals:

$$(d_i u)(z_0, \dots, z_{k-1}) = u(z_0, \dots, z_i, z_i, \dots, z_{k-1}) \ (i < k), \qquad (d_k u)(z_0, \dots, z_{k-1}) = u(z_0, \dots, z_{k-1}, z_0).$$

The HKR map extends by continuity. In local coordinates

$$\chi(u)(x) = \frac{1}{k!} \sum_{j_1, \dots, j_k} \frac{\partial^k u}{\partial z_1^{j_1} \cdots \partial z_k^{j_k}}(x, \dots, x)\, dx^{j_1} \wedge \cdots \wedge dx^{j_k}.$$

Theorem 3.15 (Connes [9])**.** *For a manifold M without boundary, compact or not, χ induces isomorphisms $\mathrm{HH}_k(C^\infty(M)) \cong \Omega^k(M)$ for all k.*

The proof, adapted from [9, 25], is in Appendix A. For example, $\mathrm{HH}_k(C^\infty(S^1))$ is $C^\infty(S^1)$, $\Omega^1(S^1)$ and 0 for $k = 0, 1, \geq 2$.

Discussion. Why complete? In $\Omega^1(\mathbb{R})$ we have $d(e^x) = e^x\, dx$. Deriving this from the Leibniz rule needs the power series of e^x, that is, infinitely many applications of the rule followed by a limit. The algebraic Hochschild complex has no limits, and its HH_1 is the algebraic Kähler module, which is not the module of smooth 1-forms. With the completed tensor product the chains are smooth functions of several variables, and Taylor expansion at the diagonal is available. This is what the proof of Connes' theorem uses.

Filtered algebras such as $\mathcal{P}$ are increasing unions $A = \bigcup_N A_N$ of Fréchet spaces with continuous multiplication $A_N \times A_{N'} \to A_{N+N'}$. Their chains are $\varinjlim_N A_N^{\widehat{\otimes}(k+1)}$; see Section 7.6 for the choices involved for $\mathcal{P}$.

We also need a version of HKR with polynomial or Laurent variables adjoined. Let S be a manifold and $\mathfrak{B} = \mathbb{C}[y_1, \dots, y_m]$ or $\mathbb{C}[r, r^{-1}]$. The algebra $C^\infty(S) \otimes \mathfrak{B}$ has chains $C^\infty(S^{k+1}) \otimes \mathfrak{B}^{\otimes(k+1)}$ (completed in the S-variables only).

Corollary 3.16 (HKR with polynomial variables)**.** *χ induces isomorphisms*

$$\mathrm{HH}_k(C^\infty(S) \otimes \mathfrak{B}) \cong \bigoplus_{p+q=k} \Omega^p(S) \otimes \Omega^q_{\mathfrak{B}},$$

which preserve the grading by degree in the variables of $\mathfrak{B}$. For $\mathfrak{B} = \mathbb{C}[r^{\pm 1}]$ the right side is $\Omega^k(S) \otimes \mathbb{C}[r^{\pm 1}] \oplus \Omega^{k-1}(S) \otimes \mathbb{C}[r^{\pm 1}]\, dr$.

Proof. The Hochschild complex of a tensor product of unital algebras is the diagonal of a tensor product of simplicial vector spaces, and by the Eilenberg–Zilber theorem (G3) the shuffle map $\mathrm{sh} : C(C^\infty(S)) \otimes C(\mathfrak{B}) \to C(C^\infty(S) \otimes \mathfrak{B})$ is a chain homotopy equivalence. Its formulas are

finite and universal, so this holds with completions in the S-variables. The homology of the tensor product of complexes over the field $\mathbb{C}$ is the tensor product of the homologies (G4). By Theorem 3.15 and Theorem 3.12 or Example 3.10 these are $\Omega(S)$ and $\Omega_{\mathfrak{B}}$. Finally, $\chi(\mathrm{sh}(\alpha\otimes\beta)) = \chi(\alpha)\wedge\chi(\beta)$: each of the $\binom{p+q}{p}$ shuffles contributes the same wedge product, and $\binom{p+q}{p}/(p+q)! = 1/(p!\,q!)$. All maps preserve the $\mathfrak{B}$-degree. □

3.8 H-unitality

Some algebras in BG have no unit: $C_c^\infty(M)$, the compactly supported symbols $\mathcal{P}_c$ on a non-compact manifold, or the smoothing operators $\Psi^{-\infty}$. For them one uses the following notion.

Definition 3.17. An algebra J, possibly without unit, is *H-unital* if its bar complex $(C_\bullet(J), b')$ is acyclic.

A unital algebra is H-unital (Lemma 3.9). The following facts are quoted (G9) [37]: $C_c^\infty(M)$ and $\Psi^{-\infty}(M)$ are H-unital (with the appropriate completed tensor products); for an H-unital ideal $J\subset A$, the sequence $0\to J\to A\to A/J\to 0$ gives a long exact sequence in Hochschild homology; and $\mathrm{HH}_*(\Psi^{-\infty}(M))\cong\mathrm{HH}_*(\mathbb{C})$ by Morita invariance. Consequently $\mathrm{HH}_k(\Psi^{\mathbb{Z}}(M))\to\mathrm{HH}_k(\mathcal{P})$ is an isomorphism for $k\geq 2$. We use H-unitality only in remarks (Remark 11.10).

4 Cyclic homology

What this section does. Hochschild homology of $C^\infty(M)$ gives all differential forms, but not de Rham cohomology. Connes' operator B plays the role of d, and cyclic homology is the homology of $b+u^{-1}B$. We prove the identities needed, derive the SBI sequence, and compute cyclic homology of $C^\infty(M)$ from Connes' theorem. This section is needed only for Sections 10 and 11.3.

4.1 The cyclic operator and Connes' B

On $C_k(A)$ let

$$t(a_0\otimes\cdots\otimes a_k) = (-1)^k\, a_k\otimes a_0\otimes\cdots\otimes a_{k-1}, \qquad N=\sum_{j=0}^{k} t^j,$$

$$s(a_0\otimes\cdots\otimes a_k) = 1\otimes a_0\otimes\cdots\otimes a_k.$$

Then $t^{k+1}=1$ on C_k, so $(1-t)N = N(1-t) = 0$.

Lemma 4.1. (1) $b(1-t) = (1-t)b'$. (2) $b'N = Nb$. (3) $b's + sb' = 1$.

Proof. Let $\tau = (-1)^k t$ be the cyclic permutation without sign. Directly from Definition 3.7,

$$d_0\tau = d_k, \qquad d_i\tau = \tau d_{i-1} \quad (1\leq i\leq k). \tag{3}$$

(1) By (3), $bt = (-1)^k\big(d_k + \sum_{i=1}^k (-1)^i \tau d_{i-1}\big) = (-1)^k d_k + tb'$, where in the last step $t = (-1)^{k-1}\tau$ on C_{k-1}. Since $b = b' + (-1)^k d_k$, we get $b - bt = b' - tb'$.

(2) Iterating (3) gives $d_i\tau^j = \tau^j d_{i-j}$ for $j\leq i$, and $d_i\tau^j = \tau^{j-1}d_{k+1+i-j}$ for $i<j$ (use $d_i\tau^i = \tau^i d_0$, then $d_0\tau = d_k$, then $d_k\tau^m = \tau^m d_{k-m}$). Now expand $b'N = \sum_{i=0}^{k-1}\sum_{j=0}^{k}(-1)^{i+jk} d_i\tau^j$ and $Nb = \sum_{j'=0}^{k-1}\sum_{l=0}^{k}(-1)^{j'(k-1)+l}\tau^{j'}d_l$. The term (i,j) of the first sum equals the term $(j',l) = (j, i-j)$ of the second if $j\leq i$, and the term $(j',l) = (j-1, k+1+i-j)$ if $i<j$. This is a bijection between the index sets, and in both cases the signs agree, since $(-1)^{i+jk} = (-1)^{l+j'+j'k} = (-1)^{j'(k-1)+l}$.

(3) $d_0 s = 1$ and $d_{i+1}s = sd_i$; this is the contraction of Lemma 3.9. □

Definition 4.2. *Connes' operator* is $B = (1-t)sN : C_k(A) \to C_{k+1}(A)$.

Proposition 4.3. $B^2 = 0$ *and* $bB + Bb = 0$.

Proof. $B^2 = (1-t)sN(1-t)sN = 0$ because $N(1-t) = 0$. By Lemma 4.1, $bB + Bb = b(1-t)sN + (1-t)sNb = (1-t)(b's + sb')N = (1-t)N = 0$. □

The term tsN puts the unit 1 into slot 1. On *normalized* chains, where chains with a 1 in some slot ≥ 1 are set to zero, it vanishes, and

$$B(a_0 \otimes \cdots \otimes a_k) = \sum_{i=0}^{k} (-1)^{ik}\, 1 \otimes a_i \otimes \cdots \otimes a_k \otimes a_0 \otimes \cdots \otimes a_{i-1}. \tag{4}$$

Proposition 4.4 (B is the de Rham differential)**.** *For a commutative algebra,* $\chi \circ B = d \circ \chi$, *where* $d(a_0\, da_1 \wedge \cdots \wedge da_k) = da_0 \wedge \cdots \wedge da_k$. *The same holds for* $C^\infty(M)$ *with completed chains.*

Proof. Since $d1 = 0$, χ vanishes on chains with a 1 in a slot ≥ 1, so χB may be computed with (4). The term i gives $\frac{1}{(k+1)!} da_i \wedge \cdots \wedge da_k \wedge da_0 \wedge \cdots \wedge da_{i-1}$. Moving the block $da_0 \wedge \cdots \wedge da_{i-1}$ of length i to the front past $k+1-i$ one-forms costs $(-1)^{i(k+1-i)}$, and $(-1)^{ik}(-1)^{i(k+1-i)} = (-1)^{i(1-i)} = 1$. So each of the $k+1$ terms equals $\frac{1}{(k+1)!} da_0 \wedge \cdots \wedge da_k$, and the sum is $\frac{1}{k!} da_0 \wedge \cdots \wedge da_k = d\chi(a_0 \otimes \cdots \otimes a_k)$. The smooth case follows by continuity. □

4.2 The (b, B) bicomplex

Following [7, §3], let u be a formal variable of degree 2, and let u^{-1} act on polynomials in u by $u^j \mapsto u^{j-1}$ for $j \geq 1$ and $u^0 \mapsto 0$. The *cyclic complex* is

$$CC_N(A) = \bigoplus_{j \geq 0} C_{N-2j}(A)\, u^j, \qquad (b + u^{-1}B)\Big(\sum_j c_j u^j\Big) = \sum_j \big(bc_j + Bc_{j+1}\big) u^j.$$

By Proposition 4.3 the square of $b + u^{-1}B$ is zero.

Definition 4.5. $\mathrm{HC}_N(A) = H_N(CC_\bullet(A), b + u^{-1}B)$.

This agrees with Connes' original definition through the cyclic complex $C^\lambda_\bullet = C_\bullet/(1-t)$, over a field of characteristic zero [22, §2.1]. We use the (b, B) form only, except in Section 14. Figure 2 shows the bicomplex.

Proposition 4.6 (The SBI sequence)**.** *There is a long exact sequence*

$$\cdots \to \mathrm{HH}_N(A) \xrightarrow{I} \mathrm{HC}_N(A) \xrightarrow{S} \mathrm{HC}_{N-2}(A) \xrightarrow{\partial} \mathrm{HH}_{N-1}(A) \to \cdots,$$

where I *is the inclusion of the column* $j = 0$, $S(\sum_j c_j u^j) = \sum_{j \geq 1} c_j u^{j-1}$, *and* $\partial[\sum_j z_j u^j] = [Bz_0]$.

Proof. $0 \to C_\bullet \xrightarrow{I} CC_\bullet \xrightarrow{S} CC_{\bullet-2} \to 0$ is a short exact sequence of complexes. For the connecting map, lift a cycle $z = \sum z_j u^j \in CC_{N-2}$ to $\sum_j z_j u^{j+1} \in CC_N$, and apply $b + u^{-1}B$. For $i \geq 1$ the coefficient of u^i is $bz_{i-1} + Bz_i$, which vanishes because it is the coefficient of u^{i-1} in $(b + u^{-1}B)z = 0$. The coefficient of u^0 is Bz_0. □

A *mixed complex* is a complex (C, b) with an operator B of degree $+1$ such that $B^2 = 0$ and $bB + Bb = 0$. The cyclic complex and the SBI sequence make sense for any mixed complex, and they are natural for maps commuting with b and B.

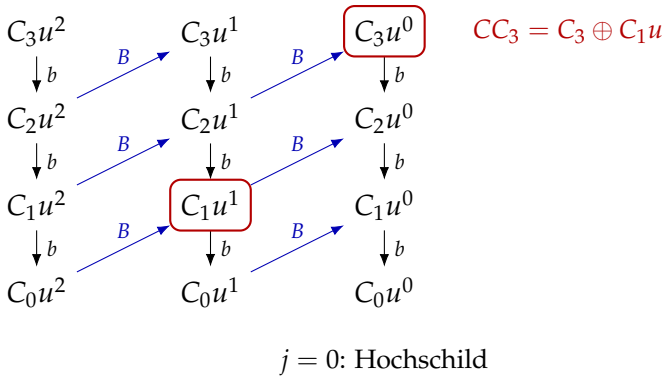


Figure 2: The (b, B) bicomplex. The column u^j is a copy of the Hochschild complex shifted by $2j$; b goes down a column, and $u^{-1}B$ goes one column to the right and one step up in Hochschild degree. Total degree is Hochschild degree plus $2j$, so $CC_3 = C_3 \oplus C_1u$ (boxed).

Lemma 4.7. *A map of mixed complexes that induces isomorphisms on the homology of b induces isomorphisms on cyclic homology.*

Proof. Induction on N, with $\mathrm{HC}_N = 0$ for $N < 0$ and $\mathrm{HC}_0 = \mathrm{HH}_0$. The map of SBI sequences and the five lemma give the step from $N-1$ and $N-2$ to N. □

Theorem 4.8 (Connes [9])**.** *For a manifold M without boundary,*

$$\mathrm{HC}_N(C^\infty(M)) \cong \Omega^N(M)/d\Omega^{N-1}(M) \oplus H^{N-2}(M) \oplus H^{N-4}(M) \oplus \cdots .$$

Proof. By Theorem 3.15 and Proposition 4.4, χ is a map of mixed complexes from $(C(C^\infty(M)), b, B)$ to $(\Omega(M), 0, d)$ that induces an isomorphism on the homology of b. By Lemma 4.7 it suffices to compute the cyclic homology of $(\Omega, 0, d)$. Its cyclic complex is $\bigoplus_j \Omega^{N-2j}u^j$ with differential $\sum_j \omega_j u^j \mapsto \sum_{j\geq 1} d\omega_j u^{j-1}$. A cycle has $d\omega_j = 0$ for $j \geq 1$ and ω_0 arbitrary, and the boundaries are the sums with $\omega_j \in d\Omega^{N-2j-1}$. This gives the stated answer. □

4.3 Periodic cyclic homology

Allow all $j \in \mathbb{Z}$ and infinite products: $CP_N(A) = \prod_{j\in\mathbb{Z}} C_{N-2j}(A)u^j$, with the same differential, now with u^{-1} invertible. Its homology $\mathrm{HP}_N(A)$ depends only on the parity of N. For $(\Omega, 0, d)$ the computation above gives $\mathrm{HP}_0 = H^{\mathrm{ev}}(M)$ and $\mathrm{HP}_1 = H^{\mathrm{odd}}(M)$, so periodic cyclic homology sees exactly de Rham cohomology. By the Milnor sequence (G10) there is an exact sequence

$$0 \to \varprojlim{}^1_S \mathrm{HC}_{N+2j+1}(A) \to \mathrm{HP}_N(A) \to \varprojlim_S \mathrm{HC}_{N+2j}(A) \to 0,$$

the limits being taken along S. If all the maps S are surjective, the $\varprojlim^1$ term vanishes. We use this only in Corollary 11.8.

5 A primer on spectral sequences

What this section does. A spectral sequence computes the homology of a complex one "order" at a time. We set up the only kind we need, explain its first pages on tiny examples, and prove two convergence results: a general criterion (Proposition 5.6) and the special case in which the second page lies on one line (Proposition 5.7). The second is what makes the computation of [7] work.

5.1 Filtered complexes of formal sums

All our complexes have the following form. For each degree k there are vector spaces $C_{p,k}$, $p \in \mathbb{Z}$, and

$$C_k = \Big\{ c = \sum_{p \le P} c_p : \; c_p \in C_{p,k}, \; P \in \mathbb{Z} \Big\},$$

formal sums with orders bounded above (or a subspace of this, containing each $C_{p,k}$). We call p the *order*. The differential has the form

$$b = \sum_{l \ge 0} b_l, \qquad b_l : C_{p,k} \to C_{p-l,k-1},$$

so it never raises order, and b_l lowers it by exactly l. The component of order p of bc is the finite sum $\sum_l b_l c_{p+l}$. The subspaces $F_pC = \{c : \; c_q = 0 \text{ for } q > p\}$ form an increasing *filtration* by subcomplexes. Comparing orders in $b^2 = 0$ gives

$$\sum_{i+j=l} b_i b_j = 0, \qquad \text{in particular} \quad b_0^2 = 0, \quad b_0 b_1 + b_1 b_0 = 0. \tag{5}$$

Definition 5.1. The *zeroth page* is $(C_{p,\bullet}, b_0)$. The *first page* is $E^1_{p,k} = H_k(C_{p,\bullet}, b_0)$. By (5), b_1 maps b_0-cycles to b_0-cycles and b_0-boundaries to b_0-boundaries, so it induces $d^1 : E^1_{p,k} \to E^1_{p-1,k-1}$ with $(d^1)^2 = 0$. The *second page* is $E^2_{p,k} = \ker d^1 \,/\, \operatorname{im} d^1$ at (p,k).

The filtration is *complete* if every series $\sum_j e_j$ with $e_j \in F_{p_j}C_k$, $\sup_j p_j < \infty$ and $p_j \to -\infty$ defines an element of C_k (each order receives finitely many contributions), and b commutes with such sums. For the full space of formal sums this is automatic. It is *not* automatic for subspaces, and this is the source of the convergence issue in Remark 11.5. If $C_{p,k} = 0$ for all $p < 0$, every series of this kind is finite, and completeness is trivial.

Discussion. Think of a cycle $c = c_P + c_{P-1} + \cdots$ as an expansion in a small parameter, with c_P the leading term. The equation $bc = 0$ at the top order is $b_0 c_P = 0$: the leading term is a cycle for the "classical" differential b_0. At the next order it reads $b_1 c_P + b_0 c_{P-1} = 0$: the first correction b_1 applied to c_P must be a b_0-boundary. So the leading term of a genuine cycle defines a class in E^1 killed by d^1. Conversely, the leading term of a boundary $b(x+y)$ is $b_1 x + b_0 y$, and so lies in the image of d^1. Thus E^2 is a first approximation to the homology: the leading terms of cycles modulo the leading terms of boundaries, computed to first order.

5.2 Three tiny examples

Example 5.2 (d^1 kills)**.** Let $C_1 = \mathbb{C}e$ with e of order 1, $C_0 = \mathbb{C}f$ with f of order 0, and $be = f$. Then $b = b_1$, $b_0 = 0$, so E^1 is spanned by $[e]$ and $[f]$, and $d^1[e] = [f]$. Hence $E^2 = 0$, which is the homology of C.

Example 5.3 (A longer differential)**.** Now give e order 2 and keep $be = f$. Then $b = b_2$ and $b_0 = b_1 = 0$, so $E^1 = E^2$ is spanned by $[e]$ and $[f]$, but the homology is zero. The second page is not the answer: a further differential $d^2 : E^2_{2,1} \to E^2_{0,0}$, induced by b_2, sends $[e]$ to $[f]$. In general there are pages E^r and differentials $d^r : E^r_{p,k} \to E^r_{p-r,k-1}$, each the homology of the previous one (G4). The differential d^r is computed by a *zig-zag*: lift a class to x with $bx \in F_{p-r}$, and take the leading term of bx.

Example 5.4 (A correction term: the Weyl algebra)**.** Filter A_1 by order in ∂, and Hochschild chains by total order. The chain $c_1 = 1 \otimes x \otimes \partial - 1 \otimes \partial \otimes x$ has order 1. Its boundary has order 0: $bc_1 = 1 \otimes [\partial, x] = 1 \otimes 1$. In the language above, c_1 is a b_0-cycle, since the associated graded algebra $\mathbb{C}[x, \xi]$ is commutative, but $b_1 c_1 = 1 \otimes 1 \neq 0$. This is a b_0-boundary, $1 \otimes 1 = b_0(1 \otimes 1 \otimes 1)$, so we correct: $c = c_1 - 1 \otimes 1 \otimes 1$ is a cycle (Example 3.14). The leading term c_1 survives to E^2, and the correction $c_0 = -1 \otimes 1 \otimes 1$ was found by solving $b_0 c_0 = -b_1 c_1$.

The spectral sequence *degenerates at* E^2 if all d^r, $r \geq 2$, are zero. It *converges* if E^∞, the page at which it stabilizes, is the associated graded of $H(C)$ for the filtration $F_p H = \mathrm{im}(H(F_p C) \to H(C))$, and if $H(C)$ is recovered from its associated graded (exhaustive, separated and complete). We need neither the general theory nor the pages E^r, $r \geq 3$: the two propositions below are proved directly from Definition 5.1.

5.3 The basic step and a convergence criterion

Lemma 5.5 (Step (P))**.** *Let $c \in F_p C_k$ with $b_0 c_p = 0$ and $b_1 c_p + b_0 c_{p-1} = 0$ (for instance $bc \in F_{p-2}$). If the class of c_p in $E^2_{p,k}$ is zero, there are $x \in C_{p+1,k+1}$ with $b_0 x = 0$ and $y \in C_{p,k+1}$ such that $c - b(x + y) \in F_{p-1} C_k$.*

Proof. The class $[c_p] \in E^1_{p,k}$ is a d^1-cycle, and since it is zero in E^2, it equals $d^1[x] = [b_1 x]$ for a b_0-cycle $x \in C_{p+1,k+1}$. So $c_p - b_1 x = b_0 y$ for some $y \in C_{p,k+1}$. The components of $b(x + y)$ of orders $p + 1$ and p are $b_0 x = 0$ and $b_1 x + b_0 y = c_p$. □

Say that *(S) holds at* (p, k) if every class in $E^2_{p,k}$ is the class of the leading term z_p of a cycle $z \in F_p C_k$. Let $F_p H_k \subset H_k(C)$ be the classes having a representative in $F_p C_k$.

Proposition 5.6 (Convergence criterion)**.** *Let the filtration be complete, and fix k. Suppose (S) holds at $(q, k+1)$ for all q and at (p, k) for all p. Then:*

1. *for each p, $\Theta_p[c] = [c_p]$, for cycles $c \in F_p C_k$, is a well-defined surjection $F_p H_k \to E^2_{p,k}$ with kernel $F_{p-1} H_k$;*
2. *$\bigcap_p F_p H_k = 0$;*
3. *$H_k = \bigcup_p F_p H_k$.*

Proof. (3) holds because every chain has bounded order.

(1) *Well defined.* Let $w = be \in F_p C_k$ with $e \in F_q C_{k+1}$. We show $[w_p] = 0$ in $E^2_{p,k}$. If $q \leq p + 1$, then $b_0 e_{p+1} = w_{p+1} = 0$ and $w_p = b_1 e_{p+1} + b_0 e_p$, so $[w_p] = d^1[e_{p+1}] = 0$. If $q > p + 1$, then $b_0 e_q = w_q = 0$ and $b_1 e_q + b_0 e_{q-1} = w_{q-1} = 0$, so e_q defines a class in $E^2_{q,k+1}$. By (S) there is a cycle $z \in F_q C_{k+1}$ with the same class, that is $z_q - e_q = b_1 x + b_0 y$ with x a b_0-cycle of order $q + 1$ and y of order q. Then $e' = e - z + b(x + y)$ lies in F_{q-1}, because its components of orders $q + 1$ and q are $-b_0 x = 0$ and $e_q - z_q + b_1 x + b_0 y = 0$; and $be' = be$. Repeating, we reach $q \leq p + 1$. *Surjective*: this is (S) at (p, k). *Kernel*: if $[c_p] = 0$, Lemma 5.5 gives a homologous cycle in F_{p-1}; conversely a class in $F_{p-1} H_k$ has a representative with $c_p = 0$.

(2) Let $c \in F_P C_k$ be a cycle whose class lies in every $F_p H_k$. By (1), $\Theta_P[c] = 0$, so Lemma 5.5 gives $e_P \in F_{P+1} C_{k+1}$ with $c^{(1)} = c - be_P \in F_{P-1}$. Its class still lies in every $F_p H_k$, so we can repeat, obtaining $e_{P-r} \in F_{P-r+1}$ with $c - b \sum_{s=0}^{r} e_{P-s} \in F_{P-r-1}$. By completeness $e = \sum_s e_{P-s}$ exists and $c - be$ lies in $\bigcap_p F_p C_k = 0$. □

5.4 When the second page lies on a line

Proposition 5.7 (Degeneration and convergence [25, Ch. 10])**.** *Let the filtration be complete, and suppose there is an integer n such that $E^2_{p,k} = 0$ whenever $p \neq k - n$. Then every class in $H_k(C)$ has a representative in $F_{k-n}C_k$, and*

$$\Theta : H_k(C) \to E^2_{k-n,k}, \qquad [c] \mapsto [c_{k-n}] \quad (c \in F_{k-n}C_k),$$

is an isomorphism.

Proof. We verify (S). Off the line $E^2 = 0$, so (S) is trivial there. Let $x \in C_{k-n,k}$ be a b_0-cycle with $d^1[x] = 0$. Then $b_1 x = -b_0 y$ for some $y \in C_{k-n-1,k}$, and $c^{(1)} = x + y$ satisfies $bc^{(1)} \in F_{k-n-2}$. Suppose $c^{(r)} \in F_{k-n}C_k$ has leading term x and $z = bc^{(r)} \in F_pC_{k-1}$ with $p = k - n - r - 1$. Then z is a cycle, and $p \neq (k-1) - n$, so $E^2_{p,k-1} = 0$ and Lemma 5.5 gives $x' \in C_{p+1,k}$, $y' \in C_{p,k}$ with $z - b(x' + y') \in F_{p-1}$. Put $c^{(r+1)} = c^{(r)} - x' - y'$. Since $p + 1 < k - n$, its leading term is still x, and $bc^{(r+1)} \in F_{p-1}$. The corrections have orders tending to $-\infty$, so by completeness $c = \lim_r c^{(r)}$ exists; it is a cycle, because bc lies in every F_p, and its leading term is x.

Now apply Proposition 5.6. Since $E^2_{p,k} = 0$ for $p > k - n$, part (1) gives $F_pH_k = F_{p-1}H_k$ for $p > k - n$, and with (3) every class lies in $F_{k-n}H_k$. Since $E^2_{p,k} = 0$ for $p < k - n$, the same argument gives $F_{k-n-1}H_k = F_pH_k$ for all $p < k - n$, and by (2) this is zero. So $\Theta = \Theta_{k-n}$ is an isomorphism $H_k = F_{k-n}H_k \to E^2_{k-n,k}$. □

Figure 3 shows the mechanism. The higher differentials d^r of Example 5.3 map (p, k) to $(p - r, k - 1)$. If E^2 lives on the line $p = k - n$, the source and target can both be on the line only if $r = 1$. So $d^r = 0$ for $r \geq 2$: the spectral sequence *degenerates*. The proposition shows, in addition, that the answer is not lost in the infinite tail of the filtration: it *converges*.

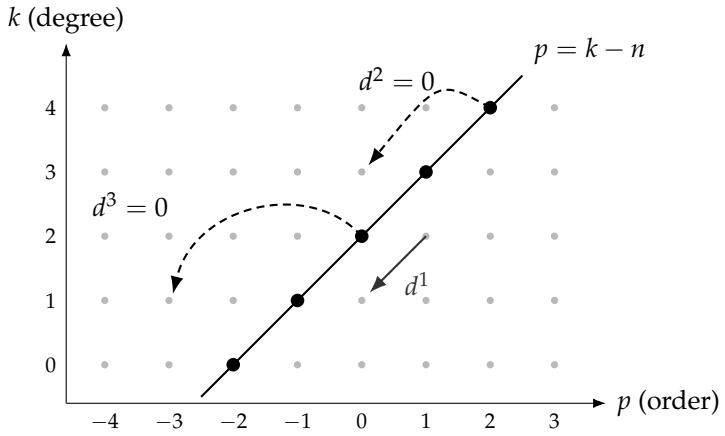


Figure 3: The spectral sequence of the order filtration for a symbol algebra, drawn for $n = 2$. Gray dots mark the positions (p, k) where $E^1_{p,k}$ (homogeneous k-forms of degree p) can be non-zero. The differential d^1 moves along lines parallel to $p = k - n$. In BG's situation E^2 lives on the line $p = k - n$ (black dots), while d^r, $r \geq 2$, maps (p, k) to $(p - r, k - 1)$, off the line, so it vanishes.

Question 5.8. *The Weyl algebra and $\mathcal{D}(\mathbb{R}^n)$ have orders ≥ 0, while symbols have orders unbounded below. Does that matter?*

Answer. Yes, for convergence. If $C_{p,k} = 0$ for $p < 0$, every chain is a finite sum, completeness is automatic, and Proposition 5.7 applies to any subcomplex of finite sums. For symbols the

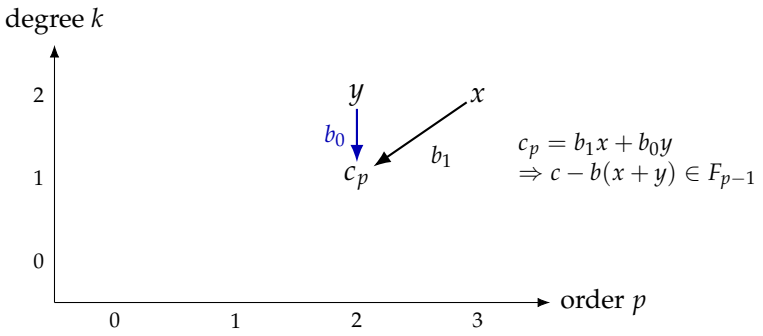


Figure 4: Step (P) of Lemma 5.5, drawn for $p = 2$, $k = 1$. If the leading term c_p of a cycle is zero in E^2, it is $b_1 x + b_0 y$ with x one order higher, and subtracting the boundary $b(x+y)$ lowers the order of the cycle. Iterating step (P) and summing the corrections is where completeness is used.

corrections in the proof form an infinite series, and one must decide which infinite sums are chains. This is the content of Section 7.6 and Appendix B. □

6 Poisson manifolds and the operator δ

What this section does. We start from the Poisson bracket that appears in the symbol of a commutator, introduce the symplectic form, Hamiltonian vector fields and the Poisson bivector on $T^*\mathbb{R}^n$, and only then the general definitions. We compute the Koszul–Brylinski operator δ on $\mathbb{R}^2$ by hand, and then prove the general results: the formula for δ, Brylinski's theorem $\delta = \pm *^{-1} d *$, and the de Rham cohomology of homogeneous forms.

6.1 From commutators to Poisson brackets

Lemma 6.1. *If $A \in \Psi^m$ and $B \in \Psi^{m'}$ have principal symbols a, b, then $[A, B] \in \Psi^{m+m'-1}$ and $\sigma_{m+m'-1}([A, B]) = -i\{a, b\}$.*

Proof. In the composition formula (1) the terms with $\alpha = 0$ cancel in $a\#b - b\#a$. The terms with $|\alpha| = 1$ give, at top order, $\sum_j (\partial_{\xi_j} a \, D_{x_j} b - \partial_{\xi_j} b \, D_{x_j} a) = -i\{a, b\}$, and the rest has lower order. □

The bracket is bilinear and antisymmetric, satisfies the Leibniz rule $\{a, bc\} = \{a, b\}c + b\{a, c\}$, and satisfies the Jacobi identity

$$\{a, \{b, c\}\} + \{b, \{c, a\}\} + \{c, \{a, b\}\} = 0.$$

The Jacobi identity can be checked directly, but Lemma 6.1 gives a proof without computation. The commutator satisfies the Jacobi identity, and $\sigma([A, [B, C]]) = (-i)^2\{a, \{b, c\}\}$. So the Jacobi identity holds for principal symbols, that is, for all homogeneous functions on $\dot{T}^*\mathbb{R}^n$. It is a differential identity in a, b, c, and the jets at a point of homogeneous functions of any fixed degree are arbitrary, so it holds for all smooth functions.

6.2 The symplectic form and Hamiltonian vector fields on $T^*\mathbb{R}^n$

On $T^*\mathbb{R}^n$ with coordinates (x, ξ) we have the symplectic form $\omega = \sum_j d\xi_j \wedge dx_j$ and, for a function f, the *Hamiltonian vector field* H_f, defined by $\iota_{H_f}\omega = -df$. In coordinates $H_f = \sum_j (\partial_{\xi_j} f \, \partial_{x_j} - \partial_{x_j} f \, \partial_{\xi_j})$,

and $\{f,g\} = H_f g = \omega(H_f, H_g)$. The Hamiltonian flow of the principal symbol of an operator is the classical motion that the operator quantizes; for instance on $\mathbb{R}^2$, $f = \frac{1}{2}(x^2+\xi^2)$ gives $H_f = \xi\partial_x - x\partial_\xi$, a rotation.

The *Poisson bivector* π is the antisymmetric form on covectors with $\pi(df, dg) = \{f,g\}$. In coordinates $\pi = \sum_j \partial_{\xi_j} \wedge \partial_{x_j}$ and $\pi(\theta,\eta) = \sum_j (\theta(\partial_{\xi_j})\eta(\partial_{x_j}) - \theta(\partial_{x_j})\eta(\partial_{\xi_j}))$. Let $\pi^\sharp\theta$ be the vector with $\eta(\pi^\sharp\theta) = \pi(\theta,\eta)$. Then $\pi^\sharp(df) = H_f$, and $\iota_{\pi^\sharp\theta}\omega = -\theta$ for every covector θ: π is the inverse of ω, up to sign.

The *radial vector field* $R = \sum_j \xi_j \partial_{\xi_j}$ generates the dilations ρ_t. Since $\rho_t^*\omega = t\omega$, we have $\mathcal{L}_R\omega = \omega$: the symplectic form is homogeneous of degree 1. Moreover $\iota_R\omega = \alpha = \sum \xi_j\, dx_j$, so by Cartan's formula $\mathcal{L}_R\omega = d\iota_R\omega = d\alpha = \omega$.

6.3 Poisson and symplectic manifolds

Definition 6.2 ([7, §1])**.** A *Poisson bracket* on a commutative algebra A is a bilinear map $\{\cdot,\cdot\}$ making A a Lie algebra and satisfying the Leibniz rule $\{f, gh\} = g\{f,h\} + h\{f,g\}$. A *Poisson manifold* is a manifold with a Poisson bracket on C^∞; the bracket is then $\{f,g\} = \pi(df,dg)$ for a bivector field π, by the Leibniz rule.

A *symplectic manifold* (X,ω) is a manifold of dimension $2n$ with a closed non-degenerate 2-form. It is Poisson: define H_f by $\iota_{H_f}\omega = -df$ and $\{f,g\} = H_f g$. By Darboux's theorem (G6), near every point there are coordinates (x,ξ) with $\omega = \sum d\xi_j \wedge dx_j$, so all the formulas of Section 6.2 hold locally. On a cotangent bundle T^*M the canonical coordinates are Darboux coordinates, and no theorem is needed. BG write $H_g f = \{f,g\}$, which is our $-H_g$; see Remark 9.5.

6.4 The operator δ on the plane

On $\mathbb{R}^2$ with $\omega = d\xi \wedge dx$, let ι_π be contraction with the bivector: $\iota_\pi = 0$ on functions and 1-forms, and $\iota_\pi(h\, d\xi \wedge dx) = h$. Put $\delta = \iota_\pi d - d\iota_\pi$. Then

$$\begin{aligned} \delta f &= \iota_\pi(df) = 0, \\ \delta(f\, dg) &= \iota_\pi(df \wedge dg) = \iota_\pi\big((f_\xi g_x - f_x g_\xi)\, d\xi \wedge dx\big) = \{f,g\}, \\ \delta(h\,\omega) &= -d\iota_\pi(h\,\omega) = -dh. \end{aligned}$$

So δ lowers degree by one. It is a differential: $\delta^2(h\omega) = -\delta(dh) = -\{1,h\} = 0$. Its homology on smooth forms is easy to compute. On 2-forms, $\ker\delta = \mathbb{C}\,\omega$. On 1-forms, $\ker\delta$ is the space of closed forms, since $\delta\theta = \iota_\pi d\theta$, and the image of δ is the space of exact forms; the quotient is $H^1(\mathbb{R}^2) = 0$. On functions, every h is $\{\xi, G\} = \partial_x G$ with $G = \int_0^x h$. Hence

$$H_k(\Omega^\bullet(\mathbb{R}^2), \delta) \cong H^{2-k}(\mathbb{R}^2) = \begin{cases} \mathbb{C}, & k = 2, \\ 0, & k \neq 2. \end{cases}$$

The duality $k \leftrightarrow 2-k$ is explained by a star operator. Define $*1 = \omega$, $*\omega = 1$, and $* = \mathrm{id}$ on 1-forms. Then $*d*(f\,dg) = *(df \wedge dg) = \{f,g\}$ and $-*d*(h\omega) = -dh$, so

$$\delta = (-1)^{k-1} * d * \qquad \text{on } k\text{-forms on } \mathbb{R}^2.$$

This is Brylinski's theorem in its smallest case. The rest of this section proves it in general.

6.5 The Koszul–Brylinski operator

Let (X, ω) be symplectic of dimension $2n$, with Poisson bivector π. Define $\iota_\pi : \Omega^k \to \Omega^{k-2}$ by

$$\iota_\pi(\theta_1 \wedge \cdots \wedge \theta_k) = \sum_{j<l} (-1)^{j+l+1} \pi(\theta_j, \theta_l)\, \theta_1 \wedge \ldots \widehat{\theta_j} \ldots \widehat{\theta_l} \cdots \wedge \theta_k.$$

In Darboux coordinates $\iota_\pi = \sum_j \iota_{\partial_{x_j}} \iota_{\partial_{\xi_j}}$. Following Koszul [19] and Brylinski [5], set

$$\delta = [\iota_\pi, d] = \iota_\pi d - d\iota_\pi : \Omega^k(X) \to \Omega^{k-1}(X).$$

Lemma 6.3. *For functions $a_0, \ldots, a_k$,*

$$\delta(a_0\, da_1 \wedge \cdots \wedge da_k) = \sum_{j=1}^{k} (-1)^{j+1} \{a_0, a_j\}\, da_1 \wedge \ldots \widehat{da_j} \cdots \wedge da_k \\ + \sum_{1 \le j < l \le k} (-1)^{j+l} a_0\, d\{a_j, a_l\} \wedge da_1 \wedge \ldots \widehat{da_j} \ldots \widehat{da_l} \cdots \wedge da_k.$$

Moreover, if β has constant coefficients in Darboux coordinates, then $\delta(f\beta) = \iota_{H_f}\beta$. Finally, for every 1-form θ,

$$[\iota_\pi, \varepsilon(\theta)] = \iota_{\pi^\sharp \theta}, \tag{6}$$

where $\varepsilon(\theta)$ is exterior multiplication by θ.

Proof. Write D_{jl} for $da_1 \wedge \cdots \wedge da_k$ with da_j and da_l omitted, and D_{0l} when only da_l is omitted. Applying ι_π to $da_0 \wedge da_1 \wedge \cdots \wedge da_k$, the pairs $(0, l)$ give $\sum_l (-1)^{l+1}\{a_0, a_l\} D_{0l}$, and the pairs $1 \le j < l$ give $\sum (-1)^{j+l+1}\{a_j, a_l\}\, da_0 \wedge D_{jl}$. On the other hand $d\iota_\pi(a_0\, da_1 \wedge \cdots \wedge da_k) = \sum_{j<l} (-1)^{j+l+1}(\{a_j, a_l\}\, da_0 + a_0\, d\{a_j, a_l\}) \wedge D_{jl}$. Subtracting gives the formula. For (6), for vector fields V, W one has $\iota_V \iota_W \varepsilon(\theta) - \varepsilon(\theta)\iota_V \iota_W = \theta(W)\iota_V - \theta(V)\iota_W$, from $\iota_W \varepsilon(\theta) + \varepsilon(\theta)\iota_W = \theta(W)$. With $V = \partial_{x_j}$, $W = \partial_{\xi_j}$, summed over j, the right side is $\iota_{\pi^\sharp\theta}$. For the middle statement, $d\beta = 0$ and $d\iota_\pi\beta = 0$, so $\delta(f\beta) = \iota_\pi(df \wedge \beta) - df \wedge \iota_\pi\beta = [\iota_\pi, \varepsilon(df)]\beta = \iota_{H_f}\beta$. □

Correction to BG 6.4 (The sign in BG's formula for δ). In [7, §1, Theorem (2)], and again on forms in [7, §2], the second sum carries the sign $(-1)^{i+j+1}$. It should be $(-1)^{i+j}$, as in Lemma 6.3. With BG's sign the operator is not well defined on forms. Take $k = 2$ and compare $\delta(1 \cdot d(bc) \wedge dg)$ with $\delta(b\, dc \wedge dg + c\, db \wedge dg)$, which must agree since $d(bc) = b\, dc + c\, db$. With the correct sign both equal $-d(b\{c, g\}) - d(c\{b, g\})$. With BG's sign the first is $+d(b\{c, g\} + c\{b, g\})$, while the second is $-\{b, g\}\, dc - \{c, g\}\, db + b\, d\{c, g\} + c\, d\{b, g\}$, and these differ by $2(\{b, g\}\, dc + \{c, g\}\, db)$. We also checked Lemma 6.3 symbolically against $\iota_\pi d - d\iota_\pi$ on $\mathbb{R}^4$ for $k = 2, 3$. The same calculation shows that the upper limit of the sums in [7, §2] should be m, not n. Since the correct formula is also what the Hochschild computation produces (Theorem 8.1), the error is a misprint.

6.6 The symplectic star operator

Fix a point and let V be the tangent space there. On $\Lambda^k V^*$ define the bilinear form

$$\Lambda(\theta_1 \wedge \cdots \wedge \theta_k, \eta_1 \wedge \cdots \wedge \eta_k) = \det\big(\pi(\theta_i, \eta_j)\big)_{i,j=1}^{k},$$

and $\Lambda(1, 1) = 1$. It is non-degenerate, because $\Lambda(\alpha, \beta) = \pm\langle \alpha, \Lambda^k \pi^\sharp \beta \rangle$ and $\pi^\sharp$ is invertible.

Definition 6.5 (Brylinski [5])**.** The *symplectic star operator* $* : \Lambda^k V^* \to \Lambda^{2n-k} V^*$ is defined by

$$\alpha \wedge *\beta = \Lambda(\alpha, \beta)\,\mathrm{vol} \qquad \text{for all } \alpha \in \Lambda^k V^*, \qquad \mathrm{vol} = \frac{\omega^n}{n!}.$$

The wedge pairing $\Lambda^k \times \Lambda^{2n-k} \to \Lambda^{2n}$ is non-degenerate, so $*$ is well defined and an isomorphism. Applied pointwise it gives isomorphisms $* : \Omega^k(X) \to \Omega^{2n-k}(X)$. In Darboux coordinates $*$ maps constant-coefficient forms to constant-coefficient forms. For example $*1 = \mathrm{vol}$, and on $\mathbb{R}^2$ we recover the operator of Section 6.4. This is the analogue of the Riemannian Hodge star of [23, §"Formal adjoints and the Hodge star"], with the metric replaced by π.

Lemma 6.6. *For $\theta \in V^*$, $\beta \in \Lambda^k V^*$ and $\gamma \in \Lambda^j V^*$,*

$$*^{-1}(\theta \wedge *\beta) = (-1)^{k-1} \iota_{\pi^\sharp \theta} \beta, \qquad *(\theta \wedge \gamma) = (-1)^{j+1} \iota_{\pi^\sharp \theta}(*\gamma).$$

Consequently $*(f_0\, df_1 \wedge \cdots \wedge df_m) = (-1)^{m(m+1)/2} f_0\, \iota_{H_{f_1}} \cdots \iota_{H_{f_m}} \mathrm{vol}$.

Proof. Expanding the determinant along the first row, $\Lambda(\theta \wedge \alpha, \beta) = \Lambda(\alpha, \iota_{\pi^\sharp\theta}\beta)$, since $\iota_v(\eta_1 \wedge \cdots \wedge \eta_k) = \sum_j (-1)^{j+1} \eta_j(v)\, \eta_1 \wedge \ldots \widehat{\eta_j} \cdots \wedge \eta_k$ and $\eta_j(\pi^\sharp\theta) = \pi(\theta, \eta_j)$. Let $\mu = *^{-1}(\theta \wedge *\beta)$. For all $\alpha \in \Lambda^{k-1} V^*$,

$$\Lambda(\alpha, \mu)\mathrm{vol} = \alpha \wedge \theta \wedge *\beta = (-1)^{k-1} \theta \wedge \alpha \wedge *\beta = (-1)^{k-1} \Lambda(\alpha, \iota_{\pi^\sharp\theta}\beta)\mathrm{vol}.$$

For the second identity expand along the first column instead: $\Lambda(\alpha, \theta \wedge \gamma) = -\Lambda(\iota_{\pi^\sharp\theta}\alpha, \gamma)$ for $\alpha \in \Lambda^{j+1}$, because $\pi(\alpha_i, \theta) = -\pi(\theta, \alpha_i)$. So $\alpha \wedge *(\theta \wedge \gamma) = -\iota_v \alpha \wedge *\gamma$ with $v = \pi^\sharp\theta$. Since $\alpha \wedge *\gamma$ has degree $2n+1$, it vanishes, and $0 = \iota_v(\alpha \wedge *\gamma) = \iota_v\alpha \wedge *\gamma + (-1)^{j+1} \alpha \wedge \iota_v * \gamma$. Hence $\alpha \wedge *(\theta \wedge \gamma) = (-1)^{j+1} \alpha \wedge \iota_v * \gamma$ for all α. The last formula follows by induction from $*1 = \mathrm{vol}$ and $\pi^\sharp df = H_f$; the signs multiply to $(-1)^{1+2+\cdots+m}$. □

Theorem 6.7 (Brylinski [5])**.** *On k-forms, $\delta = (-1)^{k-1} *^{-1} d *$. In particular $\delta^2 = 0$.*

Proof. Both sides are first-order differential operators, so it suffices to compare them on $f\beta$ with β of constant coefficients in Darboux coordinates. Then $*\beta$ has constant coefficients and $d(f * \beta) = df \wedge *\beta$. By Lemma 6.6, $*^{-1} d * (f\beta) = (-1)^{k-1} \iota_{H_f}\beta$, which is $(-1)^{k-1}\delta(f\beta)$ by Lemma 6.3. Finally $\delta^2 = \pm *^{-1} d * *^{-1} d* = \pm *^{-1} d^2 * = 0$. □

6.7 The $\mathfrak{sl}(2)$ structure and BG's star

Let $L = \varepsilon(\omega)$ and $\Lambda_\pi = \iota_\pi$, acting on $\Lambda^\bullet V^*$.

Lemma 6.8. *$[\iota_\pi, \varepsilon(\omega)] = n - k$ on $\Lambda^k V^*$.*

Proof. ι_π is even, so $[\iota_\pi, \varepsilon(d\xi_j)\varepsilon(dx_j)] = [\iota_\pi, \varepsilon(d\xi_j)]\varepsilon(dx_j) + \varepsilon(d\xi_j)[\iota_\pi, \varepsilon(dx_j)]$. By (6) with $\pi^\sharp(d\xi_j) = H_{\xi_j} = \partial_{x_j}$ and $\pi^\sharp(dx_j) = H_{x_j} = -\partial_{\xi_j}$, this is $\iota_{\partial_{x_j}}\varepsilon(dx_j) - \varepsilon(d\xi_j)\iota_{\partial_{\xi_j}} = 1 - \varepsilon(dx_j)\iota_{\partial_{x_j}} - \varepsilon(d\xi_j)\iota_{\partial_{\xi_j}}$. Summing over j gives n minus the degree operator. □

So L, Λ_π and $n - k$ span a copy of $\mathfrak{sl}(2)$, exactly as the Lefschetz operator, its adjoint and the degree do for a Kähler manifold. BG define their star as $*_{\mathrm{BG}} = \exp\big(\frac{\pi i}{2}(L + L^*)\big)$, where L^* is the symplectic adjoint of L (a multiple of Λ_π with $[L, L^*] = n - k$).

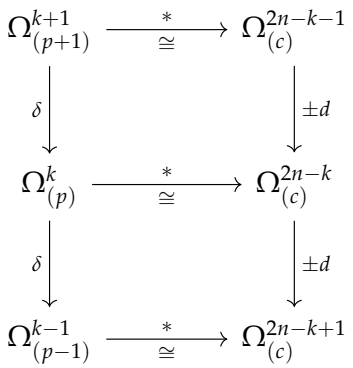


Figure 5: Brylinski's duality on a conic symplectic manifold. The star operator carries k-forms homogeneous of degree p to $(2n-k)$-forms homogeneous of degree $c = p+n-k$ (Proposition 6.11), and it intertwines δ with $\pm d$. The homogeneity c is the same in the whole right column: δ lowers both the form degree and the homogeneity by one, so c is constant along the diagonals $p-k = \text{const}$ of Figure 3.

Correction to BG 6.9 (The $\mathfrak{sl}(2)$ computation). BG deduce $*_{\mathrm{BG}}^2 = 1$ from the identity

$$\exp\left(\pi i \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\right) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

In fact, for $J = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ we have $J^2 = 1$ and $\exp(\pi i J) = \cos\pi + i \sin\pi\, J = -1$. In an irreducible representation of $\mathfrak{sl}(2)$ of dimension d, $\exp(\pi i(T_+ + T_-)) = (-1)^{d-1}$, so $*_{\mathrm{BG}}^2 = \pm 1$ on the isotypic components. For $n = 1$, on the span of 1 and ω, $*_{\mathrm{BG}} 1 = i\omega$ and $*_{\mathrm{BG}}^2 = -1$. Similarly, with the commutation relations stated in the proof of BG's Proposition (1), the series $\sum_j \frac{(ti)^j}{j!} \mathrm{ad}(L + L^*)^j$ gives hyperbolic functions of t, not $\cos t$ and $\sin t$; the factor i must be dropped (or the relations changed by a factor i). None of this affects the conclusions, since only the fact that $*$ is invertible and intertwines δ with d up to non-zero constants is used. We work with Brylinski's operator of Section 6.6, for which everything is proved above. (For it one also has $*^2 = 1$ [5]; we checked this symbolically for $n \leq 2$ and do not use it.)

6.8 Homogeneous forms and their de Rham cohomology

Now let $X = \dot{T}^*M$ with its dilations and radial field R. Choose a positive function r on $\dot{T}^*M$, homogeneous of degree 1 (for instance $|\xi|$ for a Riemannian metric). Then $\dot{T}^*M \cong S^*M \times (0,\infty)$, $(x,\xi) \mapsto ((x,[\xi]), r)$, where we identify S^*M with $\{r = 1\}$.

Lemma 6.10. $\iota_R \mathrm{vol}$ *restricts to a nowhere-vanishing form on* S^*M. *So* S^*M *is orientable, and we orient it by* $\iota_R \mathrm{vol}$.

Proof. vol is nowhere zero and R is transverse to $\{r = 1\}$, since $Rr = r$. □

Proposition 6.11. $*$ *maps* $\Omega^k_{(p)}$ *onto* $\Omega^{2n-k}_{(p+n-k)}$.

Proof. $\{a,b\}$ is homogeneous of degree $p+q-1$ if a, b have degrees p, q, so $\Lambda(\alpha,\beta)$ is homogeneous of degree $p+q-k$ for $\alpha \in \Omega^k_{(p)}$, $\beta \in \Omega^k_{(q)}$, and vol has degree n. Hence $\alpha \wedge \rho_t^*(*\beta) = \rho_t^*(t^{-p}\alpha \wedge *\beta) = t^{q+n-k}\alpha \wedge *\beta$. The values of the forms $\alpha \in \Omega^k_{(p)}$ span Λ^k at each point, so $\rho_t^*(*\beta) = t^{q+n-k} * \beta$. The same argument applies to $*^{-1}$. □

Proposition 6.12 (Homogeneous de Rham cohomology). *Let $\Omega^\bullet_{(c)}$ be the complex of forms on $\dot{T}^*M$ homogeneous of degree $c \in \mathbb{Z}$, with d.*

1. *If $c \neq 0$, the complex is acyclic: a closed θ equals $d(\iota_R\theta/c)$.*
2. *Every $\theta \in \Omega^j_{(0)}$ can be written uniquely as $\theta = \theta' + \theta'' \wedge dr/r$, with θ', θ'' pulled back from S^*M. This identifies $(\Omega^\bullet_{(0)}, d)$ with the de Rham complex of $S^*M \times S^1$, $S^1 = (0,\infty)/e^{\mathbb{Z}}$, and*

$$H^j(\Omega^\bullet_{(0)}, d) \cong H^j(S^*M) \oplus H^{j-1}(S^*M) \cong H^j(S^*M \times S^1).$$

3. *The same holds for forms that are polynomial in the fibres of T^*M, with the difference that the polynomial forms of degree 0 are the forms on M; their cohomology is $H^j(M)$.*

Proof. (1) ι_R commutes with ρ_t, so it preserves homogeneity, and $\mathcal{L}_R = c$ on $\Omega_{(c)}$. By Cartan's formula $d\iota_R + \iota_R d = \mathcal{L}_R$, a closed θ equals $d(\iota_R\theta)/c$.

(2) Since $\rho_t^*(dr/r) = dr/r$, a form is homogeneous of degree 0 exactly when θ' and θ'' are independent of r, that is, pulled back from S^*M. Then $d(\theta' + \theta'' \wedge dr/r) = d\theta' + d\theta'' \wedge dr/r$. On $S^*M \times S^1$, with $s = \log r \bmod \mathbb{Z}$ the coordinate on the circle, $dr/r = ds$, and a form on $S^*M \times S^1$ that is invariant under rotation of the circle is $\theta' + \theta'' \wedge ds$. The de Rham cohomology of $S^*M \times S^1$ is computed by invariant forms (average over the circle), and the Künneth formula gives the last isomorphism.

(3) ι_R preserves polynomial forms, so (1) applies. A polynomial form of degree 0 has no $d\xi$ (which has degree 1) and coefficients of degree 0 in ξ, so it is a form on M. □

The splitting in (2) depends on r. Replacing r by ar, with $a > 0$ homogeneous of degree 0, changes θ' by $\theta'' \wedge d\log a$, which does not change the class of θ''.

Question 6.13. *Why does the answer involve $S^*M \times S^1$ rather than $\dot{T}^*M$, which is homotopy equivalent to S^*M?*

Answer. On all smooth forms, $dr/r = d\log r$ is exact. Among *homogeneous* forms it is closed but not exact, because $\log r$ is not homogeneous. The symbol algebra only contains homogeneous functions, so it sees the extra class dr/r. Geometrically, dividing $(0,\infty)$ by the dilation $r \mapsto er$ turns the half-line into a circle, and dr/r becomes the angular form. □

7 Star products and the symbol algebra

What this section does. The composition formula for symbols is a "star product": a deformation of the pointwise product in which the order of a symbol plays the role of Planck's constant. We make this precise, work out the Moyal product on $\mathbb{R}^2$, define formal deformations and BG's quantum algebras, quote the theorem of de Wilde and Lecomte, and set up the symbol algebra $\mathcal{P}$ with its two chain models.

7.1 The composition formula is a star product

On $\mathbb{R}^n$, write the composition formula (1) by homogeneity:

$$a\#b = \sum_{l \geq 0} P_l(a,b), \qquad P_l(a,b) = \sum_{|\alpha|=l} \frac{1}{\alpha!}\, \partial_\xi^\alpha a\, D_x^\alpha b.$$

Each P_l is a *bidifferential operator*, a finite sum of products of derivatives of a and of b. If a and b are homogeneous of degrees p and q, then $P_l(a,b)$ is homogeneous of degree $p+q-l$. Moreover $P_0(a,b) = ab$ and, by Lemma 6.1, $P_1(a,b) - P_1(b,a) = -i\{a,b\}$.

Introduce a parameter $\hbar > 0$ and rescale: for a formal sum $a = \sum_j a_j$ put $L_\hbar a = \sum_j \hbar^j a_j$, and $a \#_\hbar b = L_\hbar^{-1}(L_\hbar a \# L_\hbar b)$. Since $P_l(\hbar^p a_p, \hbar^q b_q) = \hbar^{p+q} P_l(a_p, b_q)$ has degree $p+q-l$,

$$a \#_\hbar b = \sum_{l \geq 0} \hbar^l P_l(a,b) = ab - \frac{i\hbar}{2}\{a,b\} + \hbar\, S(a,b) + O(\hbar^2),$$

where S is the symmetric part of P_1. All these products are isomorphic, and at $\hbar = 0$ we recover the commutative product. This is the *semiclassical* picture of [7]: symbols are a deformation of the algebra of functions on $\dot{T}^*M$ in the direction of the Poisson bracket, and lowering the order by one corresponds to one power of $\hbar$. In particular $\hbar D_x$ has symbol $\hbar\xi$, and $[x, \hbar D_x] = i\hbar$.

7.2 The Moyal product

The left quantization is not symmetric in a and b. The Weyl quantization is, and its composition law is the Moyal product. On $\mathbb{R}^{2n}$ let $\Pi = \sum_j (\partial_{\xi_j} \otimes \partial_{x_j} - \partial_{x_j} \otimes \partial_{\xi_j})$, acting on $a \otimes b$, so that $m \circ \Pi(a \otimes b) = \{a,b\}$ with $m(a \otimes b) = ab$. Define

$$a \star b = m \circ \exp\Big(\frac{\hbar}{2i}\Pi\Big)(a \otimes b) = \sum_{l \geq 0} \frac{1}{l!}\Big(\frac{\hbar}{2i}\Big)^l m \circ \Pi^l (a \otimes b).$$

For smooth a, b this is a formal power series in $\hbar$; for polynomials it is a finite sum and we may set $\hbar = 1$.

Proposition 7.1. *The Moyal product is associative, $a \star b - b \star a = -i\hbar\{a,b\} + O(\hbar^3)$, and the term of order $\hbar^l$ is $(-1)^l$ times its transpose.*

Proof. On $a \otimes b \otimes c$ let Π_{12}, Π_{13}, Π_{23} act on the indicated factors; they commute, since they have constant coefficients. A derivative of a product fg is $(\partial \otimes 1 + 1 \otimes \partial)$ applied to $f \otimes g$, so

$$\begin{aligned}(a \star b) \star c &= m_3 \exp\Big(\tfrac{\hbar}{2i}(\Pi_{13} + \Pi_{23})\Big) \exp\Big(\tfrac{\hbar}{2i}\Pi_{12}\Big)(a \otimes b \otimes c) \\ &= m_3 \exp\Big(\tfrac{\hbar}{2i}(\Pi_{12} + \Pi_{13} + \Pi_{23})\Big)(a \otimes b \otimes c),\end{aligned}$$

and $a \star (b \star c)$ gives the same expression. Transposing $a \otimes b$ changes Π into $-\Pi$, which gives the last two statements. □

Example 7.2 (Moyal on $\mathbb{R}^2$)**.** On $\mathbb{R}^2$, $\{x,\xi\} = -1$, so $x \star \xi = x\xi + \frac{i\hbar}{2}$, $\xi \star x = x\xi - \frac{i\hbar}{2}$, and $\xi \star x - x \star \xi = -i\hbar$. Also $x \star x = x^2$, $\xi \star \xi = \xi^2$, and $x \star (x\xi) = x^2\xi + \frac{i\hbar}{2}x$. With $\hbar = 1$, the map sending x to multiplication by x and ξ to $D_x = -i\partial_x$ is an isomorphism from the polynomials with the Moyal product onto the Weyl algebra A_1 (the Weyl quantization), since $[D_x, x] = -i$. For comparison, the left-quantized product gives $x \# \xi = x\xi$ and $\xi \# x = x\xi - i$.

7.3 Formal deformations and quantum algebras

Let A be a commutative algebra, for instance $C^\infty(X)$. Let $K = \mathbb{C}((\hbar))$ be the field of formal Laurent series (finitely many negative powers), and $A((\hbar))$ the Laurent series with coefficients in A.

Definition 7.3. A *formal deformation* of A is a K-bilinear associative product on $A((\hbar))$ of the form

$$a \star b = \sum_{l \geq 0} \hbar^l P_l(a,b) \qquad (a,b \in A), \qquad P_0(a,b) = ab,$$

extended $\hbar$-linearly. It is *local* if $A = C^\infty(X)$ and the P_l are bidifferential operators. A *quantum algebra* in the sense of [7, §1] is a formal deformation of a Poisson algebra with $P_1(a,b) = \{a,b\}/2i$ and $P_l(a,b) = (-1)^l P_l(b,a)$, $P_l(1,a) = 0$.

Lemma 7.4. *For a formal deformation, $\{a,b\}_\star = i(P_1(a,b) - P_1(b,a))$ is a Poisson bracket on A.*

Proof. $a \star b - b \star a = -i\hbar\{a,b\}_\star + O(\hbar^2)$. The commutator is a derivation of $\star$ in each argument, and its leading term is a derivation of the commutative product: compare orders $\hbar$ in $[a, b \star c] = [a,b] \star c + b \star [a,c]$. The Jacobi identity for commutators, at order $\hbar^2$, gives the Jacobi identity for $\{\cdot,\cdot\}_\star$. □

With our conventions $\{a,b\}_\star$ is the Poisson bracket of Section 6.1 for both the Moyal and the left-quantized products. The Moyal product on $\mathbb{R}^{2n}$ is a quantum algebra in the sense of BG, by Proposition 7.1; the left-quantized product is not.

Remark on BG 7.5 (Quantum algebras). (1) The parity condition $P_l(a,b) = (-1)^l P_l(b,a)$ of [7, §1] is not satisfied by the product of $\mathcal{P}$ in the left quantization, which BG use in §4: there $P_1(a,b) = -i \sum \partial_{\xi_j} a \, \partial_{x_j} b$ is not antisymmetric. The condition is never used. The proof of Theorem 8.1 shows that only $P_0 = ab$ and the antisymmetric part of P_1 enter. (2) BG write that one should think of a quantum algebra as a deformation with $a \star b - b \star a = \frac{\hbar}{i}\{a,b\}$. Under the parity condition this holds only up to $O(\hbar^3)$, as Proposition 7.1 shows.

7.4 Conic star products and de Wilde–Lecomte

A *conic symplectic manifold* is a symplectic manifold with a free action ρ_t of $\mathbb{R}_+$ such that $\rho_t^* \omega = t\omega$ (Section 9.1). The example is $\dot{T}^*M$. Since $\{\rho_t^* a, \rho_t^* b\} = t\rho_t^*\{a,b\}$, the bracket lowers homogeneity by one.

Definition 7.6. A local star product on a conic symplectic manifold is *conic* if each P_l lowers homogeneity by l: $\rho_t^* P_l(a,b) = t^{-l} P_l(\rho_t^* a, \rho_t^* b)$.

On $\dot{T}^*\mathbb{R}^n$ the Moyal product is conic, since Π contains one ξ-derivative in each term, and so is the left-quantized product $\sum_l \hbar^l P_l$.

Quoted background 7.7 (De Wilde–Lecomte [10], as stated in [7, §2])**.** On every conic symplectic manifold there is a local conic star product whose first-order term has antisymmetric part $\{a,b\}/2i$. It is unique up to conjugation by a K-linear map $a \mapsto a + \sum_{l \geq 2} \hbar^l \rho_l(a)$, where ρ_l is a differential operator lowering homogeneity by l.

We do not need this theorem for $\dot{T}^*M$: there the symbol calculus gives an explicit conic star product (Corollary 7.9). The uniqueness statement explains why the Hochschild homology depends only on the symplectic cone, but we never use it.

7.5 The symbol algebra as a star-product algebra

Let M be a closed manifold and $Y = \dot{T}^*M$. Let $\widetilde{\mathcal{A}}$ be the space of formal sums $a = \sum_{j\le m} a_j$, $a_j \in C^\infty_{(j)}(Y)$.

Proposition 7.8 (Global star product)**.** *There is a linear bijection $\sigma : \mathcal{P} \to \widetilde{\mathcal{A}}$ such that $\sigma(\Psi^m/\Psi^{-\infty}) = \{a :\ a_j = 0 \text{ for } j > m\}$, whose term of degree m on Ψ^m is the principal symbol, and bidifferential operators P_l on Y lowering homogeneity by l, such that*

$$\sigma(AB) = \sum_{l\ge 0} P_l(\sigma A, \sigma B), \qquad P_0(a,b) = ab, \qquad P_1(a,b) - P_1(b,a) = -i\{a,b\}.$$

The component of degree m of the right side is a finite sum.

Proof. This follows from (G1). Choose charts U_i, a partition of unity ϕ_i subordinate to them, and $\psi_i \in C^\infty_c(U_i)$ with $\psi_i = 1$ near $\operatorname{supp}\phi_i$. For $A \in \mathcal{P}$ let $\sigma_i(A)$ be the full symbol of $\psi_i A \psi_i$ in the chart U_i, and put $\sigma(A) = \sum_i \phi_i \sigma_i(A)$, transported to Y. If $\sigma(A) = 0$, then at each order the principal symbol of A vanishes, so $A \in \Psi^{-\infty}$, that is $A = 0$ in $\mathcal{P}$. For surjectivity let $q(a) = \sum_i \phi_i \operatorname{Op}_i(a)\psi_i$ (Borel summation, G1(a)). By G1(c), $\sigma \circ q = 1 + L$, where L is a formal series of differential operators lowering homogeneity, so $\sigma \circ q$ is invertible degree by degree, and σ is onto. For the product, G1(b) and G1(d) give $\phi_j \sigma_j(AB) = \phi_j\big(\sigma_j(A)\#\sigma_j(B)\big)$, because $\phi_j A(1-\psi_j^2)B\psi_j$ is smoothing. Moreover $T_j = \sigma_j \circ \sigma^{-1} = (\sigma_j \circ q)\circ(\sigma\circ q)^{-1}$ is a formal series of differential operators with top term the identity. So $\sigma(AB) = \sum_j \phi_j\big(T_j\sigma A \# T_j \sigma B\big)$, whose part lowering homogeneity by l is a bidifferential operator P_l. The top term is $\sum_j \phi_j ab = ab$, and the antisymmetric part of P_1 is computed by Lemma 6.1. □

Corollary 7.9. *$a \star_\hbar b = \sum_l \hbar^l P_l(a,b)$ is a local conic star product on $C^\infty(Y)((\hbar))$.*

Proof. The associativity identities $\sum_{l+l'=N} P_l(P_{l'}(a,b),c) = \sum_{l+l'=N} P_l(a, P_{l'}(b,c))$ hold for homogeneous a,b,c, by associativity of $\mathcal{P}$ and comparison of degrees. They are identities between polydifferential operators, and jets at a point of homogeneous functions are arbitrary, so they hold for all smooth functions. □

7.6 Chain models for $\mathcal{P}$

Put $\mathcal{P}_N = \Psi^N/\Psi^{-\infty}$. Through σ, $\mathcal{P}_N \cong \prod_{j\le N} C^\infty_{(j)}(Y)$, a countable product of nuclear Fréchet spaces, each isomorphic to $C^\infty(S^*M)$. For a multi-index $I = (I_0,\dots,I_k) \in \mathbb{Z}^{k+1}$ let $C^\infty_{(I)}(Y^{k+1})$ be the functions on Y^{k+1} homogeneous of degree I_j in the j-th factor, and $|I| = I_0 + \cdots + I_k$. By (G2), $C^\infty_{(I_0)}(Y) \mathbin{\widehat\otimes} \cdots \mathbin{\widehat\otimes} C^\infty_{(I_k)}(Y) = C^\infty_{(I)}(Y^{k+1})$.

Definition 7.10. *Model I* (the standard continuous chains) is $C_k(\mathcal{P}) = \varinjlim_N \mathcal{P}_N^{\widehat\otimes(k+1)}$. *Model II* (BG's chains) is

$$\widehat{C}_k(\mathcal{P}) = \Big\{u = \sum_{m\le M} u_m :\ u_m \in C_{m,k}\Big\}, \qquad C_{m,k} = \bigoplus_{|I|=m} C^\infty_{(I)}(Y^{k+1}) \text{ (finite sums).}$$

In both, b is the Hochschild boundary of $\star$. In Model II, $b = \sum_l b_l$, where b_l applies P_l to adjacent variables and then restricts to the diagonal.

Lemma 7.11. *Model I is a subcomplex of Model II, and both are filtered by total order, $F_p = \{u : u_m = 0 \text{ for } m > p\}$. Model II is complete in the sense of Section 5.1. Both have the same associated graded complex $\bigoplus_m C_{m,\bullet}$ with b_0.*

Proof. By (G2), $\mathcal{P}_N^{\widehat{\otimes}(k+1)} = \prod C^\infty_{(I)}(Y^{k+1})$, the product over I with all $I_j \le N$. For fixed total order m there are finitely many such I, since $I_j \ge m - kN$. So an element of $\mathcal{P}_N^{\widehat{\otimes}(k+1)}$ is a formal sum $\sum_{m \le (k+1)N} u_m$ with $u_m \in C_{m,k}$. The face maps are given by the star product, which on homogeneous components is the formula of Proposition 7.8. In Model II the order-m component of bu is the finite sum $\sum_l b_l u_{m+l}$, and $b^2 = 0$ because it holds on algebraic tensors (by associativity), each component is continuous, and algebraic tensors are dense in each $C^\infty_{(I)}(Y^{k+1})$. Completeness of Model II is clear, since it consists of all formal sums. The graded pieces of both models are the $C_{m,k}$. □

BG define the topology of $\mathcal{P}$ as $\varprojlim_m \varinjlim_{n \ge m} \Psi^n / \Psi^m$ and assert [7, §4] that the Hochschild complex is the projective limit of its quotients by F_m. This is exactly the completion of Model I for the total order filtration, which is Model II. Figure 6 shows the difference.

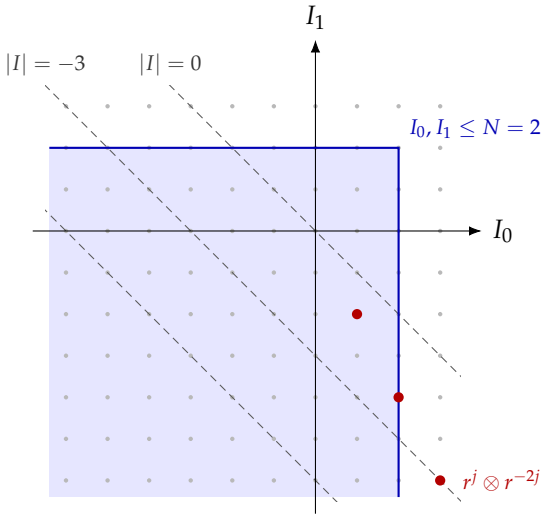


Figure 6: Multi-degrees (I_0, I_1) of Hochschild 1-chains of $\mathcal{P}$. A stage $\mathcal{P}_N^{\widehat{\otimes}2}$ of Model I contains the components in the shaded quadrant $I_0, I_1 \le N$. The total order $|I|$ is constant on the dashed anti-diagonals, and each anti-diagonal meets the quadrant in finitely many points. The series $\sum_{j \ge 1} r^j \otimes r^{-2j}$ (red, r homogeneous of degree 1) has total orders $-j \to -\infty$. It is an element of Model II, but it lies in no stage of Model I, because its first entries have unbounded order.

The convergence proof of Proposition 5.7 sums infinite series of corrections, so it applies directly only to Model II. We prove in Appendix B that Model I has the same homology (Remark 11.5).

8 The spectral sequence of a quantum algebra (BG §1)

What this section does. For a formal deformation of a commutative algebra we filter Hochschild chains by powers of $\hbar$. The first page of the resulting spectral sequence is the space of differential forms, and the first differential is $-i\hbar\,\delta$. This is the theorem of [7, §1]. We prove it, with the constant, and correct a normalization.

8.1 Setup

Let A be a commutative algebra, either a polynomial algebra or $C^\infty(X)$ for a manifold X, and let $\star$ be a formal deformation of A (Definition 7.3), local if $A = C^\infty(X)$. The Hochschild complex of $A((\hbar))$ over $K = \mathbb{C}((\hbar))$ has chains

$$C_k^K = \left(A^{\widehat{\otimes}(k+1)}\right)((\hbar)) = \Big\{\sum_{j\geq J} \hbar^j c_j : \ c_j \in C_k(A)\Big\},$$

since a K-multilinear map is determined by its values on A. Write $b = \sum_{l\geq 0} \hbar^l b_l$, where b_l is the Hochschild-type operator built from P_l: $b_l = \sum_i (-1)^i d_i^{(l)}$, with $d_i^{(l)}$ applying P_l to the entries $i, i+1$ (and $d_k^{(l)}$ to a_k, a_0, in this order). In the language of Section 5, take the order of $\hbar^j c$ to be $p = -j$. Then $C_{p,k} = \hbar^{-p} C_k(A)$, orders are bounded above, the filtration $F_p = \hbar^{-p} C_k(A)[[\hbar]]$ is complete, and b_l lowers the order by l.

The b_0-complex is the Hochschild complex of the commutative algebra A. So the first page is

$$E^1_{p,k} = \hbar^{-p}\,\mathrm{HH}_k(A) \cong \hbar^{-p}\Omega^k(A)$$

by Theorem 3.12 or Theorem 3.15. This is part (1) of BG's theorem. The isomorphism is the HKR map χ of (2).

8.2 The first differential

Theorem 8.1 ([7, §1, Theorem]). *Let $P_1(a,b) - P_1(b,a) = -i\{a,b\}$. Under χ, the differential $d^1 : E^1_{p,k} \to E^1_{p-1,k-1}$ is $-i\hbar\,\delta$, where $\delta = [\iota_\pi, d]$ is the Koszul–Brylinski operator of Lemma 6.3.*

For a quantum algebra in BG's sense, $P_1 = \{a,b\}/2i$, so $P_1(a,b) - P_1(b,a) = \{a,b\}/i = -i\{a,b\}$, and the theorem says $d^1 = (\hbar/i)\delta$, as in [7].

Proof. It suffices to compute d^1 on forms $\theta = a_0\,da_1 \wedge \cdots \wedge da_k$, which span Ω^k (for $C^\infty(X)$, use a partition of unity and coordinate functions, or an embedding of X in $\mathbb{R}^N$). By Lemma 3.11, $F = \varphi(a_0; a_1, \ldots, a_k)$ is a b_0-cycle with $\chi(F) = \theta$, so $d^1\theta = \hbar\,\chi(b_1 F)$.

Split $P_1 = S + \kappa\{\cdot,\cdot\}$ with S symmetric and $\kappa = -i/2$. Replacing P_1 by S in $b_1 F$, the inner faces cancel in pairs (σ and $\sigma \circ (i\ i{+}1)$ have opposite signs) and the first and last faces cancel as in the proof of Lemma 3.11, now because $S(a_0, a_{\sigma(1)}) = S(a_{\sigma(1)}, a_0)$. So we may replace P_1 by $\kappa\{\cdot,\cdot\}$. With the notation D_{jl} of the proof of Lemma 6.3, $(k-1)!\,\chi(b_1 F)$ is the sum of three groups of terms.

1. The face d_0: $\kappa \sum_\sigma \operatorname{sgn}(\sigma)\{a_0, a_{\sigma(1)}\}\, da_{\sigma(2)} \wedge \cdots \wedge da_{\sigma(k)}$. Grouping by $\sigma(1) = j$, whose reorderings carry the sign $(-1)^{j-1}$, this is $\kappa(k-1)! \sum_j (-1)^{j-1}\{a_0, a_j\} D_{0j}$.

2. The face $(-1)^k d_k$: $(-1)^k \kappa \sum_\sigma \operatorname{sgn}(\sigma)\{a_{\sigma(k)}, a_0\}\, da_{\sigma(1)} \wedge \cdots \wedge da_{\sigma(k-1)}$. Grouping by $\sigma(k) = j$ (sign $(-1)^{k-j}$) and using antisymmetry, this equals the expression in (1).

3. The inner faces: $\kappa \sum_\sigma \operatorname{sgn}(\sigma) \sum_{i=1}^{k-1} (-1)^i a_0\, da_{\sigma(1)} \wedge \cdots \wedge d\{a_{\sigma(i)}, a_{\sigma(i+1)}\} \wedge \cdots \wedge da_{\sigma(k)}$. Moving $d\{\cdot,\cdot\}$ to the front costs $(-1)^{i-1}$. For fixed i and $(\sigma(i), \sigma(i+1)) = (j,l)$ with $j < l$, the sign of σ is $(-1)^{j+l+1}$ times the sign of the induced ordering of the remaining indices; the pair (l,j) gives the same contribution, since both the bracket and the sign change. Each i contributes $-2\kappa(k-2)! \sum_{j<l} (-1)^{j+l+1} a_0\, d\{a_j, a_l\} \wedge D_{jl}$, and there are $k-1$ values of i.

Dividing by $(k-1)!$,

$$\chi(b_1 F) = 2\kappa\Big(\sum_j (-1)^{j+1}\{a_0, a_j\} D_{0j} + \sum_{j<l} (-1)^{j+l} a_0\, d\{a_j, a_l\} \wedge D_{jl}\Big) = -i\,\delta\theta$$

by Lemma 6.3. □

The proof shows that the symmetric part of P_1, which depends on the choice of quantization, drops out. Only the Poisson bracket survives. It also proves $\delta^2 = 0$ once more, since $(d^1)^2 = 0$.

Correction to BG 8.2 (Normalizations). BG identify chains with forms through the diagram of [7, §1]: a chain is first antisymmetrized, $E(a_0, \dots, a_m) = \frac{1}{m!}\varphi(a_0; a_1, \dots, a_m)$, and an antisymmetric chain is then sent to $a_0\, da_1 \wedge \cdots \wedge da_m$. The composite chains $\to$ forms is $\chi' = m!\,\chi$, and the inverse sends θ to $E(a_0, \dots, a_m)$. With this identification the computation above gives $d^1 = \frac{\hbar}{i\,m}\delta$ on m-forms, not $\frac{\hbar}{i}\delta$. For example, for $m = 2$ and $E(a_0, a_1, a_2) = \frac{1}{2}(a_0 \otimes a_1 \otimes a_2 - a_0 \otimes a_2 \otimes a_1)$ one finds directly

$$\chi'\big(b_1 E(a_0, a_1, a_2)\big) = -\frac{i}{2}\big(\{a_0, a_1\}\, da_2 - \{a_0, a_2\}\, da_1 - a_0\, d\{a_1, a_2\}\big) = -\frac{i}{2}\,\delta(a_0\, da_1 \wedge da_2).$$

Likewise $\chi' \circ B = (m+1)\, d \circ \chi'$ on m-chains, whereas $\chi \circ B = d \circ \chi$ (Proposition 4.4). The constants $\hbar/i$ for d^1 and 1 for B in [7, §§1,3] are correct for the normalized map χ of (2). Since the factors are non-zero, no homology computation is affected.

8.3 Examples

Example 8.3 (Moyal on $\mathbb{R}^2$)**.** For $A = C^\infty(\mathbb{R}^2)$ with the Moyal product, $E^1_{p,k} = \hbar^{-p}\Omega^k(\mathbb{R}^2)$ and $d^1 = -i\hbar\delta$. By Section 6.4, $E^2_{p,k} = \hbar^{-p} H_k(\Omega(\mathbb{R}^2), \delta)$ is $\hbar^{-p}\mathbb{C}\,\omega$ for $k = 2$ and zero otherwise. The same computation with polynomial coefficients (where $\mathbb{R}^2$ is replaced by polynomial forms and the polynomial Poincaré lemma is used) is consistent with Theorem 3.13 for $n = 1$: the class of ω corresponds, up to a non-zero factor, to the leading term $1 \otimes x \otimes \partial - 1 \otimes \partial \otimes x$ of the cycle of Example 3.14. Indeed ∂ has symbol $i\xi$, and $\chi(1 \otimes x \otimes \xi - 1 \otimes \xi \otimes x) = dx \wedge d\xi = -\omega$.

Example 8.4 (The symbol algebra in one variable)**.** On $\dot{T}^*\mathbb{R} = \{(x, \xi) : \xi \neq 0\}$, homogeneous functions of degree p are $f_\pm(x)|\xi|^p$ on the two half-lines. The 1-form $\theta = f(x)|\xi|^p\, dx$ has $\delta\theta = \{f|\xi|^p, x\} = pf|\xi|^{p-1}\operatorname{sgn}\xi$. So δ is injective on such forms for $p \neq 0$, and these 1-forms are killed at E^2. This is the mechanism behind the vanishing of E^2 off the line $p = k - n$ of Figure 3.

Discussion. BG call Hochschild homology a "semi-classical invariant": it is determined at first order in $\hbar$, by the Poisson structure alone, once the spectral sequence is known to degenerate at E^2. For a general formal deformation there is no reason for $d^2, d^3, \dots$ to vanish, and convergence must be checked separately. What makes the conic case work is homogeneity, which is the subject of the next section.

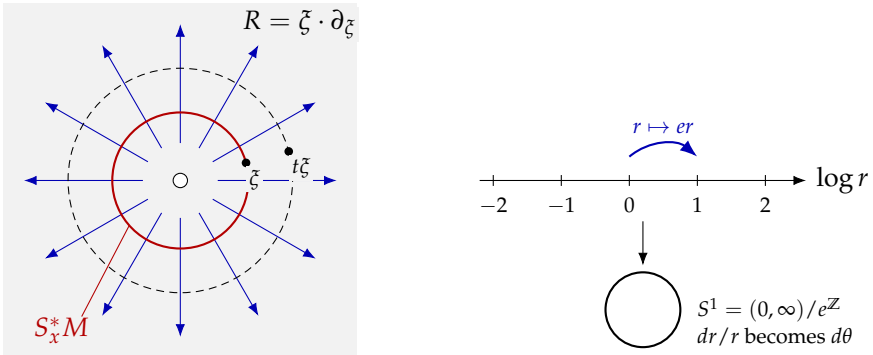
9 Conic symplectic manifolds (BG §2)

What this section does. We prove BG's commutator identity $[\delta, \varepsilon(\iota_Z\omega)] = \mathcal{L}_Z + c(n-k)$ for conformal Hamiltonian fields, and deduce from it, with Z the radial field, that the homology of δ on homogeneous forms lives in a single homogeneity. This is the step that puts E^2 on a line. Then we treat BG's theorem for smooth quantum algebras, including convergence.

9.1 Conic symplectic manifolds

Definition 9.1 ([7, §2]). A *conic symplectic manifold* is a symplectic manifold (X, ω), of dimension $2n$, with a free action ρ_t, $t > 0$, of the multiplicative group $\mathbb{R}_+$ such that $\rho_t^* \omega = t\omega$. Its generator R, defined by $\rho_t = \exp((\log t)R)$, satisfies $\mathcal{L}_R \omega = \omega$.

The basic example is $X = \dot{T}^*M$ with $\rho_t(x, \xi) = (x, t\xi)$. In general put $\alpha = \iota_R \omega$. By Cartan's formula $d\alpha = \mathcal{L}_R \omega - \iota_R d\omega = \omega$, so ω is exact; on $\dot{T}^*M$, $\alpha = \sum \xi_j \, dx_j$ is the canonical 1-form. The quotient $S = X/\mathbb{R}_+$ is a manifold, and choosing a positive function r homogeneous of degree 1 (with a partition of unity) gives $X \cong S \times (0, \infty)$. Figure 7 shows the model.



(a) fibre $T_x^* M \setminus 0$ (b) the radial direction

Figure 7: (a) A fibre of $\dot{T}^*M$, with the radial vector field R, the dilation $\xi \mapsto t\xi$, and the cosphere $S_x^* M$ (any section of the rays would do). Homogeneous functions are determined by their values on $S_x^* M$. (b) In the coordinate $\log r$ dilation by e is translation by 1. Homogeneous forms of degree 0 are invariant under it and descend to the circle $S^1 = (0, \infty)/e^{\mathbb{Z}}$, where dr/r is the angular form. This is the circle of Theorem 1.1.

Remark on BG 9.2 (Darboux and the flow). BG's conic Darboux lemma states that each point has a conic neighbourhood U and a symplectomorphism $\phi : U \to \dot{T}^*\mathbb{R}^n$. What is true and used is that ϕ maps U onto an open *conic subset* of $\dot{T}^*\mathbb{R}^n$ and commutes with the dilations (G6), [15, §21.1]; it cannot be onto $\dot{T}^*\mathbb{R}^n$ in general. Also BG write $T_t = \exp tR$; the flow at time s of R is dilation by e^s, so $T_t = \exp((\log t)R)$. For $X = \dot{T}^*M$ we never need the conic Darboux lemma: canonical coordinates are homogeneous Darboux coordinates.

A vector field Z is *conformal Hamiltonian* if $\mathcal{L}_Z \omega = c\,\omega$ for a constant c. Hamiltonian fields have $c = 0$, and R has $c = 1$. If Z has constant c, then $Z - cR$ preserves ω, so it is locally Hamiltonian, and $Z = H_f + cR$ locally, as BG observe.

9.2 The commutator identity

Theorem 9.3 ([7, §2]). *Let Z be a conformal Hamiltonian vector field with $\mathcal{L}_Z \omega = c\,\omega$. Then on $\Omega^k(X)$*

$$[\delta, \varepsilon(\iota_Z \omega)] = \delta\, \varepsilon(\iota_Z \omega) + \varepsilon(\iota_Z \omega)\, \delta = \mathcal{L}_Z + c\,(n - k).$$

The bracket is the graded commutator; both operators are odd, so it is the anticommutator. BG write $\iota^*(Z)$ for $\varepsilon(\iota_Z \omega)$.

Proof. Write $\theta = \iota_Z\omega$. We use the graded Jacobi identity for the even operator ι_π and the odd operators d and $\varepsilon(\theta)$:

$$[[\iota_\pi, d], \varepsilon(\theta)] = [\iota_\pi, [d, \varepsilon(\theta)]] - [d, [\iota_\pi, \varepsilon(\theta)]].$$

First, $[d, \varepsilon(\theta)] = d\varepsilon(\theta) + \varepsilon(\theta)d = \varepsilon(d\theta)$, since $d(\theta \wedge \beta) = d\theta \wedge \beta - \theta \wedge d\beta$. Here $d\theta = d\iota_Z\omega = \mathcal{L}_Z\omega - \iota_Z d\omega = c\,\omega$. By Lemma 6.8, $[\iota_\pi, \varepsilon(c\,\omega)] = c(n-k)$ on k-forms. Second, by (6), $[\iota_\pi, \varepsilon(\theta)] = \iota_{\pi^\sharp\theta}$, and $\pi^\sharp(\iota_Z\omega) = -Z$ because $\iota_{\pi^\sharp\eta}\omega = -\eta$ for every covector η (Section 6.2). So $[d, \iota_{-Z}] = -(d\iota_Z + \iota_Z d) = -\mathcal{L}_Z$. Putting the two together, $[\delta, \varepsilon(\theta)] = c(n-k) + \mathcal{L}_Z$. □

Corollary 9.4. (1) *For a function f, $[\varepsilon(df), \delta] = -\mathcal{L}_{H_f}$.* (2) *For the radial field, with $\alpha = \iota_R\omega$,*

$$[\delta, \varepsilon(\alpha)] = \mathcal{L}_R + (n-k) \qquad \text{on } \Omega^k.$$

Proof. (1) Take $Z = H_f$, $c = 0$, $\iota_{H_f}\omega = -df$. (2) Take $Z = R$, $c = 1$. □

Remark on BG 9.5 (The sign of the Hamiltonian identity). BG state $[\varepsilon(df), \delta] = \mathcal{L}_{H_f}$. Their Hamiltonian field is defined by $H_g f = \{f, g\}$, which is our $-H_g$, so the two statements agree. We checked Theorem 9.3 for $Z = R$ and $Z = H_f$ symbolically on $\mathbb{R}^4$, in all degrees.

BG prove Theorem 9.3 by conjugating $[d, \iota_Z] = \mathcal{L}_Z$ with their star operator and expanding with the $\mathfrak{sl}(2)$ relations; up to the issues of Correction 6.9, this is the same computation as ours, seen from the d side.

9.3 Homogeneity: the homology of δ lives in one degree

Let X be conic symplectic, and let $\Omega^k_{(p)}$ be the k-forms homogeneous of degree p. Since π is homogeneous of degree -1, δ maps $\Omega^k_{(p)}$ to $\Omega^{k-1}_{(p-1)}$, and $\varepsilon(\alpha)$ maps $\Omega^k_{(p)}$ to $\Omega^{k+1}_{(p+1)}$.

Theorem 9.6 ([7, §2])**.** *The homology of $\Omega^{k+1}_{(p+1)} \xrightarrow{\delta} \Omega^k_{(p)} \xrightarrow{\delta} \Omega^{k-1}_{(p-1)}$ vanishes unless $p = k - n$. For $p = k - n$ it is isomorphic, by $*$, to*

$$H^{2n-k}(\Omega^\bullet_{(0)}, d) \cong H^{2n-k}(S) \oplus H^{2n-k-1}(S) \cong H^{2n-k}(S \times S^1), \qquad S = X/\mathbb{R}_+.$$

Proof. By the homotopy. On $\Omega^k_{(p)}$, $\mathcal{L}_R = p$, so Corollary 9.4(2) reads $\delta\varepsilon(\alpha) + \varepsilon(\alpha)\delta = p + n - k$. If $p \neq k - n$ and $\delta\theta = 0$, then $\theta = \delta(\alpha \wedge \theta)/(p + n - k)$, with $\alpha \wedge \theta \in \Omega^{k+1}_{(p+1)}$. *By the star.* By Theorem 6.7 and Proposition 6.11, $*$ carries the complex isomorphically onto $\Omega^{2n-k-1}_{(c)} \to \Omega^{2n-k}_{(c)} \to \Omega^{2n-k+1}_{(c)}$, $c = p + n - k$, with δ going to $\pm d$. Proposition 6.12, whose proof applies to any $X \cong S \times (0, \infty)$, gives zero for $c \neq 0$ and the stated groups for $c = 0$. □

The two proofs are the same: under $*$, the homotopy $\varepsilon(\alpha)$ becomes, up to sign, the contraction ι_R used in Proposition 6.12, by Lemma 6.6, and $\pi^\sharp\alpha = -R$.

Corollary 9.7. *On all smooth forms, $\mathcal{L}_R$ acts on the homology $H_k(\Omega^\bullet(X), \delta)$ as multiplication by $k - n$.*

Proof. If $\delta\theta = 0$, then $(\mathcal{L}_R + n - k)\theta = \delta(\alpha \wedge \theta)$. □

9.4 Degeneration: BG's homogeneity argument

Let $\star$ be a local conic star product on $C^\infty(X)((\hbar))$. BG's argument for the vanishing of d^r, $r \geq 2$, runs as follows. Let Φ_t act on $C^\infty(X)((\hbar))$ by $\Phi_t(\hbar^j a) = t^j \hbar^j \rho_t^* a$. Since P_l lowers homogeneity by l, $\Phi_t(a \star b) = \Phi_t a \star \Phi_t b$: Φ_t is an automorphism over $\mathbb{C}$ (not over K, since it rescales $\hbar$). Its generator is the derivation $D = \hbar\partial_\hbar + \mathcal{L}_R$, which acts on chains, commutes with b, and preserves the filtration. On $E^1_{p,k} = \hbar^{-p}\Omega^k$ it acts by $-p + \mathcal{L}_R$, so on $E^2_{p,k}$ it is the scalar $-p + k - n$ by Corollary 9.7. The differential $d^r : E^r_{p,k} \to E^r_{p-r,k-1}$ commutes with D, and D acts on source and target by the scalars $-p + k - n$ and $-p + r + k - 1 - n$. These differ by $r - 1$, so $(r-1)d^r = 0$, and $d^r = 0$ for $r \geq 2$.

This is correct, but it uses the general pages E^r (G4), and it says nothing about convergence. For the symbol algebra, homogeneity gives more: the orders and the homogeneities are tied together, so E^2 lies on a line and Proposition 5.7 applies (Section 11). For the smooth quantum algebra we now give a direct proof of degeneration and convergence.

9.5 Smooth quantum algebras

This subsection proves BG's corollary in §2. It is not used later and can be skipped.

Theorem 9.8 ([7, §2, Theorem and Corollary])**.** *Let X be a conic symplectic manifold of dimension $2n$, $S = X/\mathbb{R}_+$, and let $\star$ be a local conic star product on $C^\infty(X)((\hbar))$ with $P_1(a,b) - P_1(b,a) = -i\{a,b\}$. Then, as K-vector spaces,*

$$\mathrm{HH}_k\big(C^\infty(X)((\hbar))\big) \cong H^{2n-k}(X)((\hbar)) = H^{2n-k}(S)((\hbar)).$$

Proof. By Section 8, with Connes' theorem for the non-compact manifold X (Theorem 3.15), $E^1_{p,k} = \hbar^{-p}\Omega^k(X)$ and $d^1 = -i\hbar\delta$. By Theorem 6.7 and de Rham's theorem, $E^2_{p,k} \cong \hbar^{-p}H^{2n-k}(X)$. The filtration is complete. By Proposition 5.6 it suffices to prove (S) at every (p,k).

The weight subcomplexes. Write $X \cong S \times (0,\infty)_r$. For $\lambda \in \mathbb{Z}$, let W^λ_k be the chains $\sum_{j \geq J} \hbar^j c_j$ with $c_j \in \bigoplus_{|I| = \lambda - j} C^\infty_{(I)}(X^{k+1})$ (finite sums of multi-homogeneous functions). Since b_l lowers total homogeneity by l and comes with $\hbar^l$, W^λ is a subcomplex, and it is complete. By Corollary 3.16 (applied to $C^\infty(S) \otimes \mathbb{C}[r^{\pm 1}]$) its first page at $\hbar^j$ is $\hbar^j \Omega^k_{(\lambda - j)}(X)$, with $d^1 = -i\hbar\delta$ by the same proof. By Theorem 9.6, its second page vanishes unless $\lambda - j = k - n$, that is, unless the order $p = -j$ equals $k - n - \lambda$. So Proposition 5.7 applies to W^λ, with n replaced by $n + \lambda$, and every E^2 class of W^λ is the leading term of a cycle of W^λ.

(S) for the smooth complex. Let $\xi \in E^2_{p,k}$, corresponding to a class in $H^{2n-k}(X)$. Represent that class by a form pulled back from S, which is homogeneous of degree 0, and let $\theta = *^{-1}$ of it; then $\theta \in \Omega^k_{(k-n)}$, $\delta\theta = 0$, and $\hbar^{-p}\theta$ represents ξ. Now $\hbar^{-p}\theta$ defines a class in the second page of W^λ, $\lambda = -p + k - n$, so there is a cycle z of W^λ with leading term representing it. Since W^λ is a subcomplex of the smooth complex and the HKR maps agree, z is a cycle of the smooth complex whose leading term represents ξ.

The answer. By Proposition 5.6, $F_pH_k/F_{p-1}H_k \cong \hbar^{-p}V$, $V = H^{2n-k}(X)$, the filtration on H_k is exhaustive and separated. Choose, for a basis of V, cycles in F_0 whose leading terms represent the basis vectors, and extend linearly to $s : V \to Z_k$. The map $\Psi(\sum_{j \geq J} \hbar^j v_j) = [\sum_j \hbar^j s(v_j)]$ is defined by completeness and K-linear. It is injective: if $v_J \neq 0$, the class lies in $F_{-J}H_k$ with Θ_{-J} equal to $[\hbar^J v_J] \neq 0$. It is surjective: for $h \in F_pH_k$, subtracting $\Psi(\hbar^{-p}v)$ with $\Theta_p(h) = \hbar^{-p}v$ moves h into $F_{p-1}H_k$, and after infinitely many steps the difference between h and Ψ of the accumulated series lies in every F_qH_k, hence vanishes. □

Remark 9.9. The weight subcomplexes W^λ are the "semiclassical" form of the symbol algebra: setting $\hbar = 1$ identifies W^0 with the Model II chains of $\mathcal{P}$ when $X = \dot{T}^*M$. Their homology contains the extra classes dr/r and is $H^{2n-k}(S \times S^1)$. These classes die in the smooth complex, where $dr/r = d\log r$ is exact. Nest and Tsygan [27] prove that HH_* of any formal deformation quantization of a symplectic manifold X is $H^{2n-*}(X)((\hbar))$, by a different method (G15).

10 Cyclic homology of quantum algebras (BG §3)

What this section does. We run the argument of the previous two sections for the cyclic complex. The point is to choose the filtration so that Connes' operator B enters only at second order. Then E^2 is the Hochschild E^2 with a polynomial variable u adjoined, and the homogeneity argument goes through.

10.1 The filtration

Let $\star$ be a formal deformation of A, as in Section 8, and consider the cyclic complex $CC_N^K = \bigoplus_{j\geq 0} C_{N-2j}^K u^j$ over K, with $b + u^{-1}B$ (Section 4.2). BG use the differential $b + i^{-1}\hbar^2 u^{-1}B$ instead; the substitution $u \mapsto i\hbar^{-2}u$ identifies the two complexes, since $i^{-1}\hbar^2$ is invertible in K. In our language this amounts to giving u the order 2:

$$\mathrm{order}(\hbar^m c\, u^j) = -m + 2j.$$

Then $b = \sum_l \hbar^l b_l$ lowers order by l as before, while B, which inserts the unit and permutes cyclically, preserves powers of $\hbar$, so $u^{-1}B$ lowers order by exactly 2. The filtration is complete. Write the total differential as $\sum_l D_l$, with D_l lowering order by l:

$$D_0 = b_0, \qquad D_1 = \hbar b_1, \qquad D_2 = \hbar^2 b_2 + u^{-1}B, \qquad D_l = \hbar^l b_l \ (l \geq 3).$$

Lemma 10.1 ([7, §3, Lemma])**.** (1) *The first page is* $E^1 = \mathrm{HH}_*(A)((\hbar))[u] \cong \Omega^*(A)((\hbar))[u]$, *and* $d^1 = -i\hbar\,\delta$ *on each coefficient of* u^j. (2) *Under the HKR map, B corresponds to the de Rham differential d.*

Proof. (1) $D_0 = b_0$ acts separately on each coefficient of u^j, and $D_1 = \hbar b_1$; apply Theorem 8.1. (2) is Proposition 4.4. □

So $E^2 = H_*(\Omega, \delta)((\hbar))[u]$, and the de Rham differential, carried by $u^{-1}B$, would first appear in d^2.

10.2 The conic case

Theorem 10.2 ([7, §3, Theorem and Corollary])**.** *In the situation of Theorem 9.8, the spectral sequence of the cyclic complex degenerates at* E^2, *converges, and*

$$\mathrm{HC}_N\big(C^\infty(X)((\hbar))\big) \cong \bigoplus_{j\geq 0} H^{2n-N+2j}(X)((\hbar))\, u^j.$$

Proof. By Lemma 10.1 and Theorem 9.6, $E^2_{q,N} = \bigoplus_j u^j \hbar^{-p_j} H^{2n-N+2j}(X)$, where $q = p_j + 2j$. The proof of Theorem 9.8 applies word for word. The weight subcomplexes are spanned by $\hbar^m c\, u^j$ with c multi-homogeneous of total degree $\lambda - m$. In the weight-λ subcomplex the second page at $\hbar^m c\, u^j$, with c of Hochschild degree $N - 2j$, is non-zero only if $\lambda - m = (N - 2j) - n$, that is, only if the order $-m + 2j$ equals $N - n - \lambda$. So it lies on a line, Proposition 5.7 gives (S), and Proposition 5.6 gives the result. □

Remark on BG 10.3. In the proof of the theorem of [7, §3] the words are exchanged: the argument given, that d^r changes homogeneity by $-r$, proves *degeneration*, and convergence is what is referred back to Section 2. Note also that, as in the Hochschild case, the homogeneity argument needs representatives of homogeneity $k-n$, which BG obtain from the homotopy $[\delta, \varepsilon(\alpha)] = \mathcal{L}_R + (n-k)$; this is Corollary 9.7.

11 The homology of the symbol algebra (BG §4)

What this section does. We prove the main theorem: $\mathrm{HH}_k(\mathcal{P}) \cong H^{2n-k}(S^*M \times S^1)$, for both chain models, and the analogous statement for cyclic homology. Then we express the answer through the cohomology of M and correct BG's formula for this. Throughout, M is closed, $n = \dim M \geq 1$, $Y = \dot{T}^*M$, and $\mathcal{P}$ carries the star product of Proposition 7.8.

11.1 The first two pages

Filter Model I or Model II by total order (Lemma 7.11). In BG's notation E^i_{kl} is our $E^i_{l,k}$: k is the form degree and l the homogeneity.

Theorem 11.1. *For both models, χ induces $E^1_{p,k} \cong \Omega^k_{(p)}(Y)$, the k-forms on Y homogeneous of degree p.*

Proof. The associated graded complex is $\bigoplus_m C_{m,\bullet}$ with b_0, the Hochschild complex of the commutative algebra $\bigoplus_j C^\infty_{(j)}(Y)$ (finite sums). Choose a positive $r \in C^\infty_{(1)}(Y)$, so that $C^\infty_{(j)}(Y) = r^j C^\infty(S^*M)$ and this algebra is $C^\infty(S^*M) \otimes \mathbb{C}[r^{\pm 1}]$, with chains $C^\infty((S^*M)^{k+1}) \otimes \mathbb{C}[r_0^{\pm 1}, \ldots, r_k^{\pm 1}] = \bigoplus_m C_{m,k}$. By Corollary 3.16, χ identifies its homology with $\Omega^k(S^*M) \otimes \mathbb{C}[r^{\pm 1}] \oplus \Omega^{k-1}(S^*M) \otimes \mathbb{C}[r^{\pm 1}]\, dr$, preserving the degree in r. The part of degree p consists of the forms $r^p(\theta' + \theta'' \wedge dr/r)$, which are exactly the homogeneous k-forms of degree p. □

Theorem 11.2. *$d^1 = -i\delta$ under χ, and*

$$E^2_{p,k} \cong \begin{cases} H^{2n-k}(S^*M \times S^1) \cong H^{2n-k}(S^*M) \oplus H^{2n-k-1}(S^*M), & p = k-n, \\ 0, & p \neq k-n. \end{cases}$$

Proof. The proof of Theorem 8.1 is a computation with the leading homogeneous terms and applies verbatim, with $\hbar = 1$. Then Theorem 9.6, with $X = Y$ and $S = S^*M$, computes E^2. □

In words: E^2 lies on the line $p = k - n$ of Figure 3. A Hochschild k-cycle is detected by its term of total order $k - n$. For $k = 0$ this is the term of order $-n$, which is the term the residue sees.

11.2 The main theorem

Theorem 11.3 ([7, §4, Theorem (1)])**.** *For both chain models,*

$$\mathrm{HH}_k(\mathcal{P}) \cong H^{2n-k}(S^*M \times S^1) \cong H^{2n-k}(S^*M) \oplus H^{2n-k-1}(S^*M).$$

More precisely, every class in $\mathrm{HH}_k(\mathcal{P})$ has a representative c of total order $\leq k-n$, and $[c] \mapsto [\chi(c_{k-n})] \in E^2_{k-n,k}$ is an isomorphism. In particular $\mathrm{HH}_k(\mathcal{P}) = 0$ for $k > 2n$.

Proof. For Model II the filtration is complete (Lemma 7.11), and Theorem 11.2 puts E^2 on the line $p = k - n$. Proposition 5.7 gives the result. For Model I, Theorem B.5 of Appendix B shows that the inclusion into Model II induces an isomorphism on homology compatible with the leading-term maps. □

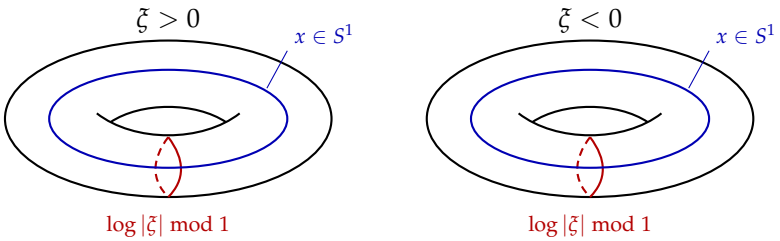


Figure 8: $S^*M \times S^1$ for $M = S^1$. The cosphere bundle $S^*S^1 = S^1 \times \{\xi > 0\} \sqcup S^1 \times \{\xi < 0\}$ is two circles (blue), and the radial circle $S^1 = (0,\infty)/e^{\mathbb{Z}}$ (red) turns each into a torus. So $\mathrm{HH}_k(\mathcal{P}(S^1)) \cong H^{2-k}(T^2 \sqcup T^2)$ has dimensions $2, 4, 2$ for $k = 0, 1, 2$ (Section 15.1).

Corollary 11.4. $\mathrm{HH}_0(\mathcal{P}) \cong H^{2n-1}(S^*M) \cong \mathbb{C}^c$, *where c is the number of components of S^*M. If M is connected and $n \geq 2$, then $c = 1$. If $n = 1$, each component of M gives two components of S^*M.*

Proof. $H^{2n}(S^*M) = 0$ for dimension reasons, and S^*M is a closed orientable manifold (Lemma 6.10), so $H^{2n-1}(S^*M)$ has dimension equal to the number of components. For $n \geq 2$ the fibre S^{n-1} is connected. □

Remark on BG 11.5 (Convergence and chain models). BG prove convergence as follows [7, §§2,4]. They observe that the second page of the quotient complexes C/F_mC stabilizes as $-m \to \infty$ (they write "m sufficiently large"), that C is the projective limit of the C/F_mC, and they invoke Grothendieck's Mittag-Leffler theorem with an induction on the page. Two points are left open.

1. The homology of a projective limit of complexes is the projective limit of the homologies only if the $\varprojlim^1$ of the homologies in one degree higher vanishes. This needs a Mittag-Leffler condition on the system $H_{k+1}(C/F_mC)$, which BG do not verify. Proposition 5.7 is a direct substitute: steps (P) and the completeness of the filtration supply exactly the needed control.

2. The statement "C is the projective limit" holds for Model II, which is the completion of Model I, but not for Model I, the chains $\varinjlim_N \mathcal{P}_N^{\widehat{\otimes}(k+1)}$ used elsewhere in the literature [26, 2]; see Figure 6. For Model I one needs a separate argument, given in Appendix B following Melrose–Nistor. Its result is that the two models have the same homology.

In later work on manifolds with boundary [26, 20] this point is a genuine difficulty: for the symbol algebras with Laurent coefficients at the boundary, the step-by-step argument inside a fixed stage, used in Appendix B, is not available, and other methods are needed. On a closed manifold there is no such problem.

11.3 Cyclic and periodic cyclic homology

For cyclic homology BG filter $\mathcal{P}[u]$ by $F_m(\mathcal{P}[u]) = \sum_j u^j F_{m+2j}(\mathcal{P})$.

Correction to BG 11.6 (The filtration of $\mathcal{P}[u]$). This is not a filtration by subcomplexes. For $c \in F_{m+2}(\mathcal{P})$, $uc \in F_m(\mathcal{P}[u])$, but $(b + u^{-1}B)(uc) = (bc)u + Bc$, and $Bc \in F_{m+2}(\mathcal{P})$ is not in $F_m(\mathcal{P})$ in general. The correct filtration, which matches the weight 2 of u in Section 10.1 and the factor $\hbar^2$ in BG's §3, is

$$F_q(CC_N(\mathcal{P})) = \bigoplus_j u^j \, F_{q-2j}\big(C_{N-2j}(\mathcal{P})\big),$$

so that $u^{-1}B$ lowers order by 2.

Theorem 11.7 ([7, §4, Theorem (2)]). *For both models,*

$$\mathrm{HC}_N(\mathcal{P}) \cong \bigoplus_{j\geq 0} H^{2n-N+2j}(S^*M \times S^1)\, u^j.$$

Proof. Use Model II and the filtration of Correction 11.6. The cyclic complex $CC_N = \bigoplus_{j=0}^{\lfloor N/2 \rfloor} \widehat{C}_{N-2j} u^j$ is a finite sum of complete complexes, hence complete. As in Lemma 10.1, the total differential is $\sum_l D_l$ with $D_0 = b_0$, $D_1 = b_1$, $D_2 = b_2 + u^{-1}B$. So $E^1_{q,N} = \bigoplus_j u^j \Omega^{N-2j}_{(q-2j)}(Y)$ and $d^1 = -i\delta$ on each coefficient. By Theorem 11.2 the coefficient of u^j in $E^2_{q,N}$ vanishes unless $q - 2j = (N-2j) - n$, that is $q = N - n$, and is $H^{2n-N+2j}(S^*M \times S^1)$ then. So E^2 lies on the line $q = N - n$, and Proposition 5.7 applies. For Model I, the inclusion into Model II is a map of mixed complexes and an isomorphism on Hochschild homology (Theorem B.5), hence on cyclic homology by Lemma 4.7. □

Corollary 11.8. *The maps $S : \mathrm{HC}_N(\mathcal{P}) \to \mathrm{HC}_{N-2}(\mathcal{P})$ are surjective, with kernel the image of $\mathrm{HH}_N(\mathcal{P})$. Hence the SBI sequence splits into short exact sequences, and $\mathrm{HP}_0(\mathcal{P}) \cong H^{\mathrm{ev}}(S^*M \times S^1)$, $\mathrm{HP}_1(\mathcal{P}) \cong H^{\mathrm{odd}}(S^*M \times S^1)$.*

Proof. S maps $u^j c$ to $u^{j-1}c$, so it lowers the filtration of Correction 11.6 by 2 and maps the line $q = N - n$ to the line $q - 2 = (N-2) - n$. The leading-term isomorphisms of Theorem 11.7 therefore intertwine S with the map that deletes the term u^0 and lowers the other powers of u. This is onto, with kernel the u^0 term, which is the image of I. For HP use the Milnor sequence of Section 4.3 (G10): the $\varprojlim^1$ term vanishes, and the inverse limit along S is $\bigoplus_i H^{N+2i}$, the even or odd cohomology. □

11.4 The answer in terms of M

BG add: "In particular, if $n > 1$, this is isomorphic to $H^{n-*-1}(M) \oplus H^{n-*}(M)$."

Correction to BG 11.9 (The cohomology of S^*M). This is false in general. For $M = S^2$, $S^*S^2 \cong SO(3) \cong \mathbb{R}P^3$, whose real cohomology is $\mathbb{C}$ in degrees 0 and 3. So $S^*S^2 \times S^1$ has cohomology $\mathbb{C}$ in degrees 0, 1, 3, 4, and $\dim \mathrm{HH}_k(\mathcal{P}(S^2)) = 1, 1, 0, 1, 1$ for $k = 0, \ldots, 4$. BG's formula gives $1, 1, 1, 0, 0$. The correct statement comes from the Gysin sequence (G5) of the sphere bundle $S^*M \to M$, with fibre S^{n-1}:

$$\cdots \to H^{j-n}(M; o) \xrightarrow{e} H^j(M) \xrightarrow{\pi^*} H^j(S^*M) \xrightarrow{\pi_*} H^{j-n+1}(M; o) \xrightarrow{e} H^{j+1}(M) \to \cdots,$$

where o is the orientation local system of M and e is the Euler class of T^*M. If $e = 0$ in real cohomology, for instance if n is odd or if M is closed and connected with $\chi(M) = 0$, the sequence splits and

$$\mathrm{HH}_k(\mathcal{P}) \cong H^{2n-k}(M) \oplus H^{2n-k-1}(M) \oplus H^{n-k+1}(M; o) \oplus H^{n-k}(M; o).$$

When $e \neq 0$, classes cancel in pairs. In all cases $\sum_k (-1)^k \dim \mathrm{HH}_k(\mathcal{P}) = \chi(S^*M \times S^1) = 0$. These statements are checked on examples in Section 15.

Remark on BG 11.10 (Hypotheses). (1) The uniqueness of the residue needs M connected and $n \geq 2$; BG say this in the text preceding their proposition. (2) For non-compact M, BG state the results for the algebra $\mathcal{P}_c$ of symbols with compact support, with $H_c^{2n-*}(S^*M \times S^1)$. The proof above goes through with compactly supported forms, provided the chains are built from the direct limit over compact sets $K \subset M$ of the corresponding Model II complexes, and Connes' theorem is used in its compactly supported form. Since $\mathcal{P}_c$ has no unit, one also needs H-unitality of $\mathcal{P}_c$ (Section 3.8) to identify the homology of the bar-type complexes. We do not write out these modifications, and we do not use the non-compact case. (3) Hochschild homology of the full algebra $\Psi^{\mathbb{Z}}(M)$ differs from that of $\mathcal{P}$ only in degrees 0 and 1, by Wodzicki excision (G9) for the H-unital ideal $\Psi^{-\infty}$.

12 The residue and the higher residues (BG §4)

What this section does. By Corollary 11.4, $\mathrm{HH}_0(\mathcal{P})$ is one-dimensional when M is connected and $n \geq 2$. We construct the trace that detects it, the non-commutative residue, prove its local formula, and prove that *every* trace on $\mathcal{P}$ is a multiple of it, by an elementary argument with commutators. Then we compare normalizations with [7], relate the residue to residues of zeta functions, and explain the higher residues as functionals on $\mathrm{HH}_m(\mathcal{P})$.

12.1 Traces and HH_0

A trace on $\mathcal{P}$ is a linear map $\tau : \mathcal{P} \to \mathbb{C}$ vanishing on commutators. In Model II, $\mathrm{HH}_0(\mathcal{P}) = \mathcal{P}/b(\widehat{C}_1)$, and $b(\widehat{C}_1)$ contains the commutators, but also limits of sums of commutators. BG call a trace *continuous* if it is continuous for their topology $\varprojlim_m \varinjlim_n \Psi^n/\Psi^m$. Concretely, such a τ factors through $\mathcal{P}/F_m$ for some m, and its restriction to each $C^\infty_{(j)}(Y)$ is continuous.

Lemma 12.1. *A continuous trace vanishes on $b(\widehat{C}_1)$. So continuous traces are the linear functionals on $\mathrm{HH}_0(\mathcal{P})$ (Model II) that factor through some $\mathcal{P}/F_m$ and are continuous on each $C^\infty_{(j)}(Y)$.*

Proof. Let $u = \sum_{q \leq Q} u_q \in \widehat{C}_1$. Then $\tau(bu) = \sum_{m<p\leq Q} \tau((bu)_p)$ is a finite sum, and $(bu)_p = \sum_l b_l u_{p+l}$ is finite. Each u_q is a finite sum of elements of $C^\infty_{(I)}(Y^2)$, on which $\tau \circ b_l$ is continuous, and on the dense subspace of algebraic tensors $\tau \circ b$ vanishes since $b(a \otimes a') = [a, a']$. □

12.2 The local residue

Let $U \subset \mathbb{R}^n$ be open. Let $\mathcal{P}_c(U) \subset \mathcal{P}(M)$, for U a chart of M, or $\mathcal{P}_c(U)$ for $U \subset \mathbb{R}^n$, be the classes of operators whose Schwartz kernel has compact support in $U \times U$. By (G1), the left quantization identifies $\mathcal{P}_c(U)$ with full symbols a with compact x-support, through $a \mapsto P_a = [\mathrm{Op}(a)\psi]$, where $\psi \in C^\infty_c(U)$ equals 1 near the x-support of a; a different ψ changes $\mathrm{Op}(a)\psi$ by a smoothing operator (G1(d)). The product is $P_a P_{a'} = P_{a\#a'}$.

Lemma 12.2. *Let $\tilde{\psi} \in C^\infty_c(U)$ be equal to 1 near* $\operatorname{supp}\psi$. *Then in $\mathcal{P}_c(U)$*

$$[\tilde{\psi} x_k, P_a] = i P_{\partial_{\xi_k} a}, \qquad [\tilde{\psi} D_{x_k}, P_a] = P_{D_{x_k} a}.$$

Proof. On kernels, $(x_k - y_k)e^{i(x-y)\cdot\xi} = -i\partial_{\xi_k} e^{i(x-y)\cdot\xi}$, and integration by parts in ξ gives $[x_k, \mathrm{Op}(a)] = i\,\mathrm{Op}(\partial_{\xi_k} a)$. Since $\tilde{\psi} = 1$ on the supports involved, $[\tilde{\psi} x_k, \mathrm{Op}(a)\psi] = [x_k, \mathrm{Op}(a)]\psi$. Similarly $D_{x_k}\mathrm{Op}(a) = \mathrm{Op}(\xi_k a + D_{x_k} a)$ and $\mathrm{Op}(a) D_{x_k} = \mathrm{Op}(a\xi_k)$, so $[\tilde{\psi} D_{x_k}, \mathrm{Op}(a)\psi] = \mathrm{Op}(D_{x_k} a)\psi + \mathrm{Op}(a)(D_{x_k}\psi)$. The last operator has kernel supported where $x \in \operatorname{supp} a$ and $y \in \operatorname{supp} D\psi$, which are disjoint, so it is smoothing. □

Lemma 12.3. *If h is smooth on $\mathbb{R}^n \setminus 0$ and homogeneous of degree $1-n$, then $\int_{S^{n-1}} \partial_{\xi_k} h \, dS = 0$.*

Proof. $\partial_k h$ is homogeneous of degree $-n$, so in polar coordinates $\int_{1\leq|\xi|\leq R} \partial_k h \, d\xi = \log R \int_{S^{n-1}} \partial_k h \, dS$. By the divergence theorem the left side is $\int_{|\xi|=R} h \, \frac{\xi_k}{R} \, dS_R - \int_{|\xi|=1} h \, \xi_k \, dS$, and since $h(R\theta) = R^{1-n} h(\theta)$ and $dS_R = R^{n-1} dS$, the two terms are equal. So the right side vanishes for all $R > 1$. (For $n = 1$, $S^0 = \{\pm 1\}$, h is constant on each half-line, and $\partial_\xi h = 0$.) □

Definition 12.4. For a with compact x-support in U, the *local residue* is

$$\tau_U(P_a) = (2\pi)^{-n} \int_U \int_{|\xi|=1} a_{-n}(x, \xi) \, dS(\xi) \, dx.$$

Proposition 12.5. *τ_U is a trace on $\mathcal{P}_c(U)$.*

Proof. For functions f, g and a first derivative ∂, $f\,\partial g = \partial(fg) - (\partial f)g$. Iterating, $f\,\partial^\beta g \equiv (-1)^{|\beta|}(\partial^\beta f)g$ modulo sums of first derivatives in the same variables. Hence, modulo sums of ξ-derivatives and x-derivatives of products of derivatives of a and a',

$$\partial_\xi^\alpha a\, D_x^\alpha a' \equiv (-1)^{|\alpha|} a\, \partial_\xi^\alpha D_x^\alpha a' \equiv (-1)^{|\alpha|}(-1)^{|\alpha|}(D_x^\alpha a)(\partial_\xi^\alpha a') = \partial_\xi^\alpha a'\, D_x^\alpha a.$$

So every term of $a\#a' - a'\#a = \sum_{\alpha\neq 0} \frac{1}{\alpha!}(\partial_\xi^\alpha a\, D_x^\alpha a' - \partial_\xi^\alpha a'\, D_x^\alpha a)$ is of the form $\sum_k \partial_{x_k} G_k + \partial_{\xi_k} H_k$, with G_k, H_k finite sums of homogeneous terms with compact x-support. The term of degree $-n$ of $\partial_{x_k} G_k$ integrates to zero over x, and that of $\partial_{\xi_k} H_k$ is ∂_{ξ_k} of the term of degree $1-n$ of H_k, which integrates to zero over the sphere by Lemma 12.3. □

12.3 Uniqueness on a chart

Proposition 12.6. *Let $U \subset \mathbb{R}^n$ be connected and $n \geq 2$. Every trace τ on $\mathcal{P}_c(U)$ is a multiple of τ_U. No continuity is assumed.*

Proof. Let $\chi(\xi)$ be smooth, 0 near 0 and 1 for $|\xi| \geq 1$. Products of χ with homogeneous functions are classical symbols, and changing χ changes them by symbols of order $-\infty$, which are 0 in $\mathcal{P}$.

Step 1: degrees $j \neq -n$. Let p_j be homogeneous of degree j with compact x-support. By Euler's identity $\sum_k \partial_{\xi_k}(\xi_k p_j) = (j+n)p_j$. So $P_{\chi p_j} = \sum_k \frac{-i}{j+n}[\tilde\psi x_k, P_{\chi\xi_k p_j}]$ by Lemma 12.2, the terms with $\partial\chi$ having order $-\infty$. Hence $\tau(P_{\chi p_j}) = 0$. The same works for a whole tail $q \sim \sum_{j\leq -n-1} q_j$: let b_k be a symbol with expansion $\sum_j \frac{-i}{j+n}\chi\xi_k q_j$ (Borel); then $P_q = \sum_k [\tilde\psi x_k, P_{b_k}]$ and $\tau(P_q) = 0$. Since an element of $\mathcal{P}_c(U)$ has finitely many terms of degree $> -n$, τ depends only on the term of degree $-n$.

Step 2: degree $-n$. Let p be homogeneous of degree $-n$ with compact x-support, and $\varrho(x) = \int_{S^{n-1}} p(x,\theta)\, dS$. Since $n \geq 2$, S^{n-1} is connected, and Hodge theory on the round sphere (G5) gives $w(x,\cdot)$, smooth in x with compact support, with $\Delta_S w = p - \varrho/|S^{n-1}|$ on S^{n-1}. Let g be the vector field on $\mathbb{R}^n \setminus 0$, tangent to the spheres and homogeneous of degree $1-n$, equal to $\nabla_S w$ on S^{n-1}. For a tangential field $g(r\theta) = r^{1-n}V(\theta)$ one has $\operatorname{div} g = r^{-n}\operatorname{div}_S V$, so $\sum_k \partial_{\xi_k} g_k = |\xi|^{-n}\Delta_S w = p - \varrho|\xi|^{-n}/|S^{n-1}|$. By Lemma 12.2, $\tau(P_{\chi p}) = \tau(P_{\chi\varrho|\xi|^{-n}})/|S^{n-1}|$. Next, fix $\chi_0 \in C_c^\infty(U)$ with $\int \chi_0 = 1$, and let $\nu = \int_U \varrho\, dx$. Since U is connected, the compactly supported function $\varrho - \nu\chi_0$ has integral zero and so is $\sum_k \partial_{x_k}\beta_k$ with $\beta_k \in C_c^\infty(U)$ (compactly supported de Rham cohomology in top degree, G5). By Lemma 12.2 again,

$$\tau(P_{\chi p}) = \nu\,\tau(P_{g_0}), \qquad g_0 = \chi_0(x)\chi(\xi)|\xi|^{-n}/|S^{n-1}|.$$

Since $\tau_U(P_{\chi p}) = (2\pi)^{-n}\nu$, we get $\tau = c_U\tau_U$ on such elements, with $c_U = (2\pi)^n\tau(P_{g_0})$. With Step 1, $\tau = c_U\tau_U$ on $\mathcal{P}_c(U)$. □

For $n = 1$ Step 2 fails: S^0 has two points. The same argument shows that every trace on $\mathcal{P}_c(U)$, $U \subset \mathbb{R}$ an interval, is a combination of $\tau_U^\pm(P_a) = (2\pi)^{-1}\int_U a_{-1}(x,\pm 1)\, dx$, each of which is a trace, because Lemma 12.3 holds on each half-line separately.

12.4 The global residue

Lemma 12.7. *Let f be homogeneous of degree $-n$ on $\dot T^*U$, with compact x-support. Then*

$$\int_U \int_{|\xi|=1} f\, dS\, dx = \int_{S^*U} f\, \iota_R \mathrm{vol},$$

where S^*U is oriented by $\iota_R\mathrm{vol}$ and identified with any section of the rays. In particular the left side does not depend on the coordinates.

Proof. The form $\iota_R(f\mathrm{vol})$ is homogeneous of degree 0 and closed: $d\iota_R(f\mathrm{vol}) = \mathcal{L}_R(f\mathrm{vol}) - \iota_R d(f\mathrm{vol}) = (-n+n)f\mathrm{vol} = 0$. So its integral over a section of the rays does not depend on the section (Stokes, with compact x-support). On $\{|\xi| = 1\}$, $|\iota_R\mathrm{vol}| = |\iota_R(d\xi_1 \wedge \cdots \wedge d\xi_n)|\,|dx| = dS\,dx$. A change of coordinates induces a map of $\dot{T}^*U$ preserving ω, R and the principal symbol. □

Theorem 12.8 (The residue)**.** *Let M be closed. There is a unique linear functional* res *on $\mathcal{P}$ with* $\mathrm{res}(\phi A) = \tau_U(\phi A)$ *whenever $\phi \in C_c^\infty(U)$ for a chart U and $\phi A \in \mathcal{P}_c(U)$ (with ϕA replaced by $\phi A\psi$, $\psi = 1$ near* $\operatorname{supp}\phi$*). It is a trace, and*

$$\mathrm{res}(A) = (2\pi)^{-n} \int_{S^*M} \sigma(A)_{-n}\, \iota_R\Big(\frac{\omega^n}{n!}\Big),$$

where σ is the global symbol of Proposition 7.8. If A has order $\le -n$, $\sigma(A)_{-n}$ is the principal symbol of order $-n$.

Proof. Invariance on overlaps. Let U, U' be charts and $B \subset U \cap U'$ a coordinate ball. Then τ_U and $\tau_{U'}$ restrict to traces on $\mathcal{P}_c(B)$ (Proposition 12.5), so $\tau_{U'} = c\,\tau_U$ there if $n \ge 2$ (Proposition 12.6). To find c, take $A \in \mathcal{P}_c(B)$ of order $-n$ with $\int_{S^*B} \sigma_{-n}(A)\iota_R\mathrm{vol} \neq 0$. In both charts the term of degree $-n$ of the full symbol is the principal symbol (G1(c)), so by Lemma 12.7 $\tau_U(A) = \tau_{U'}(A)$, and $c = 1$. For $n = 1$ apply the same argument to $\tau^\pm$. By a partition of unity inside $U \cap U'$, $\tau_U = \tau_{U'}$ on $\mathcal{P}_c(U \cap U')$.

Definition. Choose charts U_i and a partition of unity ϕ_i, and put $\mathrm{res}(A) = \sum_i \tau_{U_i}(\phi_i A)$. By invariance on overlaps, inserting a second partition of unity ϕ_j' and writing $\phi_i A = \sum_j \phi_i\phi_j' A$ shows that res does not depend on the choices. Since the full symbol of $\phi_j A$ in the chart U_j is $\phi_j\sigma_j(A)$, $\mathrm{res}(A) = (2\pi)^{-n}\sum_j \int \phi_j\sigma_j(A)_{-n} =$ the stated integral, by Lemma 12.7.

Trace. Choose the partition of unity so fine that whenever $\operatorname{supp}\phi_i \cap \operatorname{supp}\phi_j \neq \varnothing$, the union lies in one chart U_{ij} (Lebesgue number). Then $[A, A'] = \sum_{i,j}[\phi_i A, \phi_j A']$. If $\operatorname{supp}\phi_i \cap \operatorname{supp}\phi_j = \varnothing$, both products $\phi_i A\phi_j A'$ and $\phi_j A'\phi_i A$ are 0 in $\mathcal{P}$ by G1(d). Otherwise both factors lie in $\mathcal{P}_c(U_{ij})$, after cutting off on the right, and $\mathrm{res}([\phi_i A, \phi_j A']) = \tau_{U_{ij}}([\phi_i A, \phi_j A']) = 0$ by Proposition 12.5. □

Theorem 12.9 (Uniqueness)**.** *Let M be closed and connected. If $n \ge 2$, every trace on $\mathcal{P}$ is a multiple of* res*. If $n = 1$, every trace is a combination of the two traces* $\mathrm{res}^\pm$*, defined with $\xi > 0$ and $\xi < 0$ respectively.*

Proof. Let τ be a trace. With charts U_i and a partition of unity, $\tau(A) = \sum_i \tau(\phi_i A)$, and $\tau = c_i\tau_{U_i}$ on $\mathcal{P}_c(U_i)$ by Proposition 12.6 (take connected charts). If $U_i \cap U_j \neq \varnothing$, testing on an element supported in a small ball inside $U_i \cap U_j$ with non-zero residue gives $c_i = c_j$. Since M is connected, all c_i agree, and $\tau = c$ res. For $n = 1$ argue in the same way with $\tau^\pm$; on a connected 1-manifold $\xi > 0$ and $\xi < 0$ are globally defined, since 1-manifolds are orientable. □

Remark 12.10. BG prove uniqueness only among continuous traces, as a consequence of $\dim \mathrm{HH}_0(\mathcal{P}) = 1$. Theorem 12.9 is the algebraic statement of Wodzicki [35]; the proof above is the one used for the algebra of operators in [26]. Melrose [25, Ch. 10] notes that the algebraic statement needs an argument beyond the Hochschild computation, which is what Steps 1 and 2 provide; see also [21].

12.5 The residue and HH_0: normalizations

By Lemma 12.1, res defines a functional on $\mathrm{HH}_0(\mathcal{P})$. It is continuous, since it only involves the term of degree $-n$. We compare it with the isomorphism of Theorem 11.3.

Proposition 12.11. *Let $\Theta : \mathrm{HH}_0(\mathcal{P}) \to E^2_{-n,0}$ be the leading-term isomorphism, followed by $* : E^2_{-n,0} \to H^{2n}(\Omega_{(0)})$ and by $H^{2n}(\Omega_{(0)}) \cong H^{2n-1}(S^*M)$, $\theta' + \theta'' \wedge dr/r \mapsto [\theta'']$. Pairing with the fundamental class of S^*M (oriented by $\iota_R\mathrm{vol}$) gives the functional $-(2\pi)^n$ res.*

Proof. Every class has a representative a of order $\leq -n$. Then $\Theta[a] = [a_{-n}]$, and $*a_{-n} = a_{-n}\mathrm{vol} = \theta'' \wedge dr/r$. Since $\iota_R\theta'' = 0$ and $\iota_R(dr/r) = 1$, $\iota_R(a_{-n}\mathrm{vol}) = -\theta''$, so the pairing is $-\int_{S^*M} a_{-n}\iota_R\mathrm{vol} = -(2\pi)^n \operatorname{res}(a)$. □

Correction to BG 12.12 (Constants in BG's Proposition). (1) BG normalize R_M to take the value 1 on the orientation class of $H^{2n-1}(S^*M)$, derive $R(a) = c\int_{S^*M} a_{-n}\,\iota(R)\omega^n$, and state that "it can be shown that $c = 2\pi$". The constant depends on the normalizations of the HKR map (Correction 8.2), of the star operator (Correction 6.9) and of the orientation, none of which is fixed in [7]. With the conventions of these notes, Proposition 12.11 gives $R_M = -(2\pi)^n \operatorname{res} = -\frac{1}{n!}\int a_{-n}\iota_R\omega^n$, so $c = -1/n!$. We were unable to reproduce $c = 2\pi$. (2) In BG's Proposition (2), the constant in $\operatorname{Res}_{s=0}\operatorname{Tr}(PA^{-s}) = c\,R(P)$ depends on the order m of A, not only on $\dim M$: by Proposition 12.15, $\operatorname{Res}_{s=0}\operatorname{Tr}(PA^{-s}) = \frac{1}{m}\operatorname{res}(P)$. For instance, replacing A by A^2 halves the left side, since $\operatorname{Tr}(PA^{-2s})$ is the same function evaluated at $2s$. Also A must be invertible with spectrum in a sector for A^{-s} to be defined; positivity of the principal symbol alone does not ensure invertibility.

12.6 Residues of zeta functions

Lemma 12.13 (Seeley's computation [29])**.** *Let $a(s; x, \xi)$ be a holomorphic family of classical symbols of order $\mu(s) = k - ms$, $m > 0$, with compact x-support: $a = \sum_{j<J} a_{\mu-j} + r_J$, with $a_{\mu-j}$ homogeneous of degree $\mu(s) - j$ for $|\xi| \geq 1$, holomorphic in s, and $|r_J| \leq C|\xi|^{\operatorname{Re}\mu - J}$ locally uniformly. Then $I(s) = \iint a(s; x, \xi)\,d\xi\,dx$, holomorphic for $\operatorname{Re}\mu(s) < -n$, extends meromorphically to $\mathbb{C}$ with at most simple poles at the points where $\mu(s) = j - n$, and*

$$\operatorname{Res}_{s=s_j} I = \frac{1}{m}\iint_{|\xi|=1} a_{-n}(s_j; x, \xi)\,dS\,dx.$$

Proof. Split $\int_{|\xi|\leq 1} + \int_{|\xi|\geq 1}$; the first is entire. In the second use polar coordinates: $\int_1^\infty r^{\mu-j+n-1}dr = \frac{1}{j-n-\mu(s)} = \frac{1/m}{s-s_j}$, for $\operatorname{Re}\mu < j - n$, where $\mu(s_j) = j - n$. The remainder r_J contributes a function holomorphic for $\operatorname{Re}\mu < J - n$. Since J is arbitrary, this gives the continuation and the residues. □

Quoted background 12.14 (Trace class and complex powers, (G8))**.** (a) An operator in $\Psi^\mu(M)$ with $\operatorname{Re}\mu < -n$ is of trace class, and for $A = \phi\,\mathrm{Op}(a)\psi$ in a chart, $\operatorname{Tr} A = (2\pi)^{-n}\iint \phi\psi a\,d\xi\,dx$. (b) If $A \in \Psi^m(M)$, $m \geq 1$, is elliptic and invertible with principal symbol positive (more generally with spectrum and principal symbol avoiding a ray), then the complex powers A^{-s} of Seeley form a holomorphic family of order $-ms$ with $A^0 = 1$, and for $P \in \Psi^k(M)$ the local symbols of PA^{-s} are holomorphic families of classical symbols as in Lemma 12.13, up to holomorphic families of smoothing operators [29, 30].

Proposition 12.15 (Wodzicki, Guillemin)**.** *In the situation of Background 12.14(b), $\zeta(s) = \operatorname{Tr}(PA^{-s})$, defined for $\operatorname{Re}s > (k+n)/m$, extends meromorphically to $\mathbb{C}$ with at most simple poles, and $\operatorname{Res}_{s=0}\zeta = \frac{1}{m}\operatorname{res}(P)$.*

Proof. With a partition of unity and cut-offs, $\operatorname{Tr}(PA^{-s}) = \sum_i \operatorname{Tr}(\phi_i P A^{-s}\psi_i)$ plus the trace of a holomorphic family of smoothing operators, whose kernels vanish near the diagonal; that trace is entire. Apply Lemma 12.13 to the local symbol of $\phi_i P A^{-s}\psi_i$, of order $k - ms$. At $s = 0$ its term of degree $-n$ is ϕ_i times the term of degree $-n$ of the symbol of P, since $A^0 = 1$. By (G8)(a) and the lemma, the residue at $s = 0$ is $\frac{1}{m}(2\pi)^{-n}\sum_i \iint_{|\xi|=1} \phi_i p^{(i)}_{-n}\, dS\, dx = \frac{1}{m}\operatorname{res}(P)$, where $p^{(i)}$ is the full symbol of P in the i-th chart. □

The functional res is the *Wodzicki–Guillemin residue* [35, 13]; the factor $(2\pi)^{-n}$ is included by some authors and not by others. BG mention that the trace was first observed by Adler [1] for $M = \mathbb{R}$, in the theory of the KdV equation.

12.7 Higher residues

By Theorem 11.3 and Poincaré duality on the closed oriented manifold S^*M of dimension $2n-1$,

$$\mathrm{HH}_m(\mathcal{P})^* \cong \big(H^{2n-m}(S^*M) \oplus H^{2n-m-1}(S^*M)\big)^* \cong H^{m-1}(S^*M) \oplus H^m(S^*M),$$

which is BG's formula $\mathrm{HH}^*(\mathcal{P}) = H^{*-1}(S^*M) \oplus H^*(S^*M)$ for continuous Hochschild cohomology. The functionals can be written down explicitly.

Proposition 12.16. *Let $c \in F_{m-n}\widehat{C}_m$ be a cycle, $\theta = \chi(c_{m-n}) \in \Omega^m_{(m-n)}(Y)$, and $\gamma = *\theta \in \Omega^{2n-m}_{(0)}$, which is closed. For closed forms $\alpha \in \Omega^m(S^*M)$ and $\beta \in \Omega^{m-1}(S^*M)$ put*

$$T_\alpha[c] = \int_{S^*M} \iota_R\gamma \wedge \alpha, \qquad T'_\beta[c] = \int_{S^*M} \gamma|_{S^*M} \wedge \beta.$$

These are well-defined functionals on $\mathrm{HH}_m(\mathcal{P})$, depending only on the classes of α and β, and together they give the isomorphism above. Moreover, by Lemma 6.6, for $\theta = f_0\, df_1 \wedge \cdots \wedge df_m$,

$$\iota_R * \theta = (-1)^{m(m+1)/2} f_0\, \iota_R \iota_{H_{f_1}} \cdots \iota_{H_{f_m}} \mathrm{vol}.$$

Proof. Write $\gamma = \gamma' + \gamma'' \wedge dr/r$ as in Proposition 6.12. Then $\gamma|_{S^*M} = \gamma'$ and $\iota_R\gamma = \pm\gamma''$, both closed on S^*M. By Theorem 11.3, $[c] \mapsto ([\gamma'], [\gamma''])$ is an isomorphism onto $H^{2n-m}(S^*M) \oplus H^{2n-m-1}(S^*M)$. By Stokes' theorem on the closed manifold S^*M the integrals depend only on the cohomology classes, and Poincaré duality says that these pairings identify the dual space. □

For $m = 0$ and $\alpha = 1$ we recover $T_1 = \int_{S^*M} \iota_R(a_{-n}\mathrm{vol}) = (2\pi)^n \operatorname{res}$. BG's formula for Wodzicki's higher residue Res_m is the formula for $\iota_R * \theta$ above, with $\varepsilon^*(df)$ for $\pm\iota_{H_f}$ and $d\tilde{\xi}'$ for $\iota_R\mathrm{vol}$.

Remark on BG 12.17 (The higher residue cochains). BG write the Hochschild cochain attached to α as $(T\alpha)(a_0, \dots, a_m) = \int_{S^*M} \mathrm{Res}_m(a_0\, da_1 \wedge \cdots \wedge da_m) \wedge \alpha$, where Res_m keeps only the part of homogeneity $m - n$. As a multilinear functional on $\mathcal{P}$ this is not a Hochschild cocycle in general: $(T\alpha) \circ b$ involves all the P_l through the components of b that land in homogeneity $m - n$, and only the b_0 and b_1 parts are controlled (the former by $\chi \circ b_0 = 0$, the latter by $d\alpha = 0$ and Stokes). BG describe $T\alpha$ only through its image in the first page, and the precise statement is Proposition 12.16: the formula evaluates a class on its representative of order $\le m - n$. Note also that $T\alpha$ only accounts for the summand $H^m(S^*M)$; the summand $H^{m-1}(S^*M)$ comes from T'_β.

13 Differential operators

What this section does. The same spectral sequence computes the Hochschild homology of differential operators. Here orders are ≥ 0, so convergence is automatic, and the forms are polynomial in ξ, so the circle disappears: $\mathrm{HH}_k(\mathcal{D}(M)) \cong H^{2n-k}(M)$. For a ball this is BG's "Poincaré lemma", the input of the sheaf-theoretic method of Section 14. We also state the algebraic version of Kassel and Mitschi.

13.1 Differential operators on an open subset of $\mathbb{R}^n$

Let $U \subset \mathbb{R}^n$ be open and $\mathcal{D}(U)$ the algebra of differential operators with smooth coefficients, $\mathcal{D}(U) = \bigcup_N \mathcal{D}^N(U)$, with $\mathcal{D}^N(U) \cong C^\infty(U)^{\binom{n+N}{N}}$ Fréchet. The left symbol identifies $\mathcal{D}(U)$ with $C^\infty(U)[\xi_1, \dots, \xi_n]$ as a vector space, and for polynomial symbols the composition formula is exact: $a\#b = \sum_l P_l(a,b)$, a finite sum, with $P_l(a,b) = \sum_{|\alpha|=l} \frac{1}{\alpha!} \partial_\xi^\alpha a \, D_x^\alpha b$. The chains are $\varinjlim_N \mathcal{D}^N(U)^{\widehat{\otimes}(k+1)} = C^\infty(U^{k+1})[\xi^{(0)}, \dots, \xi^{(k)}]$ (G2), filtered by total degree $p \geq 0$ in the $\xi^{(j)}$. Each chain has finitely many components, so the filtration is complete, and Proposition 5.7 applies.

Theorem 13.1 ([7, §5]). $\mathrm{HH}_k(\mathcal{D}(U)) \cong H^{2n-k}(U)$. *In particular, if U is a ball (BG's "Poincaré lemma"),*

$$\mathrm{HH}_k(\mathcal{D}(U)) \cong \begin{cases} \mathbb{C}, & k = 2n, \\ 0, & k \neq 2n. \end{cases}$$

The same holds for the Weyl algebra A_n of polynomial coefficient operators, with $U = \mathbb{R}^n$, in agreement with Theorem 3.13.

Proof. The associated graded algebra is $C^\infty(U) \otimes \mathbb{C}[\xi]$. By Corollary 3.16, $E^1_{p,k}$ is the space of k-forms on T^*U that are polynomial in ξ and homogeneous of degree p (with $d\xi_j$ of degree 1). The proof of Theorem 8.1 gives $d^1 = -i\delta$. The symplectic star has constant coefficients, so it preserves polynomial forms, and by Proposition 6.12(3) and Theorem 6.7 $E^2_{p,k} \cong H^{2n-k}(U)$ for $p = k - n$ and 0 otherwise. Proposition 5.7 gives the result. For A_n replace $C^\infty(U)$ by $\mathbb{C}[x]$ and use Theorem 3.12; the polynomial Poincaré lemma gives $H^*(\mathbb{R}^n)$. □

For U a ball the class in degree $2n$ is represented, at leading order, by a chain whose HKR image is $*^{-1}1$, a multiple of vol; for $n = 1$ it is the cycle of Example 3.14.

Discussion. Compare with symbols. For $\mathcal{D}$ the first page consists of forms polynomial in ξ, defined on all of T^*U, including the zero section. A polynomial form of homogeneity 0 cannot contain dr/r, and it is just a form on U. So the answer is $H^{2n-k}(U)$ with no circle. For symbols, which are only defined on $\dot{T}^*U$ and are allowed negative homogeneity, dr/r is available and produces $H^{2n-k}(S^*U \times S^1)$. This is the contrast BG draw at the start of §4, between Laurent polynomial forms and smooth forms on the half-line.

13.2 Differential operators on a manifold

Theorem 13.2 ([7, §5]; Wodzicki [36]). *For a manifold M without boundary, assuming (G11),*

$$\mathrm{HH}_k(\mathcal{D}(M)) \cong H^{2n-k}(M), \qquad \mathrm{HC}_N(\mathcal{D}(M)) \cong \bigoplus_{j \geq 0} H^{2n-N+2j}(M)\, u^j.$$

Proof. The proof of Theorem 13.1 applies, once the first page is identified with the forms on T^*M polynomial in the fibres, homogeneous of degree p. This is the HKR theorem for the algebra of fibrewise polynomial functions on T^*M, which is (G11). In a chart it is Corollary 3.16, and the passage from local to global is the partition-of-unity argument of Appendix A, which we do not write out for this algebra. For cyclic homology use the filtration of Section 10.1; orders are still ≥ 0, and the argument of Theorem 11.7 applies. □

BG prove the Hochschild statement by the sheaf-theoretic method of Section 14, from the Poincaré lemma for balls, and remark that this gives a second proof of the degeneration. With compact supports one obtains $H_c^{2n-*}(M)$ for $\mathcal{D}_c(M)$.

13.3 Algebraic differential operators

Quoted background 13.3 (Kassel–Mitschi [18], quoted in [7, §5])**.** Let M be a smooth affine algebraic variety of pure dimension n over a field k of characteristic 0, and $\mathcal{D}_{\mathrm{alg}}(M)$ the algebra of regular differential operators. Then

$$\mathrm{HH}_*(\mathcal{D}_{\mathrm{alg}}(M)) \cong H_{\mathrm{DR}}^{2n-*}(M/k), \qquad \mathrm{HC}_*(\mathcal{D}_{\mathrm{alg}}(M)) \cong H_{\mathrm{DR}}^{2n-*}(M/k)[u].$$

This is quoted by BG from a private communication; no proof is given in [7].

BG sketch a proof for $k = \mathbb{C}$. The map $\mathcal{D}_{\mathrm{alg}}(M) \to \mathcal{D}_{\mathrm{an}}(M)$ into analytic differential operators on the Stein manifold M respects the order filtrations, so it induces a map of spectral sequences. On the second pages it is the map from algebraic to analytic de Rham cohomology, which is an isomorphism by Grothendieck's theorem [12]. Granting the convergence of both spectral sequences, the two algebras have the same homology, and the analytic case is computed as in Theorem 13.2. The Lefschetz principle extends the statement to fields of characteristic 0. All of this is (G14).

Example 13.4. For the affine line, $\mathcal{D}_{\mathrm{alg}}(\mathbb{A}^1) = A_1$ and $H_{\mathrm{DR}}^*(\mathbb{A}^1) = k$ in degree 0. The proposition gives $\mathrm{HH}_2(A_1) = k$ and $\mathrm{HH}_j(A_1) = 0$ otherwise, in agreement with Theorem 3.13. For $\mathbb{G}_m = \operatorname{Spec} k[x, x^{-1}]$, $H_{\mathrm{DR}}^0 = H_{\mathrm{DR}}^1 = k$, so the proposition predicts $\mathrm{HH}_2 = \mathrm{HH}_1 = k$ for the algebra generated by $x^{\pm 1}$ and ∂.

Remark 13.5. Over $\mathbb{C}$ the spectral sequence of this section also applies directly to $\mathcal{D}_{\mathrm{alg}}(M)$, with the algebraic HKR theorem for the smooth affine variety T^*M, and gives the Kassel–Mitschi statement without passing through analytic operators. We have not written out the details, and we do not use the statement.

14 The sheaf-theoretic approach (BG §5)

What this section does. BG's second method imitates Weil's proof of de Rham's theorem: a Poincaré lemma on small open sets, plus a gluing argument. We first carry out Weil's proof completely on the circle, then introduce sheaves only as far as needed, then explain BG's argument for differential operators and its cyclic version, and finally the microlocal version for symbols. The microlocal inputs are quoted.

14.1 Weil's proof of de Rham's theorem on the circle

Cover S^1 by three open arcs U_0, U_1, U_2 such that each $U_{ij} = U_i \cap U_j$ $(i < j)$ is a non-empty arc and $U_0 \cap U_1 \cap U_2 = \varnothing$ (Figure 9). Two arcs would not do: their intersection would be two arcs, which is not connected. A cover by contractible sets with contractible (or empty) intersections is called *good*.

Čech cohomology. Let $\check{C}^0 = \mathbb{C}^3 \ni (f_0, f_1, f_2)$ (a constant on each arc), $\check{C}^1 = \mathbb{C}^3 \ni (g_{01}, g_{02}, g_{12})$ (a constant on each intersection), and $\delta(f)_{ij} = f_j - f_i$. The kernel of δ consists of the constant triples, and the image is two-dimensional. So $\check{H}^0 = \mathbb{C}$ and $\check{H}^1 = \mathbb{C}^3/\operatorname{im}\delta = \mathbb{C}$: the same as $H^*_{\mathrm{dR}}(S^1)$. Weil's theorem explains why.

The double complex. Put $K^{p,q} = \bigoplus_{|I|=p+1} \Omega^q(U_I)$ for $p = 0, 1$: forms on the arcs ($p = 0$) and on the intersections ($p = 1$). There are two differentials, d on forms and δ (restriction and difference).

Lemma 14.1. (Rows) *For each q, $0 \to \Omega^q(S^1) \to K^{0,q} \xrightarrow{\delta} K^{1,q} \to 0$ is exact.* (Columns) *For each p, $K^{p,\bullet}$ has cohomology $\check{C}^p$ in degree 0 (locally constant functions, which are constant) and 0 in degree 1.*

Proof. (Rows) Exactness at $\Omega^q(S^1)$ and at $K^{0,q}$ says that forms agreeing on overlaps glue. For surjectivity, take a partition of unity ρ_k subordinate to the cover, and given $\omega = (\omega_{ij})$, put $\eta_i = \sum_k \rho_k \omega_{ki}$ on U_i, with $\omega_{ii} = 0$ and $\omega_{ki} = -\omega_{ik}$. On U_{ij} the terms with $k \notin \{i, j\}$ vanish, because $U_k \cap U_{ij} = \varnothing$, so $\eta_j - \eta_i = \rho_i \omega_{ij} - \rho_j \omega_{ji} = (\rho_i + \rho_j)\omega_{ij} = \omega_{ij}$. (Columns) This is the Poincaré lemma on arcs: a closed 1-form on an interval is exact, and a function with $df = 0$ on an interval is constant. □

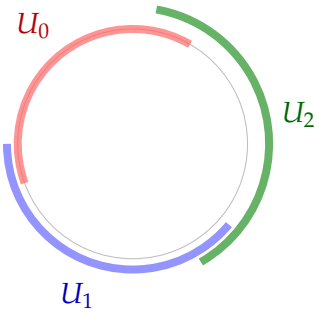
Theorem 14.2 (Weil, on the circle)**.** $H^*_{\mathrm{dR}}(S^1) \cong \check{H}^*(\{U_i\}; \mathbb{C})$.

Proof. Let T be the total complex, $T^m = K^{0,m} \oplus K^{1,m-1}$, with differential $(x, y) \mapsto (dx, \delta x - dy)$. This is the mapping cone of the chain map $\delta : K^{0,\bullet} \to K^{1,\bullet}$, shifted. By the rows, δ is onto with kernel $\Omega^\bullet(S^1)$, so the inclusion $\Omega^\bullet(S^1) \to T$, $x \mapsto (x, 0)$, is a quasi-isomorphism (a cone of a surjection is quasi-isomorphic to its kernel). By the columns, the inclusions of constants $\check{C}^p \to K^{p,\bullet}$ are quasi-isomorphisms, so the induced map from the cone of $\delta : \check{C}^0 \to \check{C}^1$, which computes $\check{H}^*$, to T is a quasi-isomorphism by the five lemma applied to the long exact sequences of the cones. □

14.2 Sheaves in a nutshell

A *presheaf* $\mathcal{F}$ on M assigns a vector space $\mathcal{F}(U)$ to each open set and restriction maps $\mathcal{F}(U) \to \mathcal{F}(V)$ for $V \subset U$, compatible with composition. It is a *sheaf* if sections that agree on overlaps glue uniquely. Examples: Ω^q $(U \mapsto \Omega^q(U))$, the constant sheaf $\mathbb{C}_M$ (locally constant functions), and $\mathcal{D}_M$ $(U \mapsto \mathcal{D}(U))$. The *stalk* at x is $\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U)$. Every presheaf has a *sheafification* with the same stalks. A sheaf is *fine* if it admits partitions of unity, as all $C^\infty(M)$-modules do; fine sheaves behave like the rows of Lemma 14.1. A morphism of complexes of sheaves is a *quasi-isomorphism* if it induces isomorphisms on the stalks of the cohomology sheaves; Lemma 14.1(columns) says that $\mathbb{C}_M \to \Omega^\bullet$ is one. The *hypercohomology* $\mathbb{H}^*(M, \mathcal{F}^\bullet)$ of a complex of sheaves is the cohomology of the total complex of the Čech double complex for a good cover (in general a limit over covers), as in Theorem 14.2.

Quoted background 14.3 (Weil's theorem in general, (G5), (G12))**.** (a) Quasi-isomorphic complexes of sheaves have isomorphic hypercohomology. (b) If every $\mathcal{F}^q$ is fine, $\mathbb{H}^*(M, \mathcal{F}^\bullet) = H^*(\Gamma(M, \mathcal{F}^\bullet))$. (c) $\mathbb{H}^*(M, \mathbb{C}_M[m]) = H^{*+m}(M; \mathbb{C})$, where $[m]$ shifts degrees down by m. In particular $H^*_{\mathrm{dR}}(M) = \mathbb{H}^*(M, \Omega^\bullet) = \mathbb{H}^*(M, \mathbb{C}_M) = H^*(M; \mathbb{C})$ [3, 34].



(a) a good cover of S^1

$$\begin{array}{ccccc} \check{C}^0 & \xrightarrow{\delta} & \check{C}^1 & & \text{columns exact:} \\ \downarrow & & \downarrow & & \text{Poincaré lemma} \\ \Omega^0(S^1) & \longrightarrow & K^{0,0} & \xrightarrow{\delta} & K^{1,0} \\ \downarrow d & & \downarrow d & & \downarrow d \\ \Omega^1(S^1) & \longrightarrow & K^{0,1} & \xrightarrow{\delta} & K^{1,1} \end{array}$$

rows exact:
partition of unity

(b) the Čech–de Rham double complex

Figure 9: Weil's proof of de Rham's theorem on S^1. (a) Three arcs with connected pairwise intersections and empty triple intersection. (b) Exact rows (gluing with a partition of unity) identify the total cohomology with de Rham cohomology; exact columns (the Poincaré lemma on arcs) identify it with Čech cohomology of constants. BG replace forms by Hochschild chains of differential operators, and the Poincaré lemma by Theorem 13.1.

14.3 Differential operators via sheaves

BG's analogue [7, §5]: let $\mathcal{C}^{-k}$ be the sheaf associated with the presheaf $U \mapsto C_k(\mathcal{D}(U))$, so that $\mathcal{C}^\bullet$ is a complex of sheaves in non-positive degrees, with differential b. Its cohomology sheaves have stalks $\varinjlim_{U \ni x} \mathrm{HH}_k(\mathcal{D}(U))$. Balls are cofinal among neighbourhoods of x, so by Theorem 13.1 (the Poincaré lemma) the only non-zero cohomology sheaf is $\mathbb{C}_M$ in degree $-2n$:

$$\mathcal{C}^\bullet \simeq \mathbb{C}_M[2n].$$

Assuming that the global Hochschild complex computes the hypercohomology, $\mathrm{HH}_k(\mathcal{D}(M)) = \mathbb{H}^{-k}(M, \mathcal{C}^\bullet) = \mathbb{H}^{-k}(M, \mathbb{C}_M[2n]) = H^{2n-k}(M)$, which is Theorem 13.2.

14.4 Cyclic homology via sheaves

Let $\mathcal{C}^{\lambda,\bullet}$ be the analogous complex built from Connes' cyclic complex. By Theorem 13.2 for balls (or Theorem 13.1 and the cyclic version of its proof), $\mathrm{HC}_N(\mathcal{D}(U)) = \mathbb{C}$ for $N = 2n + 2j$, $j \geq 0$, and 0 otherwise, so the cohomology sheaves of $\mathcal{C}^{\lambda,\bullet}$ are $\mathbb{C}_M$ in degrees $-2n, -2n-2, \ldots$ and zero otherwise.

Proposition 14.4 ([7, §5]). *In the derived category of sheaves on M, $\mathcal{C}^{\lambda,\bullet} \simeq \bigoplus_{j \geq 0} \mathbb{C}_M[2n + 2j]$. Consequently, granting the identification of global chains with hypercohomology, $\mathrm{HC}_N(\mathcal{D}(M)) \cong \bigoplus_{j \geq 0} H^{2n-N+2j}(M)$.*

Argument (in the derived category, (G12)). Since $-2n$ is the highest degree with a non-zero cohomology sheaf, there is a truncation morphism $t : \mathcal{C}^{\lambda,\bullet} \to \mathbb{C}_M[2n]$. The composite $t \circ I : \mathcal{C}^\bullet \to \mathcal{C}^{\lambda,\bullet} \to \mathbb{C}_M[2n]$ is a quasi-isomorphism: on stalks in degree $-2n$ it is $I : \mathrm{HH}_{2n}(\mathcal{D}(U)) \to \mathrm{HC}_{2n}(\mathcal{D}(U))$, which is an isomorphism for balls by the SBI sequence, since HC_{2n-1} and HC_{2n-2} vanish there. Hence I has a left inverse up to quasi-isomorphism, and $\mathbb{C}_M[2n]$ is a direct summand of $\mathcal{C}^{\lambda,\bullet}$, with complement identified by S with $\mathcal{C}^{\lambda,\bullet}[2]$. Induction on j gives the statement. □

Remark on BG 14.5 (Gaps in the sheaf-theoretic argument). (1) The presheaf $U \mapsto C_k(\mathcal{D}(U))$ assigns to U functions on U^{k+1}, not on U, so its sections over $U \cup V$ are not determined by those over U and V. The step "the global Hochschild complex computes the hypercohomology of $\mathcal{C}^\bullet$" therefore needs a *localization* theorem: chains supported near the diagonal compute Hochschild homology, and they form a complex of fine sheaves. This is true for the algebras considered here (Teleman [31]; Brylinski [6]), but it is not proved in [7]; it is (G12). (2) BG write "Hochschild cohomology" where homology is meant. (3) In the cyclic argument BG say that I is "a left inverse to the map $\mathbb{C}_M[2n] \to \mathcal{C}^{\lambda,*}$". The morphisms go the other way, as above: the composite $\mathcal{C}^\bullet \to \mathcal{C}^{\lambda,\bullet} \to \mathbb{C}_M[2n]$ is a quasi-isomorphism. (4) Theorem 13.2 gives an independent proof of both statements, modulo (G11).

14.5 Microdifferential operators

For symbols, the Poincaré lemma must live on $S^*M \times S^1$. BG obtain the circle from a complexification. We describe the objects and quote the results.

Microdifferential operators [28, 16]. Let X be a complex manifold of dimension n. On T^*X there is a sheaf of rings $\mathcal{E}_X$, locally constant along the orbits of $\mathbb{C}^\times$ on $\dot{T}^*X$. A section over a conic open set is a formal sum $\sum_{j \le m} p_j(z,\zeta)$ of holomorphic functions homogeneous of degree j in ζ, subject to a growth condition as $j \to -\infty$ (on compact sets $|p_j| \le CR^{-j}(-j)!$). The product is the composition formula (1), with ∂_ζ and D_z. So $\mathcal{E}_X$ is an analytic version of $\mathcal{P}$. There are larger sheaves $\mathcal{E}_X^\infty$ (operators of infinite order) and $\mathcal{E}_\mathbb{R}$ (defined through cohomology with supports; BG describe its stalks at $(0, dz_1)$ by holomorphic kernels on wedge-shaped regions modulo kernels that extend across one of the walls). The stalks of $\mathcal{E}_\mathbb{R}$ carry a product only modulo such kernels; $\mathcal{E}^\infty = \pi'_*\mathcal{E}_\mathbb{R}$.

Real symbols. Let M be real analytic (every smooth closed manifold has a real analytic structure) and $M \hookrightarrow M_\mathbb{C}$ a complexification. Then Figure 10 commutes; i' is a closed immersion followed by an étale map of degree 2. After a Riemannian metric is chosen, the $\mathbb{C}^\times$-bundle $\pi' : \dot{T}^*M_\mathbb{C} \to \dot{T}^*M_\mathbb{C}/\mathbb{C}^\times$ trivializes over S^*M, so $\pi'^{-1}(S^*M) \cong S^*M \times \mathbb{C}^\times$, which is homotopy equivalent to $S^*M \times S^1$.

$$\begin{array}{ccc}
S^*M \times \mathbb{C}^\times & \longrightarrow & \dot{T}^*M_\mathbb{C} \\
\downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \pi'} \\
S^*M & \xrightarrow{\;i'\;} & \dot{T}^*M_\mathbb{C}/\mathbb{C}^\times \\
\downarrow & & \downarrow \\
M & \xrightarrow{\;i\;} & M_\mathbb{C}
\end{array}$$

Figure 10: The diagram of [7, §5]. The circle S^1 of Theorem 1.1 reappears as $\mathbb{C}^\times$, the fibre of π' over the real cosphere bundle.

Quoted background 14.6 (Microlocal inputs, (G13))**.** The following are quoted in [7, §5].

1. The dual of the $\mathcal{E}(U \times U)$-module $\mathcal{E}(U)$ is $\mathcal{E}(U)[-2n]$ (Brylinski [6, Lemma 2.2.1]); $\mathcal{E}_\mathbb{R}(U)$ is faithfully flat over $\mathcal{E}(U)$ [28]. Hence $\mathrm{HH}_*(\mathcal{E}_\mathbb{R}(U)) \cong \mathrm{Ext}^{2n-*}_{\mathcal{E}_\mathbb{R}(U\times U)}(\mathcal{E}_\mathbb{R}(U), \mathcal{E}_\mathbb{R}(U))$.

2. $\mathcal{E}(U)$ is a holonomic module over $\mathcal{E}(U \times U)$, supported on the diagonal, of multiplicity one, and [16, Thm. 3.2.1] computes the Ext groups. The result is BG's microlocal Poincaré lemma: for $U \subset \dot{T}^*M_\mathbb{C}$ diffeomorphic to a ball, $\mathrm{HH}_q(\mathcal{E}_\mathbb{R}(U)) = \mathbb{C}$ for $q = 2n$ and 0 otherwise.

3. $\mathcal{E}_{\mathbb{C}^n}$ is a holonomic $\mathcal{E}_{\mathbb{C}^n\times\mathbb{C}^n}$-module with regular singularities [28, Thm. 5.4.1].

4. For holonomic $\mathcal{E}_X$-modules $\mathcal{M}, \mathcal{N}$ with regular singularities, the map
$$\mathrm{Ext}^i_{\mathcal{E}_X}(\mathcal{M},\mathcal{N}) \to \mathrm{Ext}^i_{\mathcal{E}^\infty_X}(\mathcal{M}^\infty,\mathcal{N}^\infty)$$
is an isomorphism (Kashiwara–Kawai [17, Thm. 6.1.3]). With (3), the inclusion $\mathcal{E} \subset \mathcal{E}^\infty$ induces isomorphisms on Hochschild and hence on cyclic homology.

With these inputs, and "without giving all the detailed sheaf-theory arguments required", BG state:

Quoted background 14.7 ([7, §5, Theorem])**.** If U is an open subset of $\dot{T}^*X/\mathbb{C}^\times$, X a complex manifold of dimension n, then $\mathrm{HH}_*(\mathcal{E}(U)) \cong \mathrm{HH}_*(\mathcal{E}^\infty(U)) \cong H^{2n-*}(\pi'^{-1}(U))$ and $\mathrm{HC}_*(\mathcal{E}(U)) \cong H^{2n-*}(\pi'^{-1}(U))[u]$.

Taking $X = M_{\mathbb{C}}$ and the limit over neighbourhoods of S^*M in $\dot{T}^*M_{\mathbb{C}}/\mathbb{C}^\times$, one obtains $H^{2n-*}(S^*M \times \mathbb{C}^\times) = H^{2n-*}(S^*M \times S^1)$. BG assert that a comparison of the spectral sequences of $\mathcal{P}$ and of $\mathcal{E}$ identifies this with $\mathrm{HH}_*(\mathcal{P})$. We have not checked these arguments and treat the whole of this subsection as quoted background. The point of the method is structural: the circle in the answer for symbols is the fibre $\mathbb{C}^\times$ of the complexified cotangent bundle, and the local computation is a statement about holonomic modules.

15 Worked examples

What this section does. We compute HH_*, HC_* and the residue for the circle, the torus and spheres, check the dimension counts against the Gysin formula of Correction 11.9 and against the Euler characteristic, and record a few explicit classes and traces.

15.1 The circle

Let $M = S^1 = \mathbb{R}/2\pi\mathbb{Z}$, $n = 1$. Then $\dot{T}^*S^1 = S^1 \times (\mathbb{R} \setminus 0)$ has two components, $\xi > 0$ and $\xi < 0$, and $S^*S^1 \times S^1$ is two tori (Figure 8). So
$$\dim \mathrm{HH}_k(\mathcal{P}(S^1)) = 2,\ 4,\ 2 \qquad (k = 0, 1, 2), \qquad \mathrm{HH}_k = 0\ (k \geq 3).$$
Let $\chi_\pm$ be the characteristic function of $\pm\xi > 0$; it is homogeneous of degree 0 and smooth on $\dot{T}^*S^1$ (it is the symbol of the projection onto non-negative or negative Fourier modes, modulo smoothing).

Degree 0. The leading term of a 0-cycle has order $k - n = -1$. The two classes are detected by the traces $\mathrm{res}^\pm(A) = (2\pi)^{-1}\int_0^{2\pi} a_{-1}(x, \pm 1)\,dx$. For example $A = (1 + D^2)^{-1/2}$ has $a_{-1} = |\xi|^{-1}$, so $\mathrm{res}^\pm(A) = 1$ and $\mathrm{res}(A) = 2$. By Proposition 12.15 with $P = A$ and A itself as the operator of order $m = 1$, $\mathrm{Res}_{s=0}\mathrm{Tr}(A^{-1-s}) = \mathrm{res}(A) = 2$. Directly, $\mathrm{Tr}(A^{-1-s}) = \sum_{k\in\mathbb{Z}}\langle k\rangle^{-1-s}$, and $\sum_k \langle k\rangle^{-1-s} = 2\int_1^\infty t^{-1-s}\,dt + O(1) = 2/s + O(1)$. The difference $\mathrm{res}^+ - \mathrm{res}^-$ is a trace that is not a multiple of res, which shows that $n \geq 2$ is needed in Theorem 12.9.

Degree 1. The leading term has order 0, and $E^2_{0,1} = H^1(\Omega_{(0)})$ after $*$, which is the identity on 1-forms in dimension 2 (Section 6.4). On each component the closed homogeneous 1-forms of degree 0 are spanned, modulo exact ones, by dx and $d\xi/\xi = dr/r$. So HH_1 has the four classes with leading terms
$$-i\,\chi_\pm e^{-ix} \otimes e^{ix} \ \mapsto\ \chi_\pm\,dx, \qquad \chi_\pm \xi^{-1} \otimes \xi \ \mapsto\ \chi_\pm\,\frac{d\xi}{\xi}.$$

(Here ξ is the symbol of D, and $e^{\pm ix}$ are multiplication operators.)

Degree 2. The leading term has order 1, and $*^{-1}(\chi_\pm) = \chi_\pm\omega$. The classes have leading terms $-i\varphi(\chi_\pm e^{-ix}; \xi, e^{ix})$, whose HKR image is $-i\chi_\pm e^{-ix}\, d\xi \wedge (ie^{ix}dx) = \chi_\pm\omega$.

Cyclic. By Theorem 11.7, $\dim \mathrm{HC}_N = 2, 4, 4, 4, 4, \ldots$ for $N = 0, 1, 2, 3, \ldots$ (for $N \geq 2$, HC_N is $H^0 \oplus H^2$ or H^1 of the two tori), and $\dim \mathrm{HP}_0 = \dim \mathrm{HP}_1 = 4$.

15.2 The torus

Let $M = T^2 = \mathbb{R}^2/2\pi\mathbb{Z}^2$, $n = 2$. Then $S^*T^2 = T^2 \times S^1 = T^3$, and $S^*T^2 \times S^1 = T^4$. So

$$\dim \mathrm{HH}_k(\mathcal{P}(T^2)) = \binom{4}{4-k} = 1,\ 4,\ 6,\ 4,\ 1 \qquad (k = 0, \ldots, 4).$$

Gysin check. Since $\chi(T^2) = 0$, the formula of Correction 11.9 applies: $\mathrm{HH}_k \cong H^{4-k} \oplus H^{3-k} \oplus H^{3-k} \oplus H^{2-k}$ of T^2, with dimensions $0+0+0+1, 0+1+1+2, 1+2+2+1, 2+1+1+0, 1$, as it should be.

The residue. $\mathrm{HH}_0 = \mathbb{C}$, and every trace is a multiple of res. Let Δ be the Laplacian and $P = \Delta^{-1}$ on functions orthogonal to the constants, 0 on constants. Then $p_{-2} = |\xi|^{-2}$, and

$$\mathrm{res}(P) = (2\pi)^{-2} \int_{T^2} dx \int_{|\xi|=1} dS = (2\pi)^{-2}(2\pi)^2\, 2\pi = 2\pi.$$

With $A = \Delta + \Pi_0$ (Π_0 the projection onto constants), of order $m = 2$, Proposition 12.15 predicts $\mathrm{Res}_{s=0} \mathrm{Tr}(PA^{-s}) = \frac{1}{2}\,\mathrm{res}(P) = \pi$. Directly, $\mathrm{Tr}(PA^{-s}) = \sum_{k \in \mathbb{Z}^2 \setminus 0} |k|^{-2-2s}$, an Epstein zeta function, and comparing the sum with $\int_{|t|\geq 1} |t|^{-2-2s} dt = 2\pi \int_1^\infty r^{-1-2s} dr = \pi/s$ shows that the residue at $s = 0$ is π. The factor $\frac{1}{2} = \frac{1}{m}$ is the one missing from BG's Proposition (2) (Correction 12.12).

15.3 Spheres

By the Gysin sequence, the real cohomology of S^*S^n is:

- for n even, $\mathbb{C}$ in degrees 0 and $2n-1$ (the Euler class is $\chi(S^n) = 2$ times a generator, so $H^0(S^n) \to H^n(S^n)$ is onto and the middle classes cancel);
- for n odd, $\mathbb{C}$ in degrees 0, $n-1$, n, $2n-1$ (the Euler class vanishes).

Multiplying by S^1 and using $\mathrm{HH}_k = H^{2n-k}$:

- n even: $\mathrm{HH}_k(\mathcal{P}(S^n)) = \mathbb{C}$ for $k \in \{0, 1, 2n-1, 2n\}$, and 0 otherwise;
- n odd, $n \geq 3$: $\dim \mathrm{HH}_k = 1$ for $k \in \{0, 1, n-1, n+1, 2n-1, 2n\}$, $\dim \mathrm{HH}_n = 2$, and 0 otherwise.

For S^2 this gives $1, 1, 0, 1, 1$, and BG's formula $H^{1-k}(S^2) \oplus H^{2-k}(S^2)$ gives $1, 1, 1, 0, 0$ (Correction 11.9). For $S^3 = SU(2)$, which is parallelizable, $S^*S^3 = S^3 \times S^2$ and $S^*S^3 \times S^1$ has Poincaré polynomial $(1+t^3)(1+t^2)(1+t)$, giving $1, 1, 1, 2, 1, 1, 1$, in agreement with the list.

15.4 Dimension checks

Table 2 collects the dimensions. In each row the alternating sum is $0 = \chi(S^*M \times S^1)$, HH_0 has dimension equal to the number of residue traces, and HH_{2n} has dimension equal to the number of components of S^*M.

For the torus T^n, $S^*T^n = T^n \times S^{n-1}$, so the Poincaré polynomial of $S^*T^n \times S^1$ is $(1+t)^{n+1}(1+t^{n-1})$, and $\dim \mathrm{HH}_k$ is the coefficient of t^{2n-k}. For $n = 2$ this is $(1+t)^3(1+t) = (1+t)^4$, as above.

M	n	$\dim \mathrm{HH}_k(\mathcal{P}(M))$, $k = 0, 1, \dots, 2n$
S^1	1	2, 4, 2
T^2	2	1, 4, 6, 4, 1
S^2	2	1, 1, 0, 1, 1
S^3	3	1, 1, 1, 2, 1, 1, 1
T^3	3	1, 4, 7, 8, 7, 4, 1
S^4	4	1, 1, 0, 0, 0, 0, 0, 1, 1

Table 2: Dimensions of $\mathrm{HH}_k(\mathcal{P}(M))$.

15.5 Differential operators

By Theorem 13.1, $\mathrm{HH}_*(A_n)$ and $\mathrm{HH}_*(\mathcal{D}(\mathbb{R}^n))$ are $\mathbb{C}$ in degree $2n$. For the circle, Theorem 13.2 (which for $M = S^1$ can be proved by the local computation and Fourier series, but we rely on (G11)) gives $\mathrm{HH}_1(\mathcal{D}(S^1)) = \mathrm{HH}_2(\mathcal{D}(S^1)) = \mathbb{C}$ and $\mathrm{HH}_0(\mathcal{D}(S^1)) = 0$. The last statement can be checked by hand: $[\partial, f] = f'$, so functions with mean zero are commutators, and $1 = -i[e^{-ix}\partial, e^{ix}]$; then $[\partial, f\partial^k] = f'\partial^k$ and $[e^{-ix}\partial, e^{ix}\partial^k] = i\partial^k$ show that every differential operator is a sum of commutators. So there is no trace on $\mathcal{D}(S^1)$, in contrast with $\mathcal{P}(S^1)$.

A Proof of Connes' theorem

What this section does. We prove Theorem 3.15: for a manifold M without boundary, compact or not, χ induces $\mathrm{HH}_k(C^\infty(M)) \cong \Omega^k(M)$. The argument follows Connes [9] as presented in [25, Ch. 10]: reduce to normalized chains, expand them in Taylor series at the diagonal, lift the Taylor series with the polynomial case, and divide the flat remainder.

By (G2), $C_k = C^\infty(M^{k+1})$, with the face maps of Section 3.7.

A.1 Normalized chains

Let $D_i = \{z_i = z_{i+1}\} \subset M^{k+1}$ for $0 \le i < k$, and $D_k = \{z_k = z_0\}$. Put

$$N_k = \{u \in C_k : \ u|_{D_i} = 0, \ 0 \le i < k\}, \qquad M_k = \{u \in N_k : \ u|_{D_k} = 0\}, \qquad B_k = d_{k+1}(N_{k+1}),$$

with $N_0 = M_0 = C_0$. If $u \in N_k$, then $d_i u = 0$ for $i < k$, so $bu = (-1)^k d_k u$, and $d_k u \in N_{k-1}$. So N is a subcomplex; its k-cycles are M_k, and its k-boundaries are $B_k \subset M_k$. The same definitions apply to polynomial algebras, with $C^\infty(M^{k+1})$ replaced by polynomials in $z_0, \dots, z_k$.

Proposition A.1. *$N \subset C$ is a quasi-isomorphism. Hence $\mathrm{HH}_k = M_k / B_k$, and an element of M_k that is a boundary in C lies in B_k.*

Proof. For $j \ge 0$ let $J^j_m = \{u \in C_m : \ u|_{D_i} = 0 \text{ for } i < \min(j, m)\}$. One checks from the definitions that each J^j is a subcomplex, $J^0 = C$, and $J^j_m = N_m$ for $m \le j$. For $m \ge j+1$ and $u \in J^j_m$ put $u'(z) = u(z_0, \dots, z_j, z_j, z_{j+2}, \dots, z_m)$ and $u'' = u - u'$. Then u'' vanishes on D_j, so $u'' \in J^{j+1}_m$, and u' lies in the space V^j_m of elements of J^j_m independent of z_{j+1} ($V^j_m = 0$ for $m \le j$). The decomposition is unique, so $J^j = J^{j+1} \oplus V^j$. On V^j, $d_i = 0$ for $i < j$ and $d_j = d_{j+1}$, so $bu = \sum_{i \ge j+2} (-1)^i d_i u \in V^j$: V^j is a subcomplex. Let $(Hu)(z_0, \dots, z_{m+1}) = u(z_0, \dots, z_{j+1}, z_{j+3}, \dots, z_{m+1})$, inserting a dummy variable in position $j+2$. Then $d_{j+2}H = 1$, $d_i H = H d_{i-1}$ for $i \ge j+3$, and the faces d_i, $i \le j+1$,

of Hu vanish or cancel as before. So $bH + Hb = (-1)^j$ on V^j, V^j is contractible, and $J^{j+1} \subset J^j$ is a quasi-isomorphism. Hence $H_k(C) \cong H_k(J^{k+1}) = H_k(N)$. □

Lemma A.2. (i) *B_k is a $C^\infty(M^{k+1})$-submodule of M_k; more precisely, if $v \in N_{k+1}$ and $\psi \in C^\infty(M^{k+1})$, then $\psi\, d_{k+1}v = d_{k+1}(\tilde{\psi}v)$ with $\tilde{\psi}(z_0,\dots,z_{k+1}) = \psi(z_0,\dots,z_k)$, and $\tilde{\psi}v \in N_{k+1}$.* (ii) *If $f \in C^\infty(M)$ and $g \in M_k$, then $(f(z_j) - f(z_{j+1}))g \in B_k$ for $0 \le j < k$, and $(f(z_k) - f(z_0))g \in B_k$.*

Proof. (i) is immediate. (ii) For $j < k$ let $w = (-1)^{j+1}(f(z_j) - f(z_{j+1}))\, g(z_0,\dots,z_j,z_{j+2},\dots,z_{k+1}) \in C_{k+1}$. The face d_j kills the first factor; the faces d_i, $i \neq j, j+1$, restrict g to one of its diagonals D_l, $l \le k$, where it vanishes since $g \in M_k$. So $bw = (-1)^{j+1}d_{j+1}w = (f(z_j) - f(z_{j+1}))g$, which lies in M_k and is a boundary, hence in B_k by Proposition A.1. The last case is the sum of the others. □

A.2 Taylor expansion at the diagonal

Near a point $(x_0,\dots,x_0)$ of the diagonal Δ choose a chart around x_0 whose image U is a ball, and use the coordinates $x = z_0$, $y_j = z_j - z_{j-1}$ ($1 \le j \le k$) on U^{k+1}. Then $D_{j-1} = \{y_j = 0\}$, $D_k = \{y_1 + \cdots + y_k = 0\}$, and $\Delta = \{y = 0\}$.

Lemma A.3. *A k-linear form q on $\mathbb{R}^n$ with $q(y_1,\dots,y_{k-1},-(y_1+\cdots+y_{k-1})) \equiv 0$ is alternating, and conversely.*

Proof. If q is alternating the expression is $-\sum_{j<k} q(y_1,\dots,y_{k-1},y_j) = 0$. Conversely the terms $q(y_1,\dots,y_{k-1},y_j)$ have distinct multidegrees, so each vanishes; thus q is alternating in the arguments j, k, and these transpositions generate $\mathfrak{S}_k$. □

Lemma A.4. *Let $u \in N_k$. Each monomial of the Taylor series of u in y at Δ contains a component of every y_j, so the series starts in degree k, with $u_{(k)} = \sum_I w_I(x)\, y_{1,i_1}\cdots y_{k,i_k}$, and $\chi(u) = \frac{1}{k!}\sum_I w_I\, dx^{i_1}\wedge\cdots\wedge dx^{i_k}$. If $u \in M_k$, then $u_{(k)}(x;\cdot)$ is alternating, and $\chi(u)(x) = 0$ if and only if $u_{(k)}(x;\cdot) = 0$.*

Proof. The Taylor series vanishes on each $\{y_j = 0\}$. Since $z_l = x + y_1 + \cdots + y_l$, $\partial_{z_l} = \partial_{y_l} - \partial_{y_{l+1}}$ (and $\partial_{z_k} = \partial_{y_k}$), and in $\partial_{z_1^{i_1}}\cdots\partial_{z_k^{i_k}}u$ at $y = 0$ only the term in which each y_l is differentiated once survives. If $u \in M_k$, $u_{(k)}$ vanishes on $\{y_1 + \cdots + y_k = 0\}$; apply Lemma A.3. □

Lemma A.5. *Let $k \ge 1$ and $m \ge k+1$. Let $\mathcal{M}_m$ be the homogeneous polynomials of degree m in $(y_1,\dots,y_k)$ vanishing on each $\{y_j = 0\}$ and on $\{y_1 + \cdots + y_k = 0\}$, and $\mathcal{V}_m$ those in $(y_1,\dots,y_{k+1})$ vanishing on each $\{y_j = 0\}$. Then $R : \mathcal{V}_m \to \mathcal{M}_m$, $(RV)(y) = V(y_1,\dots,y_k,-(y_1+\cdots+y_k))$, is onto.*

Proof. Given $q \in \mathcal{M}_m$, the polynomial chain $u(z) = q(z_1 - z_0,\dots,z_k - z_{k-1})$ lies in M_k of the polynomial algebra, and $\chi(u) = 0$ because $m > k$. By Theorem 3.12 and Proposition A.1 for polynomials, $u = d_{k+1}v$ with $v \in N_{k+1}$. In the coordinates $x, y_1,\dots,y_{k+1}$ the component of v of degree 0 in x and m in y lies in $\mathcal{V}_m$ (the conditions are homogeneous and linear), and R maps it to q. □

Lemma A.6. *Let $k \ge 1$ and $u \in M_k$ with $\chi(u) = 0$. Every point $p \in M^{k+1}$ has a neighbourhood on which u agrees with an element of B_k.*

Proof. If $p \notin \Delta$, then $p_j \neq p_{j+1}$ for some $j < k$. Choose $f \in C^\infty(M)$ with $f(p_j) \neq f(p_{j+1})$ and $\psi \in C^\infty(M^{k+1})$ equal to 1 near p and supported where $f(z_j) \neq f(z_{j+1})$. Then $g = \psi u/(f(z_j) - f(z_{j+1})) \in M_k$, and near p, $u = (f(z_j) - f(z_{j+1}))g \in B_k$ by Lemma A.2(ii).

Let $p = (x_0,\dots,x_0)$. By Lemma A.4 and $\chi(u) = 0$, the Taylor series of u at Δ is $\sum_{m\ge k+1} u_{(m)}$ with $u_{(m)} \in C^\infty(U)\otimes\mathcal{M}_m$. Using a linear right inverse of R, choose $V_{(m)} \in C^\infty(U)\otimes\mathcal{V}_m$ with

$RV_{(m)} = u_{(m)}$. By Borel's lemma, $V = \sum_m \chi_0(\lambda_m|y|)V_{(m)}$, with $\chi_0 \in C_c^\infty(\mathbb{R})$ equal to 1 near 0 and $\lambda_m \to \infty$ fast enough, is smooth for $x \in U$ and small y, has Taylor series $\sum V_{(m)}$, and vanishes on each $\{y_j = 0\}$. Let Θ be a cut-off near $(x_0, \dots, x_0) \in M^{k+2}$, and $v = \Theta V$, written in the variables z. Then $v \in N_{k+1}$, and $s = d_{k+1}v \in B_k$ has the same Taylor series as u at Δ near p, because $z_{k+1} = z_0$ means $y_{k+1} = -(y_1 + \cdots + y_k)$.

So $g = u - s \in M_k$ vanishes to infinite order at Δ near p. Since U^{k+1} is convex, Taylor's formula along segments gives $|\partial^\alpha g| \le C_{\alpha,N}|y|^N$ for all N. Hence $g_{j,i} = y_{j,i}g/|y|^2$, extended by 0 on Δ, is smooth and vanishes where g does. With $f_i \in C^\infty(M)$ equal to the i-th coordinate near x_0 and a cut-off ψ equal to 1 near p, $g = \sum_{j,i}(f_i(z_j) - f_i(z_{j-1}))\psi g_{j,i}$ near p, which lies in B_k by Lemma A.2(ii). So $u = s + g$ agrees with an element of B_k near p. □

A.3 The theorem

Proof of Theorem 3.15. For $k = 0$, $b = 0$ on C_1, and $\mathrm{HH}_0 = C^\infty(M)$. Let $k \ge 1$. χ_* is onto by Lemma 3.11: every k-form is a finite sum $\sum a_0\, df_1 \wedge \cdots \wedge df_k$, using the coordinate functions of an embedding $M \subset \mathbb{R}^N$. By Proposition A.1, a class is represented by $u \in M_k$, and it vanishes if and only if $u \in B_k$. Let $\chi(u) = 0$. By Lemma A.6 there are a locally finite partition of unity $\sum_\beta \psi_\beta = 1$ on M^{k+1} and elements $s_\beta = d_{k+1}v_\beta \in B_k$ with $u = s_\beta$ on $\operatorname{supp}\psi_\beta$. By Lemma A.2(i), $u = \sum_\beta \psi_\beta s_\beta = d_{k+1}\big(\sum_\beta \tilde\psi_\beta v_\beta\big)$. The last sum is locally finite on M^{k+2}, since $\tilde\psi_\beta$ depends only on the first $k+1$ variables, and each term lies in N_{k+1}. So $u \in B_k$. □

B The uncompleted chain model

What this section does. We show that Model I, $C_k(\mathcal{P}) = \varinjlim_N \mathcal{P}_N^{\widehat\otimes(k+1)}$, has the same Hochschild homology as Model II, its completion. Model I is not complete (Figure 6), so Proposition 5.7 does not apply to it directly. Following Melrose and Nistor [26], we compare it with a "mixed" complex whose stages are complete. Throughout, M is closed.

B.1 The mixed complex

Let $\mathfrak{B} = \mathcal{P}_0 \subset \mathcal{P}$ be the subalgebra of symbols of order ≤ 0, and $\mathcal{A}_N = \mathcal{P}_N$. Fix an elliptic $T \in \mathcal{P}_{-1}$ with principal symbol r^{-1}, $r > 0$ homogeneous of degree 1. It is invertible in $\mathcal{P}$ with inverse of order 1 (parametrix construction). Put $u_N = T^{-N}$. Then $\mathcal{A}_N = \mathfrak{B}u_N = u_N\mathfrak{B}$, since $au_N^{-1} = aT^N \in \mathfrak{B}$ for $a \in \mathcal{A}_N$. The *mixed chains* are

$$C_k(\mathfrak{B}, \mathcal{P}) = \varinjlim_N S_{N,k}, \qquad S_{N,k} = \mathcal{A}_N \mathbin{\widehat\otimes} \mathfrak{B}^{\widehat\otimes k},$$

with the Hochschild boundary of $\mathcal{P}$ as a $\mathfrak{B}$-bimodule: d_0 and d_k put a_0b_1 and b_ka_0 in slot 0, and the inner faces multiply in $\mathfrak{B}$.

Lemma B.1. *Each S_N is a subcomplex. With the total order filtration it is complete, and its associated graded at total order q is a subspace of $C_{q,k}$.*

Proof. $\mathcal{A}_N$ is a $\mathfrak{B}$-bimodule, so S_N is a subcomplex. By (G2), $S_{N,k}$ is the product over multi-degrees $(j_0, \dots, j_k)$ with $j_0 \le N$ and $j_i \le 0$ of the spaces $C^\infty_{(j)}(Y^{k+1})$. At a fixed total order q only finitely many multi-degrees occur ($j_i \ge q - N$), so S_N consists of all formal series in the total order, bounded above by N, with components in the corresponding finite sums. □

Lemma B.2. *The inclusion $C(\mathfrak{B},\mathcal{P}) \to C(\mathcal{P})$ (Model I) is a quasi-isomorphism.*

Proof. This is [26], specialized to the closed case. Let $Q_k = \varinjlim_N \mathcal{A}_N^{\widehat{\otimes}(k+2)}$ and $P_k = \varinjlim_N \mathcal{A}_N \mathbin{\widehat{\otimes}} \mathfrak{B}^{\widehat{\otimes}k} \mathbin{\widehat{\otimes}} \mathcal{A}_N \subset Q_k$ for $k \geq 0$, with the bar differential b', and $P_{-1} = Q_{-1} = \mathcal{P}$, with $b' = \mu$ the multiplication $Q_0 \to \mathcal{P}$. These are complexes of $\mathcal{P}$-bimodules (acting on the outer slots). The contraction $s(a_0 \otimes \dots) = 1 \otimes a_0 \otimes \cdots$ shows that Q is acyclic.

For P, let $P^{(N)} \subset P$ be the subcomplex of chains whose last slot lies in $\mathcal{A}_N$ (b' maps that slot into $\mathfrak{B}\mathcal{A}_N = \mathcal{A}_N$), with $P^{(N)}_{-1} = \mathcal{P}$. Define

$$h_N(a_0 \otimes \beta \otimes a) = (-1)^{k+1} a_0 \otimes \beta \otimes a u_N^{-1} \otimes u_N, \qquad h_N(a) = a u_N^{-1} \otimes u_N,$$

for $\beta \in \mathfrak{B}^{\widehat{\otimes}k}$, $a \in \mathcal{A}_N$. Since $a u_N^{-1} \in \mathfrak{B}$, h_N maps $P^{(N)}_k$ to $P^{(N)}_{k+1}$. In $b'h_N + h_N b'$, the faces not involving the last two slots cancel in pairs, the face merging $a u_N^{-1}$ with u_N returns $a_0 \otimes \beta \otimes a$, and the face merging the last entry b_k of β with $a u_N^{-1}$ cancels h_N of the last face of b', since $(b_k a)u_N^{-1} = b_k(a u_N^{-1})$. So $b'h_N + h_N b' = 1$ on $P^{(N)}$, and every cycle of P lying in some $P^{(N)}$ is a boundary there.

Index the stages by $N \geq 0$. Let $V_0 = \mathfrak{B}$ and $V_N = C^\infty_{(N)}(Y)$ for $N \geq 1$, so that $\mathcal{A}_N = V_0 \oplus \cdots \oplus V_N$, a finite sum of closed subspaces. Put $W_{k,N} = \bigoplus_{\max N_i = N} V_{N_1} \mathbin{\widehat{\otimes}} \cdots \mathbin{\widehat{\otimes}} V_{N_k}$; then $\varinjlim_N \mathcal{A}_N^{\widehat{\otimes}k} = \bigoplus_N W_{k,N}$ algebraically. We construct a bimodule chain map $g : Q \to P$ over the identity of $\mathcal{P}$, by induction on k, with $g_{-1} = g_0 = 1$. If g_{k-1} maps the generators $1 \otimes W_{k-1,N} \otimes 1$ into $P^{(\nu(k-1,N))}$, then for $w \in W_{k,N}$ the element $z(w) = g_{k-1} b'(1 \otimes w \otimes 1)$ is a cycle of the augmented complex lying in $P^{(\nu)}$, $\nu = N + \max_{N' \leq 2N} \nu(k-1, N')$, and we set $g_k(1 \otimes w \otimes 1) = h_\nu(z(w))$, extended bimodule-linearly. This is continuous on each stage and satisfies $b' g_k = g_{k-1} b'$ on generators, hence everywhere. In the same way one builds bimodule homotopies $K : Q \to Q$ with $b'K + Kb' = \iota g - 1$, using s, and $H : P \to P$ with $b'H + Hb' = g\iota - 1$, using h_ν, where $\iota : P \to Q$ is the inclusion.

Finally, $\rho(a_0 \otimes w \otimes a) = a a_0 \otimes w$ maps Q_k onto Model I chains $C_k(\mathcal{P})$ and P_k onto $C_k(\mathfrak{B},\mathcal{P})$, with $\rho b' = b\rho$ and $\rho(aX) = \rho(Xa)$. For a bimodule map F between P and Q, $\bar{F}(a_0 \otimes w) = \rho F(a_0 \otimes w \otimes 1)$ satisfies $\bar{F}\rho = \rho F$. So $\bar{g}$ is a chain map $C(\mathcal{P}) \to C(\mathfrak{B},\mathcal{P})$, $\bar{\iota}$ is the inclusion, and $\bar{K}$, $\bar{H}$ are homotopies $\bar{\iota}\bar{g} \simeq 1$, $\bar{g}\bar{\iota} \simeq 1$. □

B.2 Convergence for the mixed complex

Lemma B.3. *For $q \leq N-1$, the b_0-homology of S_N at total order q is, via χ, the space $\Omega^k_{(q)}(Y)$ of all homogeneous k-forms of degree q.*

Proof. The associated graded algebra of $\mathfrak{B}$ is $R = C^\infty(S^*M)[t]$, $t = r^{-1}$ of degree -1, and the graded coefficient module is $t^{-N}R$, free of rank one with equal left and right actions. Multiplication by t_0^{-N} identifies the Hochschild complex of R with coefficients in $t^{-N}R$ with that of R, so by Corollary 3.16 the homology is t^{-N} times $\Omega^k(S^*M)[t] \oplus \Omega^{k-1}(S^*M)[t]\, dt$. A homogeneous form of degree q is $t^{-q}(\alpha - \beta \wedge dt/t)$, and it lies in this space if $N - q \geq 1$. □

Theorem B.4. *$\Theta : H_k(C(\mathfrak{B},\mathcal{P})) \to E^2_{k-n,k}$, the leading term of a representative of order $\leq k-n$, is well defined and an isomorphism.*

Proof. The first and second pages of the direct limit are those of Model II, by Lemma B.3 and the proof of Theorem 11.2, so E^2 lies on the line $p = k - n$. The existence of representatives in F_{k-n} and the well-definedness of Θ use finitely many steps (P) (proofs of Propositions 5.6(1) and 5.7, which off the line need no completeness), so they hold in the direct limit.

Steps inside a stage. Let c be a chain in S_N whose leading term c_p, of order $p \leq N-2$ and $p \neq k-n$, represents a d^1-cycle. Then $\theta = \chi(c_p)$ satisfies $\delta\theta = 0$, and by Corollary 9.4 $\theta = -i\delta\zeta$ with $\zeta = i\,\alpha \wedge \theta/(p+n-k)$, homogeneous of degree $p+1 \leq N-1$. Write ζ as a sum of forms $a_0\,da_1 \wedge \cdots \wedge da_k$ with $a_1, \ldots, a_k$ of order ≤ 0 (functions of degree 0 and r^{-1}, using $dr = -r^2 d(r^{-1})$) and a_0 of degree $\leq N$. Then $x = \sum \varphi(a_0; a_1, \ldots, a_k)$ lies in S_N, and the computation of Theorem 8.1 gives $\chi(b_1 x) = -i\delta\zeta = \theta$. By Lemma B.3, $c_p - b_1 x = b_0 y$ with y in S_N. So step (P) can be carried out inside S_N.

Injectivity and surjectivity. Given a cycle $c \in F_{k-n}$ with $[c_{k-n}] = 0$ in E^2, perform one step (P) in the direct limit, and choose N so that S_N contains the remainder and $N \geq k-n+2$. All later steps are off the line at orders $\leq k-n-1 \leq N-2$, so they stay in S_N, and the infinite sum of corrections converges in S_N by Lemma B.1. So c is a boundary. Surjectivity is proved in the same way, following the proof of Proposition 5.7. □

Theorem B.5. *The inclusion of Model I into Model II induces an isomorphism on Hochschild homology, compatible with the leading-term maps Θ. It also induces isomorphisms on cyclic homology.*

Proof. The composite $C(\mathfrak{B}, \mathcal{P}) \to C(\mathcal{P}) \to \widehat{C}(\mathcal{P})$ preserves the filtrations and the leading terms, so it intertwines the isomorphisms Θ of Theorems B.4 and 11.3 (Model II); hence it is an isomorphism on homology. The first map is an isomorphism by Lemma B.2, so the second is too. Both models are mixed complexes and the inclusion commutes with B, so Lemma 4.7 gives the statement for cyclic homology. □

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